Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics

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Abstract

In Observer Patch Holography (OPH) a world is a finite network of observer patches that repair disagreements on their overlaps. This paper is the mathematical layer of the OPH tower that treats that repair. A repair law whose accepted moves satisfy a transactional local diamond, strictly decrease a declared descent measure and are complete reaches one quotient normal form whatever the order of accepted repairs. This is Newman's lemma for a precisely stated acceptance filter, and explicit countermodels show that atomic commits, disjoint write sets or termination alone do not suffice. Canonical adaptive repair is eventually constant, pathwise weak fairness makes its stable state a normal form, and a stutter allowance q gives the sharp horizon of $(q + 1)$ times the initial mismatch count. Hidden local representation is gauge: the normal form descends to the quotient by changes that preserve overlap data.

The residue that local repair cannot remove is cycle holonomy. For affine abelian overlap constraints, global consistency holds exactly when every cycle holonomy vanishes, and with fixed edge offsets and positive weights a nonzero holonomy forces a strictly positive minimum mismatch. This frustration theorem is the mathematical seed of the OPH matter hypothesis, which reads stable massive matter as localized defects of native overlap transport; the theorem supplies neither a native defect nor its localization, motion or energy. Authenticated read-from provenance generates a finite informational order, a fair-block theorem controls noisy repair, and a fixed input-independent repair federation computes every finite Boolean function under fair scheduling. The bare consensus reduct determines no Lorentzian metric or Einstein equation; geometry, matter and dynamics enter through the higher layers under their own premises.

Interpretation and state variables. OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed. Its realization map must specify whether it matches one actual history, a family of preparation and intervention responses, or every independently allowed microstate. These are different requirements. A candidate must specify the history domain and closure operator; neither the schematic equation nor the finite constructions establish this physical realization or select a unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence, geometric domain, action, representation, or clock is a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

Basis, premise labels and paper set. OPH has one hypothesis, closure: the universe is a timeless self-referential structure whose observers recover its specification from their records and realize it. Its internal-realization policy requires its operations and information access to be accounted within the structure; operational agreement on accepted shared data defines what is public. The executable specification of carrier, repair law, observer algebra and closure inputs is implemented and runs; it is a candidate judged by whether it reproduces what observers measure. The labels A1, A2 and A3 name theorem premises: the carrier, federation and response layer of that specification; the agreement condition with an additional response-endogeneity principle; and an adopted least-informative state-selection principle. Closure alone implies neither the entropy minimizer nor inner implementation of rechartings on the connected response algebra. The basis reference states this formally.[1] A flagship synthesis gives the account, an executable observer construction states the specification with its acceptance criteria and premise map, and the theorem library carries conditional implications whose antecedents that premise map tracks.

Time conventions. Host runtime, solver counters, native attempt counts, local event-clock readings, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks are declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

One theory and its executable hypothesis. OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

> *A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed* $3+1$*-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.*

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Coherence, protected codes, radial directions, fields, clocks and couplings may be specified as part of its design; their

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/docs/BASIS.md

implementation and observable consequences must be established on that process. A failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

The tower. The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

0. **Closure and agreement.** One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.

1. **Repair mathematics.** Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.

2. **Carrier and repair law.** Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.

3. **Executable observer.** Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.

4. **Clocks and fields.** Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A declared radial extension gives a conditional smooth $1+3$ scalar description. Complete finite source factors join spatial CAR dynamics, particle creation, repair and a common calibrated clock. Physical validation of that clock and energy normalization, and an interacting chiral $3+1$ limit, remain open.

5. **Gauge structure and matter.** Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. A specified coherent ordering memory protects native exchange-history sectors, and a colored extension implements their fermionic exchange. A localized physical defect, its maintenance-to-mass law and a predicted measured spectrum remain unestablished; the earlier double-load candidate is transient.

6. **Gravity.** Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.

7. **Cosmology and tests.** The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology papers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

Completion and stopping. The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning. The ordered joins, retained foundations and stopping rules are recorded with the executable model in the public repository.[2]

Fixed physical boundaries. Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a specified and implemented clock calibration relates it to duration. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

From repair topology to a field experiment. A specified repair process admits protected quantum history sectors, fermionic exchange and an executable coupling to field and clock observables. Its ingredients are specified observer, scheduler, repair and memory laws; they need not be uniquely selected to define a mathematical model. What must be established is that the resulting operations share a carrier, retain complete quantum outputs and reproduce the intended observables. The constructions below establish several such joins, including a joint source experiment. They do not identify a complete physical theory.

Protected transport sectors. On the native 66-state binary two-load configuration complex, an exact noncommutative reduction of the face and disjoint-move relations gives $\pi_1 = \langle g \mid g^2 \rangle = \mathbb{Z}/2$. The surviving loop has odd ordered-cover parity. This is topology of the declared transport and deformation domain, not topology of a localized field defect. The complete active branch $*$-algebra on either the unextended 78-state source or its binary shell is the full matrix algebra. A Hermitian source charge commuting with every branch is therefore scalar. This excludes that charge class, not every invariant subspace, expectation-conserved observable or controlled lifetime.

A coherent ordering bit changes the algebra. If a native branch maps x to $f_h(x)$, let $p_h(x)$ be the parity of the transported order and define

$$K_h = \frac{1}{\sqrt{2}} \sum_{x \in D_h} |f_h(x)\rangle\langle x| \otimes X^{p_h(x)}, \qquad Q = I_{66} \otimes X.$$

Every branch commutes with Q, so its two sharp sectors survive arbitrary finite record-adaptive schedules. An actual odd exchange loop returns the source and applies X to the bit: the memory participates in transport. Sector preservation does not assert preservation of every coherence between the sectors. For the all-seam C or E family, effect separation, a connected movement tree and the odd loop generate exactly $M_{66}(\mathbb{C}) \oplus M_{66}(\mathbb{C})$, with commutant $\mathrm{span}\{I, Q\}$, although the classical ordered cover is connected. The states $I_{66}/66 \otimes |\pm\rangle\langle\pm|$ have identical source and complete classical-record marginals; their distinction is quantum memory. Computational-basis reset

[2] https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/docs/COMPLETION_PROGRAM.md

or dephasing destroys it. Protection thus depends on the specified memory law preserving these sectors.

Literal source words implement this lift. Complete execution covers all 132 input columns on seam $(0, 5)$, and exact word audits cover all 1024 addresses of the 90 seam/current feedback words. Native outcomes, padding and source-service failures remain in the accounting. The result is a protected sector of the enlarged process; a spatially localized source realization of its charge remains an additional construction.

This same carrier also separates operator action from scalar repair cost. Writing $P_- = (I-X)/2$ and $M_h = \sum_{x\in D_h} |f_h(x)\rangle\langle x|$,

$$A_h = \pi \sum_{x\in D_h} p_h(x)|f_h(x)\rangle\langle f_h(x)| \otimes P_-, \qquad K_h = e^{-iA_h}(M_h \otimes I)/\sqrt{2}.$$

Two closed histories with identical multisets of complete branch labels and identical compiled resource totals induce respectively I and X. An additive scalar cost of those labels cannot determine this response; the displayed operator action does. Its phase is not physical energy per unit time. Nor is the reduced memory autonomous: its one-step flip probability is $26/495$, while the two-step Z multiplier differs from the squared one-step multiplier by $398/245025$. The continuing source must be retained in any proposed Hamiltonian or memory description.

Internal states and fermionic exchange. Give each occupied port an independent internal space V. The map

$$|\{x < y\}; a, b\rangle \longmapsto c^\dagger_{x,a} c^\dagger_{y,b}|0\rangle$$

fixes the CAR convention. Passing the occupied spectator reverses the internal slots and crosses one creation operator, hence

$$K_h^V = \frac{1}{\sqrt{2}} \sum_{x\in D_h} |f_h(x)\rangle\langle x| \otimes (-\mathrm{SWAP})^{p_h(x)}.$$

The complete effects remain the original domain projectors divided by two. Independent CAR calculations check every branch for $\dim V = 2, 3$; the crossing argument applies to every finite V. The odd loop acts as $-$SWAP, giving -1 on two equal internal states. Global Fock parity on this two-particle sector is $+1$ and cannot replace that exchange.

The ordering memory implements the same operation: for $J = -\mathrm{SWAP}$,

$$E_V\psi = (|0\rangle\psi + |1\rangle J\psi)/\sqrt{2}, \qquad (X \otimes I)E_V = E_V J.$$

Every native branch, and every common $U \otimes U$, intertwines through E_V. The negative exchange character acts on $\mathrm{Sym}^2 V$; the positive one acts on $\Lambda^2 V$. Both are compatible with identical fermions carrying independent internal states.

The finite algebra constrains but does not fix internal encodings. An untouched fiber of dimension m enlarges the repair commutant to $M_m \oplus M_m$. A separate collective-label encoding that puts all fifteen states of $\Lambda^2\mathbb{C}^5 \oplus \overline{\mathbb{C}^5}$ in the negative sector requires $m \geq 15$ and has an explicit inclusion at equality. This is not a universal memory bound for the two-particle construction. No element of the supplied $S(U(3) \times U(2))$ gauge group acts as $-I$ on the whole faithful fifteen-state module: its nonidentity involutions have signature $(7, 8)$. The smaller weak-parity encoding is consequently an internal-grading control, not an all-fermion exchange assignment.

An exact basis permutation joins that exterior module and all twelve currents to the retained chiral action. Specified two-component and three-family factors give $\dim V = 90$; neither family

count nor physical Lorentz spin is derived by this basis change. On twelve ports the hard-core two-particle dimension is 534,600, whereas the complete 1080-mode two-particle space has dimension 582,660. The remaining 48,060 states are same-port collisions and must be included when propagation or interactions can reach them.

One calibrated action and its observables. The Weyl–Higgs action uses explicit charged Yukawa, Higgs-background and CKM inputs with recorded matching conventions. Recovering them from its response verifies implementation; it is not an independent mass prediction. The creation-pairing matrix $\Delta = \epsilon_{\text{spin}} \otimes \overline{K}$ is antisymmetric on 90 modes. At rest,

$$B_0 = \begin{pmatrix} 0 & \Delta \\ \Delta^\dagger & 0 \end{pmatrix}, \qquad H = E_{\text{vac}} + \sum_{a=1}^{90} E_a \gamma_a^\dagger \gamma_a.$$

The canonical chart retains 84 positive charged directions, six zero-energy neutrino annihilators and every particle–hole partner. Choosing the annihilator half of the zero block specifies the static reference; it does not supply a positive-energy continuation at nonzero momentum. This minimal action has massless neutrinos; a separate neutrino mass extension is not silently included.

Copying this internal system to twelve ports gives 1080 CAR modes. The static hard-core two-quasiparticle sector supports the colored construction, whereas a bare two-particle sector need not survive pairing. The compiled static convention subtracts $11E_{\text{vac}}$ from the replicated vacuum scalar $12E_{\text{vac}}$; the remaining common E_{vac} does not affect fixed-sector Ramsey gaps. Every extension across particle-number sectors must specify the same reference convention explicitly.

Complete source-controlled Ramsey observations recover an energy difference between equal-charge modes, with an opposite-charge partner. A separate neutral-ancilla phase probe reads the same single-particle charge, retaining all seven charge values and all pointer outcomes. Neither measures an absolute rest energy. The common electromagnetic phase cancels on the neutral pair and is not an electromagnetic-response signal. These executions retain all 128 padded internal addresses, native records and quantum output factors.

Currents are transformed with the Nambu chart, including their anomalous blocks. The neutral Higgs background has a nine-dimensional unbroken stabilizer, color and electromagnetism; three complementary gauge directions are broken. General weak operations cannot therefore be represented solely by $U \otimes U$ within fixed quasiparticle number. A time-dependent Higgs background requires the complete CAR domain.

Spatial dynamics, clock and particle creation. A Hamiltonian cycle of the actual twelve-port seam graph supplies a finite spatial ring. With specified spacing and $c = 1$ units, its bare action is

$$h = -D \otimes \sigma_z \otimes I_{45}, \qquad D_{r,r+1} = -i/(2a), \quad D_{r+1,r} = i/(2a),$$

with the same local pairing matrix. The static symbol satisfies $E(k)^2 = \sin^2(k)/a^2 + E(0)^2$. All twelve momenta, all 2160 Nambu directions and both kinetic zeros remain. This is a one-dimensional witness, not a chiral $3+1$-dimensional limit. Its 1080 CAR modes retain every occupation sector, including collisions. The pairing graph splits into closed internal components of sizes one, two and six, so the largest spatial component has 72 modes. Complete four-state Fock source gates and fermionic routing determine their full Gaussian unitary maps without allocating arbitrary 2^{1080}-component vectors. A source-prepared localized excess and its bond current have complete occupation readouts, with purifiers retained. The executed finite symmetric product is specified as such; its error relative to the summed Hamiltonian is separate from its source-gate error.

Two source-prepared clock hands, each with faithful eigenvalues $1/6, 5/6$ and a retained purifier, share the reference energy $\Omega_{\mathrm{ref}} = 1/4\,\mathrm{GeV}$. Both hands and every field factor use $\lambda_j = 1 + j/4$ for the actual native current. Complementary readouts infer $\tau_{\mathrm{eff}} = \sum_h \lambda_{j_h} \Delta\tau$ on a declared unaliased interval. The inverse receives complete probabilities and error bounds, not the programmed duration or a species mass. Rational enclosures and finite-shot bounds distinguish ensemble inference from a single clock reading. The energy-unit calibration and $\hbar = h/(2\pi)$ set seconds; repair counts and host runtime do not set this clock.

The particle-changing drive follows the actual common Higgs field $H = (0, ve^{i\phi})$ with fixed gauge background. Its creation phases have opposite signs in the up and down/charged-lepton channels, according to their conjugate-Higgs tensors. In a compatible static chart it comprises 42 neutral two-mode pairs and six massless spectators per site, with

$$H_{E,\phi} = E\cos\phi\,\mathrm{diag}(-1,0,0,1) + E\sin\phi\big(|00\rangle\langle 11| + |11\rangle\langle 00|\big).$$

Literal source gates retain each complete four-state space and all nonadjacent CAR signs. Across twelve sites this gives 504 pairs and 72 spectator modes. An externally driven quadratic field is not a quantized interacting gauge–Higgs theory.

Repair itself extends to every occupation sector. Binary endpoint occupancies follow the original C/E rule; nonbinary endpoints are held. A coherent extension includes that identity block in the existing zero-current Kraus operator. The binary and nonbinary output ranges are orthogonal, so the complete partial-injection effects sum to identity. On states with even occupancy at both endpoints the unread channel is exactly the identity, including coherences and references. The extension is unital before arbitrary record-conditioned controls; its faithful tracial reference has trivial modular flow. General completeness is algebraic. Literal source dilations verify the bounded collision and spectator-crossing cases, not a full 1080-mode dilation.

A separately flagged hold is retained as a control. It measures a vacuum/pair distinction that the coherent hold preserves. If A is the probability of no created pair on the repaired sites, a drive–repair–inverse echo returns $A^2 + (1-A)^2$ for the flagged policy and one for the coherent policy on this even-endpoint domain. For the stated drive at $\tau = 0.002\,\mathrm{GeV}^{-1}$, $j = 0$ and $\phi = \pi/2$, the flagged return is 0.63176674. Coherent collision holding does not remove backaction from the original current or E-parity records.

A joint source experiment. Selected source factors test repair, propagation, particle creation and clock readout together. Four active CAR modes on two sites carry the same charged component. A spatial superposition is source-prepared with a path reference and an opposite-charge spectator at a third site. The quasiparticle-to-bare chart is fixed by both static and driven terms; diagonalizing the static mass alone does not suffice. Each of two events applies complete native repair, routed bare hopping in that chart, and the local Higgs factors. The same current advances the clock without reset. All 16 active states, spectator states, references and clock purifiers remain, including states with three active particles.

For both C/E laws and both collision policies, all $32^2 \times 4$ outcomes are retained: 16,384 in total. Complete effects are checked on the 64-dimensional field/clock input algebra, together with the untouched reference marginal. The coherent C process has collision population 0.23404814 before the second repair and final three-particle probability 0.53649765, versus 0.36223079 for the flagged control. Their unread final-state trace distance is 0.20709787. E distinguishes the relevant odd binary occupancies, so the two policies agree on this witness within numerical admission. The composed all-outcome total-variation bound is 4.200002×10^{-7}, conditional on the full-column numerical source checks. It is not an interval-certified hardware bound. Boundary hops out of

the selected sites are omitted: this experiment joins actual source factors but is not an invariant two-site restriction of the full spatial field.

Energy and modular observations on the same field. The selected four-mode factors admit energy and reference-state calculations on their complete 16-state algebra. Let H be the sum of the static hopping and Higgs terms, with its vacuum scalar retained. The executed ordered product is not silently replaced by e^{-itH}. An explicit piecewise control schedule reproduces that product, separates switching work from energy transfer during complete repair, and uses the same clock duration. The latter transfer is not assigned to heat or controller work without an apparatus model. On the spatial action, onsite terms and half of each incident bond define

$$H = \sum_x h_x, \qquad J_{xy} = i[h_x, h_y] = -J_{yx}, \qquad \frac{dh_x}{dt} = -\sum_y J_{xy} + \partial_t h_x.$$

These identities hold on all occupation sectors. Equality of full unitaries after a scalar energy-reference shift requires the corresponding global source phase; density evolution, currents and gaps do not determine that scalar. Complete, charge-preserving source instruments on all 64 apparatus input columns reconstruct energy, drive-power and ring-current expectations from separate quadratic-group settings. This is expectation readout, not a projective measurement of the summed H.

For the declared $\beta = 1\,\mathrm{GeV}^{-1}$, the same static H gives a faithful $\rho = e^{-\beta H}/Z$. A complete eight-qubit source word prepares its field–purifier state. Signed CAR restrictions and compatible Petz observations compose on every matrix unit, and a native recovery isometry retains all four regional inputs, sixteen field states and four environment states. This is a new Gibbs-input experiment with the same field instrument, not a replacement of the earlier localized input. The modular convention is

$$\rho^{is} A \rho^{-is} = e^{-i\beta s H} A e^{i\beta s H} = \tau_{-\beta s}(A);$$

the dimensionless parameter s does not supply a second physical clock.

The observation is not a regional retraction. The cross-boundary hopping has vanishing partial traces and squared Hilbert–Schmidt norm exactly 4 in the specified units. Its nonadditivity makes the Gibbs density nonproduct for $\beta > 0$, obstructing a reference-preserving UCP retraction onto the full site factor. Compatible non-idempotent Petz observations remain valid. Nor is this Gibbs density stationary under the driven, repaired history. The complete quantum entropy balance is

$$S(\sigma) - S(\rho) = \beta \operatorname{Tr}[H(\sigma - \rho)] - D(\sigma\|\rho)$$

for normalized states; retaining classical records gives an identity involving conditional relative entropies. Numerical trace drift contributes $(\operatorname{Tr}\sigma - 1)\log Z$, and the quadratic Bregman remainder subtracts $\epsilon \operatorname{Tr}\delta$ from $D(\rho + \epsilon\delta\|\rho)$. These reported entropies concern the unnormalized continuing component. First-export abort weights and output families are retained separately; their entropy is not included. A geometric cap kernel, local stress dictionary and generalized-entropy stationarity do not follow from these finite identities.

The Weyl/contact branch and its finite premise closures. The following constructions use the supplied Weyl/contact action with $c = 1/8$ and $U = 1$, separately from the calibrated Weyl–Higgs field above. Its dimer and plaquette restrictions retain their actual reference states. The charged-block service is first defined on decoupled dimers. The three-dimensional cube below instead uses bipartite eta symmetry to join a larger protected interacting sector to the complete field. The common local coefficients do not identify distinct global domains.

A complete charged internal sector. The same lower charged dimer doublets admit a sharper finite native-record attachment than the fixed-column dictionary. Retain both internal columns at each of two occupied blocks among twelve identical dimers, with the same even half-filled ground at every other block. Their carrier has dimension $66 \cdot 4 = 264$. For the declared graded whole-block repair, a movement that crosses the spectator acts internally as minus the swap; a hold acts as the identity. In the singlet–triplet basis every complete native branch is therefore

$$K_{\text{spinful}} = K_+ \otimes I_1 \ \oplus \ K_- \otimes I_3.$$

An integer certificate checks all 720 raw C/E slots and 31,680 internal input columns, including zero slots. The triplet qutrit and its untouched reference correlations survive every admitted complete history without intermediate recovery. For initially correlated spatial/internal inputs, this assertion concerns the complete internal marginal; conditioning on a record can still steer it.

The raw effects separate all 66 positions. A connected transition tree and an odd exchange loop then classify the entire passive source commutant as $I_{66} \otimes (M_1 \oplus M_3)$, of complex dimension ten. Thus the three-dimensional factor is maximal under the full branch grammar; singlet–triplet coherences do not give a protected four-state factor. Positive all-seam coverage and the checked invariant maximally mixed state supply the corresponding fixed-algebra premises. This enlarged field carrier is distinct from the bare H66 source with scalar fixed algebra.

Complete extension and an exact spectral exclusion. Let P be the sum of the orthogonal two-doublet/remaining-ground block projectors. The native outcomes have total effect P; adjoining the outside outcome $I-P$ gives a trace-preserving instrument on the full 48-mode Fock space. This is a formal operator construction, with bounded checks on all 4,096 states of three dimers, not a 48-mode source execution. Every definite occupied pair has the unchanged ground on its entire exterior. With the original supplied $K_j = H_j - N_j/2$, $c = 1/8$ and $U = 1$, the same defect has

$$\Delta = \frac{\sqrt{17}}{8} - \frac{1}{16}, \qquad (K_{\text{tot}} - 12E_g I)|_P = 2\Delta I_P.$$

Its excess CAR number is $q = -1$ for the hole and $q = +1$ for the particle; native currents transport q times the defect count. All three protected triplet states have the same gap and total charge $2q$. A common scale or chemical potential cannot split them into three different masses. The triplet is an internal spin-one representation, not three generations.

The same sector is not invariant under the joined uniform four-site Weyl/contact field. For its four charged-pair columns V, exact dyadic arithmetic gives

$$\|(I - VV^*)H_{\text{uniform}}V\|_F^2 = \frac{3}{256} > 0$$

for both particle and hole pairs, whereas the decoupled sum gives zero. Indeed, with $L = (I - VV^*)H_{\text{uniform}}V$, the stronger identity $L^*L = (3P_{\text{singlet}} + P_{\text{triplet}})/512$ excludes every nonzero nonleaking input, including every triplet state. This generator leakage preserves the distinction between the supplied protecting block service and the interacting field law. No binding, topological boundary charge, physical clock, Lorentz spin, observed species or absolute mass follows; the physical particle and common-field gates remain open.

A protected interacting sector on the original three-dimensional cube. The source-bound `cube_eta_protection` packet retains the original open $2\times2\times2$ Weyl/contact action, $c = 1/8$, $U = 1$, on all $2^{16} = 65\,536$ occupation states. In the physical onsite basis define

$$\eta^+ = \sum_x (-1)^{x_1+x_2+x_3} c^\dagger_{x\uparrow} c^\dagger_{x\downarrow}, \qquad \eta^- = (\eta^+)^\dagger, \qquad \eta_z = (N-8)/2.$$

Onsite singlets are invariant under each supplied $i\sigma_i$ link; the bipartite signs cancel the hopping commutator. Thus $[H,\eta^+]=U\eta^+$, checked exactly on the complete field action. For $X=\eta^+|_{N=4}:\mathcal{H}_4\to\mathcal{H}_6$,

$$X^\dagger X=(\eta^-|_{N=4})^\dagger(\eta^-|_{N=4})+4I,\qquad H_6X=X(H_4+UI).$$

Consequently X is injective and its polar isometry intertwines the full interacting four-particle parent with a protected candidate of dimension $\binom{16}{4}=1820$. Its range is precisely the $\eta\geq 2$ subspace at $N=6$.

The executed repair domain is a finite four-block service: the four x_1-directed dimers, native ports $(0,1,5,7)$, and all five inherited seams $(0,1),(0,5),(0,7),(1,5),(1,7)$. Two blocks carry the unchanged lower-hole doublet and two the unchanged dimer ground. This gives a 24-dimensional $N=6$ support P. Its physical columns are expanded exactly as $A+\sqrt{17}B$, with integer Gaussian matrices A,B; $\eta^-A=\eta^-B=0$, so the support has $\eta=1$. The columns are also bound to the consumed normalized analytic dictionary: its exact normalization is $n=136-32\sqrt{17}$, with $(A+\sqrt{17}B)^\dagger(A+\sqrt{17}B)=n^2I$. Every native C/E coin and raw slot is checked against its actual graded whole-block permutation. With the complete outside outcome $Q=I-P$, all 130 slots are retained: 120 native and ten outside over the ten seam/law choices. Orthogonality gives

$$PX=0,\qquad L_rX=0\quad (r\text{ native}),\qquad QX=X.$$

Every admitted complete repair is therefore exactly the identity on this sector, including arbitrary reference cross terms; interleaved field evolution remains unitary and interacting. This mechanism requires no alignment of the transverse dimer spin frames. It does not assert a twelve-block execution, arbitrary repair invariance, or maximality of the protected subspace.

Exact interacting spectral census and a local fixed-number probe. Let $v=\eta^+|0\rangle$, $v^\dagger v=8$, and $\Pi_3=I-vv^\dagger/8$ on the 120-dimensional $N=2$ space. For $T=(\eta^+)^2|_{N=2}$,

$$T^\dagger T=60I+24vv^\dagger,\qquad H_6T=T(H_2+2UI).$$

Hence $T/\sqrt{60}$ is isometric on the rank-119 $\eta=3$ parent. The hidden-spin census is 35 singlets and 84 triplet states; this is neither Lorentz-spin identification nor a family count. Exact integer characteristic polynomials in the actual onsite Hubbard frame give, with $z=16(E_{N=6}-2)$,

$$\begin{aligned}p_{119}(z)&=z^{57}(z^2-12)^{21}q(z)c_{64}(z)^3c_{128}(z)^3,\\ q(z)&=z^2-16z-12,\qquad c_a(z)=z^3-16z^2-12z+a.\end{aligned}$$

These are eleven distinct energies: multiplicity 57 at zero, 21 at each $\pm\sqrt{12}$, one at each quadratic root and three at each of the six cubic roots. The last twenty states have contact-shifted energies. Squarefreeness, coprimality and rational root isolation are exact; approximate displayed response weights are separate diagnostics.

The $N=6$ reference and one-site density probe are

$$|R\rangle=\frac{(\eta^+)^3|0\rangle}{\sqrt{2016}},\qquad H|R\rangle=3|R\rangle,\qquad |s\rangle=\frac{4n_0-3}{\sqrt{15}}|R\rangle.$$

Here $\langle n_0\rangle_R=3/4$, $\mathrm{Var}_R(n_0)=15/16$; $|s\rangle$ is an $\eta=3$ singlet with mean energy 3 and variance 3/112. Its normalized two-particle parent is $(8|\uparrow\downarrow\rangle_0-v)/\sqrt{56}$. The exact parent Krylov Gram determinant is $14883950729121848391888005329065541569085 44>0$. Together with the annihilator $qc_{64}c_{128}$

this proves positive response weight at all eight interacting energies and zero weight at the three free-energy values.

The local number-preserving phase $F_0 = e^{i\pi n_0/2}$ satisfies

$$F_0|R\rangle = \frac{|R\rangle - \sqrt{15}|s\rangle}{4}, \qquad \mathrm{Tr}_0(F_0|R\rangle\langle R|F_0^\dagger) = \mathrm{Tr}_0\,|R\rangle\langle R|.$$

Thus the entire exterior reduced state is unchanged initially, while the global pure-state trace distance is $\sqrt{15}/4$ and the disturbed state has energy variance $45/1792$. This is a finite local unitary disturbance of an entangled reference, not yet a boundary charge or an asymptotic localized particle. Both components remain protected. A fixed-$N = 6$ pointer Ramsey construction using controlled F_0, common field evolution and controlled $F_0^\dagger$ has

$$C(t) = \langle R|F_0^\dagger e^{-it(H-3)}F_0|R\rangle = \frac{1}{16} + \frac{15}{16}\sum_{j=1}^{8} w_j e^{-it(z_j/16-1)}, \qquad w_j > 0, \quad \sum_j w_j = 1.$$

No vacuum/charged-sector phase reference is needed in this protocol. The specified H and reference energy fix the frequency origin; these finite spectral gaps are not physical masses.

Native preparation and the remaining detector premise. The actual source prepares the 56 signed three-doublon amplitudes using an eight-bit, 5379-pulse amplitude tree on the even modes, with the odd modes blank. Eight native CNOT words copy into the odd modes in 24 pulses, followed by the three-pulse one-site phase including its scalar phase. The compact tree evaluation is an exact spectator blank isometry; all 65 536 rows are retained for copying and the phase. The independent exact trigonometric and literal-port certificate bounds the complete source-state error by 8.48×10^{-13}, including every source export. All 130 following repair records are retained without success normalization. The native seventeen-bit time-dependent detector and native field continuation are not executed here: the fixed-number response above is analytic, while the reference preparation and local kick are executed. Hardware precision, growing-volume localization, deformed excitation transport, a continuum limit and empirical species identification remain open. The earlier small-carrier leakage and transverse hidden-spin obstruction remain valid; the larger eta sector supplies a different common protection mechanism.

The preceding preparation packet stops before the time-dependent detector. The following extension executes that detector and field continuation on the same cube, closing this finite readout premise.

An internal observer reads the protected response. The `protected_particle_readout` construction attaches the preceding cube spectrum to complete native pointer records. Its identified object is the eight-dimensional cyclic response generated by the centered local density probe, inside the 119-state $\eta = 3$ sector and the larger 1820-dimensional protected space. It is a neutral, fixed-$N = 6$ response; these three dimensions are not interchangeable particle counts. The construction uses Ramsey interference to read a correlation function, a standard measurement principle, and supplies its actual source implementation for this field.[3]

The apparatus retains sixteen field bits and one pointer, hence all $2^{17} = 131\,072$ field/pointer occupation rows. The unchanged source prepares $|R\rangle$ and a pointer $|+\rangle$, then applies controlled $F_0 = \exp(i\pi n_0/2)$. There are two field segments of duration $1/2$: the first is followed by complete C

[3] For many-body Ramsey correlation measurements, see Knap et al., *Phys. Rev. Lett.* 111, 147205 (2013), https://arxiv.org/abs/1307.0006.

repair on all five admitted seams, and the second by complete E repair on those seams. Controlled $F_0^\dagger$, a pointer phase $\phi \in \{0, \pi/2\}$, and the inverse preparation rotation precede the complete six-outcome pointer export. Every source first-failure record and all 130 repair records remain present. An outside repair outcome continues; a native repair outcome or source failure terminates in its own retained record. No probability is conditioned on survival.

With exact field evolution, each repair acts identically on both ideal arms. Immediately before the final pointer rotation their field vectors are

$$b_0 = e^{-itH}|R\rangle, \qquad b_1 = e^{i\phi}F_0^\dagger e^{-itH}F_0|R\rangle.$$

Consequently the unconditional ideal pointer-zero probability is

$$p_0(t,\phi) = \tfrac{1}{2}\{1 + \mathrm{Re}[e^{i\phi}C(t)]\},$$
$$C(t) = \frac{1}{16} + \frac{15}{16}\sum_{j=1}^{8} w_j e^{-it\omega_j}, \qquad \omega_j = z_j/16 - 1.$$

Thus the two settings read $\mathrm{Re}\, C = 2p_0(t,0) - 1$ and $\mathrm{Im}\, C = 1 - 2p_0(t,\pi/2)$. The phase $3t$ follows from the reference eigenenergy; it is not an adjustable frequency fit. A common scalar added to H, or a common number chemical potential on this fixed number domain, cancels between the arms. Absolute energy zero is therefore not determined by this experiment.

Complete source and field error, without a sector projection. The executor composes success and every first-export failure map of each local word on at most three work bits, then applies these maps to the full field/pointer array. This is finite block contraction of the same native word, not replacement by an ideal gate. Independent tests compare the contracted maps with direct source execution, including order-one failure controls. The exact $N = 6$ restriction is used only by the independent original-H comparator, not by the native execution.

Each half-time field segment uses 32 symmetric product steps. The actual ordered support list has 24 hopping terms of norm $1/16$ and eight contact terms of norm one. Counting only possibly nonzero nested commutators gives the all-Fock coefficient $K = 629/256$. The two segments therefore contribute

$$\varepsilon_{\mathrm{split}} = 2\,\frac{629}{256}\,\frac{(1/2)^3}{32^2} = \frac{629}{1048576}.$$

Independent rational trigonometric and literal-port bounds add less than 8.56×10^{-11} for the preparation, controlled pointer operations, and both complete source field words. The complete output-vector error is consequently $\varepsilon < 6.00 \times 10^{-4}$. Every halted record is part of this norm. Complete repair and export preserve the comparison by isometric extension or contraction; the all-input field bounds also hold with untouched references. The initial preparation bound consumes the specified initialized blank.

For normalized complete outputs, half trace distance is at most their vector distance. Each unconditional pointer probability therefore differs from its ideal value by at most ε, and each reconstructed quadrature by at most 2ε. Rational root, residue and trigonometric enclosures at $t = 1$ imply, for the actual specified source,

$$0.993649 < p_0(1,0) < 0.994850, \qquad 0.501379 < p_0(1,\pi/2) < 0.502580.$$

Both intervals exclude the corresponding no-response values, one and one half. Numerical source residuals have their separate declared admission; they are not substituted for these proved source

bounds. The executed construction uses 681 283 native pulses per setting. It gives pointer-zero probabilities approximately 0.9942497011 and 0.5019798894, versus the independent original-H values 0.9942494663 and 0.5019797169. The complete output-vector discrepancy is approximately 2.27×10^{-6}, below the rigorous bound.

The imaginary fringe also distinguishes the contact interaction from the same free cube. Set $U = 0$ while retaining the hopping, reference, local probe and ideal repair schedule. Bipartite occupation parity $\Gamma_{\mathrm{sub}} = (-1)^{\sum_{x\,\mathrm{odd}} n_x}$ satisfies

$$\Gamma_{\mathrm{sub}} H_0 \Gamma_{\mathrm{sub}} = -H_0, \qquad H_0|R\rangle = 0, \qquad \Gamma_{\mathrm{sub}} F_0|R\rangle = F_0|R\rangle.$$

The free characteristic function is consequently real at every time, so $p_{0,\mathrm{free}}(t, \pi/2) = 1/2$ exactly. Both free arms remain in the protected sector. The certified interacting interval therefore excludes this specified contact-free response. This is discrimination between two internal dynamics, not a fit of a physical particle mass.

For independent reset repetitions, Hoeffding's inequality and a union bound give simultaneous sampling error at most 1/2048 in these two probabilities using 2^{24} trials per setting, except with probability less than $4(3/8)^8$.[4] The sampling allowance still separates both fringes from no response. Independent resets are the sampling-service premise; a seeded numerical sample is an illustration of that model, not laboratory data. The recorded PCG64 illustration, with seed 20261101, gives respectively 16 680 454 and 8 423 380 pointer-zero outcomes out of 2^{24} trials; the complementary count includes every other complete outcome.

What eight-line resolution would additionally require. The executed two settings attach the known protected spectral measure to an internally accessible record. They do not reconstruct eight unknown frequencies or individually resolve the eight lines. There is, however, a certified prospective experiment on the source-specified support. At times $t_n = 39n$, $n = 0, \ldots, 7$, use both pointer quadratures and set

$$y_n = \frac{16C(39n) - 1}{15}, \qquad V_{nj} = e^{-i39n\omega_j}, \qquad y = Vw.$$

Exact interval sine bounds in the Vandermonde product show

$$|\det V| = \prod_{j<k} 2\left|\sin\frac{39(\omega_k - \omega_j)}{2}\right| > 1400.$$

Hadamard's inequality bounds every 7×7 cofactor by $7^{7/2} < 343(8/3)$, so

$$\|V^{-1}\|_\infty < \frac{2744}{525}, \qquad \|\widehat{w} - w\|_\infty < \frac{43904}{2625}\, e,$$

where e bounds the total error of each pointer probability. The latter includes both quadratures and removal of the known elastic atom. Finite precision evaluation of an implemented inverse would need its own error allowance.

A sufficient prospective budget is systematic probability error 1/65536 and sampling error 1/32768 at each of the sixteen settings. With $9 \cdot 2^{29}$ independent trials per setting, the simultaneous failure probability is less than $32(3/8)^9 < 1/100$ and the weight error is less than 343/448000. Twice that error is smaller than the least certified weight, so every line is separated from zero. The

[4] The bounded-independent-sum inequality is imported from Hoeffding, *J. Amer. Statist. Assoc.* 58 (1963), 13–30, https://doi.org/10.1080/01621459.1963.10500830.

total is 77 309 411 328 trials and the largest model time is 273. This long-time accuracy and sampling schedule is not executed here. It identifies weights on the supplied support, not an arbitrary unknown spectrum from finitely many samples.

The finite result is an internally observable protected excitation response of the admitted simulator, with complete-history and finite-time errors. It does not require an external particle name or an SI calibration. It also does not establish that these neutral modes are single odd fermions, that the reference is a vacuum ground state, or that finite spectral lines are relativistic rest masses. Transported excitations, many-body continuum dynamics and comparison with nature remain separate constructions.

The one-time readout above leaves its long-time reconstruction prospective. The following certified accelerated calculation closes that known-support weight-resolution premise while retaining a bound to the full native word.

Full weight resolution by certified accelerated emulation. The preceding sixteen-setting proposal can now be executed as a certified emulation of the same source word. The result resolves all eight weights on the source-specified support; it does not infer arbitrary unknown frequencies or identify physical species. The observable remains the eight-dimensional neutral cyclic response of the excited reference, with elastic weight $1/16$.

The actual chronological field word places the two disjoint spin hoppings of each spatial edge consecutively. They commute, and their sum is the complete Weyl edge commuting with physical η^+. The complete contact layer satisfies $D_6T = TD_2 + 2T$, where $T = (\eta^+)^2$. Thus every ideal Strang step has the exact identity

$$S_6(h)T = e^{-2ih}TS_2(h), \qquad S_6(h)|R\rangle = e^{-3ih}|R\rangle.$$

The complete protected repairs act identically on these ideal states. This is a property of the grouped ideal word, not an exact projection identity for its rounded native implementation.

The actual Hubbard-frame singlet chart has dimension36, retaining all35 $\eta = 3$ singlets and the $\eta = 4$ parent. If U is the spatial forward half-hop matrix and $W = \mathrm{Sym}^2 U$ on normalized unordered pairs, the step is $W^T \exp(-ihD)W$. Transpose reverses the hopping order because every individual Hubbard hopping matrix is symmetric. Integer generator checks bind this chart to every original edge and the contact layer. Outward interval square roots supply the distinct-site normalization.

Use the common step $h = 39/2^{19}$. Time $39j$ takes $j2^{19}$ steps, with half before C repair and half before E repair. At time273 this is 3 670 016 steps. Their 38 755 368 960 field pulses are represented by the repeated source word; they are not individually executed. Instead, outward160-bit step entries are rounded to128-bit dyadics and powered by exact Gaussian-integer matrix multiplication. For approximate unitaries with errors a, b, one rounded product has error at most $a+b+ab+36\,2^{-128}$. The longest power needs24 products and has operator error $249561087/2^{128} < 7.34 \times 10^{-31}$.

Independent literal-source templates bound the complete 10 560-pulse step error by 1.323294×10^{-12}. Telescoping includes all native first-export failures, and the full-Fock splitting error is $(629/256)t^3/n^2$. With preparation, pointer operations, complete repairs, certified arithmetic and exact binary64 probability-conversion error, every setting has systematic probability error below $8.568121 \times 10^{-6} < 1/65536$. Individual long-word failure weights are not materialized; their record addresses and complete-output norm bound are retained. No survival normalization is used.

The emulator draws counts from a binary64 proxy to the certified powered ideal-word probabilities, with PCG64 seed20261102. These are surrogate counts, not exact samples of the unmaterialized native failure instrument. There are $9\,2^{29}$ independent reset trials at each of sixteen settings, or77 309 411 328 overall. Zero is contrasted with all other complete outcomes. The stated

independent binomial service gives simultaneous sampling error 1/32768 except with probability below $32(3/8)^9 < .01$. The same conditional recovery theorem applies to genuine native counts under that sampling event and the proved native systematic bounds.

Only the source frequency intervals enter the inverse. Rational Lagrange polynomials give a96-bit matrix B for $V_{nj} = e^{-i39n\omega_j}$; outward arithmetic certifies $\|B\|_\infty < 2$ and $\|BV - I\|_\infty < 10^{-18}$. Counts determine $\widehat{y}$ and $\widehat{w} = \mathrm{Re}(B\widehat{y})$. Since the true source weights form a probability vector, the conditional error is bounded by

$$\|\widehat{w} - w\|_\infty \le \|B\|_\infty \frac{16}{5}(\varepsilon_{\text{systematic}} + \varepsilon_{\text{sampling}}) + \|BV - I\|_\infty < 0.000258393828.$$

In ascending frequency order the recovered weights are

$$(.002457894673,\ .005641397395,\ .005890881572,\ .007131698528,$$
$$.004528321839,\ .421578515341,\ .415796456292,\ .136974834359).$$

Every resulting interval is strictly positive and contains the independent exact residue, checked after inference. This closes weight resolution on the given support in the certified emulator. It does not turn $E_H - 3$ into an absolute rest mass, identify an unknown support, or establish physical particle species or native hardware execution.

Comparing the actual interacting regional clocks. The complete four-mode Weyl/contact Gibbs source supplies a useful finite alternative to assuming regional modular covariance. Let $W_S(s) = \exp(isH)$ be its state modular unitary at $\beta = 1$, and let $W_A(s) = \rho_A^{-is}$ be the embedded unitary of an actual CAR-reduced reference, with fixed scalar gauges understood. The relative operations

$$C_{AB}(s) = W_A(s)W_B(s)^*, \qquad C_{AB}(s)C_{BD}(s) = C_{AD}(s)$$

compare these clocks on every input and untouched reference. They obey the time-cocycle identity $C_{AS}(s+t) = C_{AS}(s)W_S(s)C_{AS}(t)W_S(s)^*$. The existing source words implement each global-to-regional correction at $s = 1/2$ with 2,826 pulses and ideal operator splitting bound 11/32768; the regional-to-regional comparison needs twenty pulses. All sixteen Fock columns and first export failures remain present. Such corrections are active: the first changes the common Gibbs reference by half trace distance about 0.00444. They do not assert that both regional flows preserve the same global state.

The stronger identification fails on this reference. For a linear modular intertwiner, an eigenoperator coefficient satisfies $(\nu_j - \omega_i)L_{ji} = 0$. The actual dimer at speed 1/8, contact $U = 1$, $\beta = 1$ and $\mu = 0$ has three positive regional frequencies, approximately 0.0001258906, 0.9988218345 and 0.9989477251. An exact source characteristic polynomial and nine partial-trace moment identities determine its reduced spectral weights. Rational bounds on the square root, exponential and logarithm then separate all three frequencies from every global energy difference by more than 10^{-4}, in all 243 comparisons. Consequently every same-parameter linear intertwiner has range in the six-dimensional regional modular centralizer and cannot transmit a moving local clock hand. The group and coefficient implications are Lean checked; the concrete spectral and interval attachment is checked separately by exact arithmetic. This excludes a strong identification on one finite reference, while the explicit relative operations provide a functioning comparison. The supplied action, coherent services, reference preparation and dimensionless parameter remain visible inputs. No physical duration, common energy calibration, geometric modular flow or Einstein premise is discharged.

Controlled regional modular energy. The same complete-Fock Weyl/contact action now supplies a quantitative regional modular attachment on its four-site open plaquette. All 256 field states and all sixteen states of the two-site region are retained. For the normalized signed partial trace E, put $H_c = H - I$ and $X(\beta) = E(e^{-\beta H_c})$. Exact Gaussian-integer and rational arithmetic constructs the first eight coefficients of $-\log X$. In particular, with $M_j = E(H_c^j)$,

$$C_1 = M_1, \qquad C_2 = \tfrac{1}{2}(M_1^2 - M_2), \qquad C_3 = \tfrac{1}{6}M_3 - \tfrac{1}{4}(M_1M_2 + M_2M_1) + \tfrac{1}{3}M_1^3.$$

For $0 \le x = \beta B < \log 2$, where $B \ge \|H_c\|$, the positive coefficient majorant $f(x) = -\log(2 - e^x) = \sum a_j x^j$ gives

$$\left\| -\log X - \sum_{j=1}^{p} \beta^j C_j \right\| \le f(x) - \sum_{j=1}^{p} a_j x^j.$$

This follows from the norm-convergent noncommutative logarithm series, without replacing the reduced reference by a local Gibbs ansatz. At the declared $\beta = 1/32$ and exact bound $B = 3$, an outward rational evaluation gives eighth-order error below 1.917×10^{-9}. Hence every Hermitian trace-zero regional variation obeys the entropy-tangent error bound $1.917 \times 10^{-9}\|\delta\|_1$. The normalization scalar in $K_A = -\log X + \log \operatorname{Tr}_A X$ cancels only on that tangent space.

The first nonscalar boundary correction is cubic and does not commute with the bare regional energy. Removing the contact interaction removes its interaction-dependent terms; removing the crossing bonds removes all nonscalar corrections. A full-versus-empty regional matrix-element witness and the same rational tail prove distance greater than 2.7×10^{-8} from $\beta C_1 + aI$ for every scalar a on this instance. This supplies both a corrected finite modular generator and a certified boundary to its bare-energy identification. The source preparation below supplies this plaquette reference to a separate controlled error. The action, temperature and lattice remain declared; the norm bound can grow with volume. Geometric cap weights, null stress, fixed-volume entropy–area stationarity, a physical scaling limit and positive Newton normalization remain additional obligations.

A separately prepared interacting Gibbs reference. On the complete 256-state plaquette at $\beta = 1/32$, the degree-eight Gibbs ensemble retains all 256 preparation records and every export abort. Independent outward arithmetic certifies its literal-source preparation error below 1.700×10^{-12} on the initialized input. The actual regional success logarithm, including the exact normalization scalar, is within 2×10^{-9} of the preceding P_8 comparison. The all-input continuation has a separate source and splitting bound. The proof in `model/certified_reference/README.md` supplies finite Gibbs preparation; it does not assert invariance under the complete repair. The following stationary reference has a different modular generator.

The actual stationary source weight. One identical literal native $Y(-\pi/3)$ request and complete dephasing export per occupation mode prepare $\sigma_M = \operatorname{diag}(a, b)^{\otimes M} = a^M e^{\gamma N}$, where $a = |U_{00}|^2$, $b = |U_{10}|^2$ for the actual represented source unitary and $\gamma = \log(b/a)$. The exact one-mode error is below 1.894×10^{-16}, proving $a, b > 0$ and $b > a$; abort blocks remain separate. Number conservation and complete unital repair preserve this weight exactly. Tracing complete outside outputs before regional compression gives a k-mode logarithm error at most $k\delta/(1/4 - \delta)$, with $\delta = (2 + \epsilon)\epsilon$. This charge clock is trivial within fixed number. On the earlier six-site $N = 4, S = 1$ active sector, every common stationary modular generator stays at distance at least $|\beta|\sqrt{13/48}$ from βK, even allowing scalar and protected-spin terms. Thus the stationary reference and interacting Gibbs energy remain distinct. The exact source and reference proofs are

in `model/continuing_reference/README.md`; the following construction closes its previously restricted scheduler premise.

A stationary finite controller for native feedback. The reference can now continue under an explicit record-adaptive scheduler. Use two classical working bits (c, d); $c = 0$ selects the complete native C repair on seam $(0, 1)$, and $c = 1$ the complete E repair on $(0, 5)$. Append every raw record r, including zero slots and the outside outcome $Q = I - P$, and update

$$(c, d) \longmapsto (d \,\mathrm{XOR}\, f(r), c), \qquad f(r) = \text{coin XOR (raw mod 2)}, \quad f(Q) = 0.$$

At output (u, v) every incoming term uses the same instrument v. Completeness and unitality of the unchanged repair therefore give

$$\sum_r L_r^{(v)} \frac{\sigma_M}{4} L_r^{(v)*} = \frac{\sigma_M}{4}, \qquad \Omega = \sigma_M \otimes I_4/4, \quad \sigma_M = \mathrm{diag}(a, b)^{\otimes M}.$$

Here a, b are the actual literal-source weights, so the successful block still has trace $(a + b)^M$. All initial export-failure sectors are retained and frozen, emitting a bypass record. Separate unitary layers selected by the current working memory and generated by the original number-conserving field Hamiltonian also preserve this marginal. The controller is a declared reversible classical law on $\mathbb{C}^4$, not a claimed quantum pulse circuit.

Two unchanged native events on an explicitly initialized binary spectator, occupied at ports $2, 3$ and repaired on $(0, 1)$, leave it fixed and supply two exact fair coins. Their retained records seed $I_4/4$ without an angle-error assumption. This uses the supplied simulator basis initialization; no hidden interacting-field preparation of that spectator is asserted. The working memory remains two bits, while the archive grows by at most five bits per tick, besides the two seed records and all 6^M source readouts. Stationarity traces these records; it never normalizes the successful field block or declares the growing archive stationary. The modular generator remains $-\gamma N - M \log a + \log 4$, $\gamma = \log(b/a)$, and every complete step intertwines the charge flow $e^{it\gamma N}$. No physical energy-temperature identification follows. Earlier local literal-source bounds remain uniform in volume and ideal adaptive history length; implemented per-step errors must still be added.

A complete finite mixing certificate and its boundary. On the invariant three-position highest-spin repair corner, with no joined Hamiltonian step inserted, the field/controller algebra is $\bigoplus_4 M_3$. Each E outcome has rank at most one. Exact integer enumeration finds at least one eight-tick history encountering E for every one of the 12^2 input/output population pairs, with common probability denominator 2^8. The corresponding rank-one Kraus histories prove

$$\Phi^8 = \frac{3}{64}\mathcal{D} + \frac{61}{64}\mathcal{R}, \qquad \mathcal{D}(X) = \mathrm{Tr}(X) I_{12}/12,$$

where $\mathcal{R}$ is completely positive, trace preserving and unital. Thus $I_{12}/12$ is the unique stationary density on this CQ algebra, and the archive-traced error contracts by $(61/64)^{\lfloor n/8 \rfloor}$, including arbitrary untouched quantum references and all field coherences. This is neither whole-Fock mixing nor mixing with interleaved Hamiltonian evolution. Indeed the full retained history recovers the initial corner basis state and memory. As a negative control, the recordwise reversible one-bit policy $c \mapsto c \,\mathrm{XOR}\, f(r)$ sends the uniform six-state input to $(3, 1, 2, 1, 3, 2)/12$, a trace distance $1/6$ from uniform. The two-register incoming-row balance is a substantive premise, not a consequence of memory reversibility alone.

Exact energy Ward cells under coarse-graining. The same complete-Fock Weyl/contact action supplies a conserved energy column on its standard three-dimensional lattice. With the declared half-bond convention $H = \sum_x h_x$, its range-two currents are $J_{xy} = i[h_x, h_y]$. Distributing each current equally over all shortest nearest-neighbour paths preserves its incidence $\delta_x - \delta_y$: intermediate vertices cancel on every path. The resulting oriented face current $F_{x,i}$ therefore satisfies $\dot{h}_x + \sum_i (F_{x,i} - F_{x-e_i,i}) = 0$ exactly. The rational Clifford certificate checks the actual $3 \times 3 \times 3$ action at $c = 1/8, U = 1$: all 351 site pairs, 81 nonzero currents, 54 faces and 27 local equations, on the complete 54-mode algebra. Its coverage of all 2^{54} Fock states is a symbolic operator identity, not execution of that many amplitudes. Independent complete small-Fock matrices attach all three hopping axes and the interaction to the existing source action.

For a finite time cell, the exact flux is $I_{x,i}(n) = \int_{t_n}^{t_{n+1}} F_{x,i}(t)\, dt$. Hence

$$h_x(t_{n+1}) - h_x(t_n) + \sum_{i=1}^{3} \big(I_{x,i}(n) - I_{x-e_i,i}(n)\big) = 0.$$

For any normalized input and untouched reference, setting $T_{00}(n, x) = \langle h_x(t_n) \rangle$ and $T_{i0}(n, x) = \langle I_{x-e_i,i}(n) \rangle$ supplies the $j = 0$ forward-difference Ward column of the finite tensor interface. These are cell energy and integrated flux, not pointwise continuum stress. Summing spatial blocks cancels internal faces, so $D_{\text{coarse}} P_{\text{faces}} = P_{\text{sites}} D_{\text{fine}}$. Successive block maps compose exactly; successive time cells telescope. All 27 rectangular width choices, including ragged blocks, pass the exact source certificate. Generic finite-sum divergence, continuity, block composition and time telescoping are also Lean-checked; the concrete CAR source binding remains the exact rational certificate.

The existing local source bounds control the approximation uniformly in ambient volume. With $A = U + 3c$, one has $\|h_x\| \le A$ and $\|F_{x,i}\| \le B = 14A^2$: a face receives absolute shortest-path weight at most seven. Its support fits a radius-three cube. If L_h, L_F are the existing normalized locality tails and ϵ is the patch unitary error, an endpoint energy error is at most $A \min(2, L_h + 2\epsilon)$ and an integrated-face error at most $tB \min(2, L_F + 2\epsilon)$. Replacing the integral by m midpoint samples adds the explicit bound $t^2 B\, 343(U + 6c)/(2m)$; endpoint sampling is not exact. A cell residual is bounded by two endpoint errors plus six face errors, with supplied readout errors added separately. The receipt evaluates these resource bounds without claiming to execute the large sample count. Input preparation, hardware errors and controller/failure-space energy remain separate. This closes a finite same-action energy-column attachment, not the three momentum columns, a symmetric Lorentz tensor, geometric modular weights, area stationarity, a continuum scaling theorem or the full Einstein premise register.

Local momentum balance retains its force. On the open Weyl/contact box, the three local central-difference momenta obey exact balances with kinetic face stress and quartic contact force. All spatial blocks and time cells retain that force. On the 3^3 box at $U = 1$ its total-force Gram matrix is $(9/4)I_3$, so an internal closed divergence cannot absorb it. Conserved number-preserving quadratic charges of the connected action reduce to number and hidden spin. These exact results in `model/lattice_momentum_balance/README.md` concern local quadratic candidates. The following periodic construction supplies conserved global momenta with a separate locality boundary.

Global periodic momentum and the local question. On a declared periodic box, graded translations T_j commute with the full interacting action. Their principal Hermitian logarithms P_j are exactly conserved. They differ from additive one-body momentum by integer many-body winding. On odd-prime rings every number-preserving translation logarithm has nonzero one-particle spin-trace kernels at all nonzero separations. For the principal branch, every range-R

density sum obeys $\|P_j - A_R\| \geq 1/(R+1)$ when $L > 2(R+1)$. These full-space statements, proved in `model/periodic_momentum/README.md`, leave controlled restricted state domains available. The following bounds quantify one such domain and retain its actual interaction leakage.

A collision-suppressed local Ward bound. The same periodic Weyl/contact action supplies a controlled local approximation beyond its nonlocal conserved momentum. Let B be a spin-complete set of spatial Fourier modes, $f = |B|/V$, and let Ω restrict the initial state to that band and $N \leq N_{\max}$. Write $D = \sum_x n_{x\uparrow} n_{x\downarrow}$ and $d = \lfloor N_{\max}/2 \rfloor$. Band annihilators have norm $\sqrt{f}$, so $\Omega D \Omega \leq f \min(N_\uparrow, N_\downarrow) \leq fd$. Since $D^2 \leq dD$ on this number sector,

$$\|UD\Omega\| \leq Ud\sqrt{f}, \qquad \|(I-\Omega)e^{-itH}\Omega\| \leq \min\{1, |t|Ud\sqrt{f}\}.$$

The second bound uses free invariance of Ω and Duhamel's formula; the full interacting action is never projected. For two particles, the exact contact principal angles are $\lambda_K = |B \cap (K-B)|/V$, giving escape norm squared $\max_K \lambda_K(1-\lambda_K)$ before multiplication by U^2. Thus projecting the outgoing contact back into B would miss its leading square-root leakage.

Let $s_j(x)$ be the local sine-momentum density. Write $\Pi_{xy,j}$ for the antisymmetric kinetic stress across a nearest-neighbor face. The exact identity is $i[H, s_j(x)] + \sum_y \Pi_{xy,j} = F_j(x)$, with $F_j(x) = i[UD, s_j(x)]$. Normal ordering its quartic contact terms gives $\|F_j(x)\| \leq U$ and $\|F_j(x)\Omega\| \leq 4Uf$. For preparation at zero and $0 \leq t_0 \leq t_1$, the actual integrated Ward residual therefore obeys

$$\left\| \left[s_j(x,t_1) - s_j(x,t_0) + \sum_y \int_{t_0}^{t_1} \Pi_{xy,j}(t)\, dt \right] \Omega \right\|$$
$$\leq \min\left\{ U(t_1-t_0),\ 4Uf(t_1-t_0) + \frac{U^2 d\sqrt{f}}{2}(t_1^2 - t_0^2) \right\}.$$

The preparation clock is not restarted in later cells. Spatial blocks cancel their internal faces and sum the site bounds; time cells telescope. The complete 54-mode 3^3 source certificate checks all 81 local equations, 81 kinetic faces per column and 27 rectangular coarse partitions per column. Finite-sum transport of the errors is also Lean-checked. This is symbolic full-Fock coverage, not execution of all its amplitudes.

The two-particle total sine momentum has multiplier $2\sin(K_j/2)\cos(k_j - K_j/2)$. The uniform contact vector gives the sharp complete-fiber force norm $\|i[UD, S_j]\| = \sqrt{2}U|\sin(K_j/2)|$. If initially $|k_j| \leq \kappa$ and $2\kappa < \pi$, total momentum stays within $|K_j| \leq 2\kappa$ even after relative-momentum scattering. Combining this conserved domain with the sine remainder gives

$$\|(P_j - S_j)e^{-itH}\Omega\| \leq \frac{\kappa^3}{3} + 4\kappa \min\{1, |t|U\sqrt{f}\}.$$

For a 64^3 counting example with $B = \{-1,0,1\}^3$, $N_{\max} = 2$, $U = 1$ and $t = 1/8$, the certified leakage is below 0.00127, the local Ward defect below 0.000132, and the momentum error after division by κ below 0.0083. The separate eight-site exact-Hamiltonian diagnostic retains a nonzero unforced local defect, so the contact force has not been discarded. These are fixed-number dilute bounds. The critical band-edge scaling $f \sim a^3$, $U \sim a^{-2}$ instead makes $U\sqrt{f} \sim a^{-1/2}$, for which this estimate supplies no vanishing leakage. A finite-density or critically scaled local stress limit, Lorentz symmetry, physical units and the remaining Einstein premises remain separate. Proofs, finite domains and replay are in `model/dilute_momentum_ward/README.md`.

Interacting lattice scattering and its zero threshold. The same Weyl/contact action now admits an on-shell two-fermion scattering theorem. On the physical exchange subspace of $L^2(\mathbb{T}^3, \mathbb{C}^4)$ at total momentum K, its free multiplication operator $H_{0,K}$ is bounded self-adjoint and the contact perturbation is $U|d\rangle\langle d|$, with $\|d\| = 1$. The difference is trace class with norm U. The Kato–Rosenblum theorem therefore gives complete wave operators on the absolutely continuous subspaces and a unitary scattering operator.[5] With $g_+(E) = a(E) - i\pi\rho_0(E)$, its contact-channel eigenvalue is

$$S_c(E) = \frac{1 - Ua(E) - i\pi U\rho_0(E)}{1 - Ua(E) + i\pi U\rho_0(E)}.$$

The orthogonal reducing complement of the contact-generated subspace has identity scattering. Point spectrum, including the isolated pair and the free dark zero modes, remains outside the incoming scattering domain. Completeness here concerns two-body absolutely continuous scattering; it is not many-body or chiral-continuum completeness.

At $K = 0$, the singlet resolvent is $z/(z^2 - 4c^2|\sin k|^2)$. Coarea with all eight sine-map preimages gives, for $0 < |E| < 2c$,

$$\rho_0(E) = \frac{E^2}{4\pi^2 c^3} \sum_{n\geq 0} A_n \left(\frac{E}{2c}\right)^{2n}, \qquad A_0 = 1, \quad A_1 = \tfrac{1}{2}, \quad A_2 = \tfrac{11}{40}, \quad 0 < A_n \leq 1.$$

The coefficients are exact sphere moments of the binomial Jacobian. After degree M, the dimensionless remainder is at most $r^{2M+2}/(1 - r^2)$, $r = |E|/(2c)$; rational arithmetic and a Machin bound for π give certified density intervals. Local analyticity and evenness give $g_+(E) \to 0$ at zero. Hence $\rho_U(E) = \rho_0(E)/|1 - Ug_+(E)|^2 \sim E^2/(4\pi^2 c^3)$. This supplies nontrivial scattering and a gapless contact spectral support on one interacting lattice action while retaining its isolated pair. Zero model energy is not an identified physical vacuum threshold, so no massless physical particle is assigned. The source-bound proof and imported scattering theorems are recorded in `model/contact_scattering/ONSHELL.md`.

Interacting edge scattering and a moving bound pair. The same two-body source family now has a controlled moving bound band. Set $c_a = \sqrt{3}/a^2$ and retain the repulsive coupling

$$U_a^{-1} = \frac{a^2 W_0}{2} - \frac{a^3\gamma}{\pi}, \qquad W_0 = \left\langle \frac{1}{\sum_j \cos^2 k_j} \right\rangle, \qquad \gamma > 0.$$

The normalized full-torus integral satisfies $2/3 \leq W_0 < 5$; $a\gamma \leq 1$ makes the coupling positive. The single global comparison is $J_a = 6/a^2 - H_a$, explicitly shifting energy and reversing time. Neither γ nor this clock/energy choice is inferred from physical data.

At zero total momentum the threshold estimate is

$$\left| \frac{g_0(6/a^2 + z) - a^2 W_0/2}{a^3} + \frac{\sqrt{z}}{\pi} \right| \leq 19a|z|$$

for real $|a^2 z| \leq 1/8$, using the upper boundary square root; the exterior constant improves to 11. Coarea for all eight valleys, an inner ball of radius $1/2$ and its complementary measure give the

[5] The imported trace-class and finite-rank scattering results are reviewed in https://arxiv.org/abs/1711.00528 and https://arxiv.org/abs/0902.3568; the attachment here verifies their premises for the stated Weyl/contact exchange fiber.

respective Green remainders 18η and $36|\eta|$ before rescaling. Thus, for $0 \le s \le S$, $a^2 S \le 1/8$ and $836aS \le 7\gamma$,

$$S_J(s) \longrightarrow \frac{\gamma - i\sqrt{s}}{\gamma + i\sqrt{s}}, \qquad \sup_{[0,S]} |S_J - S_{J,\mathrm{lim}}| \le \frac{1672aS}{7\gamma}.$$

This is the earlier nonidentity scalar contact-channel limit; its orthogonal free channels remain present. Lattice contact renormalization is an established mechanism, not a new particle identification; compare Koerber–Berkowitz–Luu (2019) and Winkler et al. (2006).

Nonzero total momentum has a controlled pole. Write $K = aP$, $q = k - K/2$, $d_j = \sin q_j \cos(K_j/2)$, $e_j = \cos q_j \sin(K_j/2)$ and $R^2 = \sum_j \cos^2(K_j/2)$. For $|K_j| \le \pi/2$, concavity gives the exact complete free edge $B_a(K) = 2c_a R$, attained at all eight $q_j = \pm\pi/2$. The four-spin singlet resolvent is

$$G_K(q, E) = \frac{E(E^2 - 4c_a^2 e^2)}{E^4 - 4c_a^2(d^2 + e^2)E^2 + 16c_a^4 (d \cdot e)^2}.$$

At $E = B_a(K) + z$, put $t = E/(2c_a)$, $\eta = t^2 - R^2$ and $V = \sum_j \cos K_j \cos^2 q_j$. Equivalently,

$$g_K(E) = \frac{t}{2c_a} \left\langle \frac{1 - e^2/t^2}{V + \eta + (d \cdot e)^2/t^2} \right\rangle.$$

Let $\kappa = \max |K_j| \le 1/2$ and $h = a^2 z \le 1/8$, with $z \ge 0$. Writing $Q = \sum_j \cos^2 q_j$, the denominator lies between $(1 - \kappa^2/2)(Q + h + h^2/12)$ and $Q + h + h^2/12$; $e^2 \le \kappa^2 Q/4$ and $(d \cdot e)^2/t^2 \le e^2$. The complete-torus integral and prefactor therefore give

$$|g_K(B_a(K) + z) - g_0(6/a^2 + z)| \le 3a^2\kappa^2.$$

Combining this with the exterior threshold estimate proves

$$\left| \frac{g_K(B_a(K) + z) - a^2 W_0/2}{a^3} + \frac{\sqrt{z}}{\pi} \right| \le a(11z + 3P_{\max}^2), \qquad |P_j| \le P_{\max}.$$

Under $256a(\gamma^2 + P_{\max}^2) \le \gamma$, the pole denominator has opposite signs at $z = \gamma^2/4$ and $z = 4\gamma^2$. Strict monotonicity of the exterior Stieltjes transform supplies exactly one upper pole $E_a(K) = B_a(K) + \delta_a(K)$, uniformly on that momentum cube:

$$\frac{\gamma^2}{4} \le \delta_a(K) \le 4\gamma^2, \qquad |\delta_a(K) - \gamma^2| \le \frac{22}{7} a\gamma(132\gamma^2 + 9P_{\max}^2).$$

This does not invoke the older, stronger sufficient condition $U > B_a$.

An explicit bound-band wavepacket comparison. The same global J_a has bound energy $\epsilon_a(P) = 6/a^2 - B_a(aP) - \delta_a(aP)$. Rationalizing B_a and bounding the sine remainder gives

$$\sup_{|P_j| \le P_{\max}} \left| \epsilon_a(P) - \left(\frac{|P|^2}{4} - \gamma^2 \right) \right| \le \mathcal{E}_a := \frac{a^2 P_{\max}^4}{4} + \frac{22}{7} a\gamma(132\gamma^2 + 9P_{\max}^2).$$

In particular, subtracting $B_a(K)$ separately at each momentum would incorrectly remove the center-of-mass kinetic term.

Let $R_K = (E_a(K) - H_{0,K})^{-1}$ and $\psi_{a,K} = R_K d/\langle d, R_K^2 d\rangle^{1/2}$. Its positive contact overlap fixes its phase, and the pole equation makes it a normalized, measurable, exchange-antisymmetric eigenvector. With consistent measures $dK/(2\pi)^3$ and $dP/(2\pi)^3$, the formula

$$(W_a f)(K,k) = a^{-3/2} f(K/a)\psi_{a,K}(k)$$

defines an isometry on packets supported in the stated cube. Its range reduces the two-body source action; the remaining source states are retained. If h_{pair} multiplies by $|P|^2/4-\gamma^2$, then for arbitrary untouched references,

$$\|e^{-itJ_a}W_a - W_a e^{-ith_{\mathrm{pair}}}\| \le \min\{2, |t|\mathcal{E}_a\}.$$

This proves coherent moving-pair propagation with an explicit rational spacing schedule. The limiting inertial mass is twice that of a single free edge in these model coordinates; neither an SI mass nor a relativistic rest energy is identified. The exact bounds, independent four-spin and torus checks, and wrong-subtraction/time-orientation controls are retained in `model/moving_bound_pair/README.md`.

The result concerns one composite sector. Nonzero-K on-shell scattering, a physical vacuum/detector map, many-pair interactions, relativistic massive/massless sectors and Einstein dynamics remain separate. The earlier fixed-zero $c = v/a$ result also remains valid: $\int_{-M}^{M} |S-I|^2 dE \le 1024M^2/(63c)$ for arbitrary real finite $U(a)$ and $M \le c/4$, so every fixed energy-normalized packet becomes transparent while tuned shrinking resonances can persist. Its scaling and energy comparison differ explicitly from the bound-band construction.

Identified model energies and the physical particle boundary. The model energies and inertial mass are identifiable. The cube has eight exactly characterized interacting energies, each reached by the local probe with positive weight. The moving pair has $\epsilon(P) = |P|^2/4 - \gamma^2$ for the reflected generator $J_a = 6/a^2 - H_a$, hence binding energy γ^2 and inertial mass 2 in those model units. If physical energy and length units $E_{\mathrm{u}}, L_{\mathrm{u}}$ are independently attached and $p_{\mathrm{phys}} = \hbar P/L_{\mathrm{u}}$, then $M = 2\hbar^2/(E_{\mathrm{u}}L_{\mathrm{u}}^2)$ and $E_{\mathrm{bind}} = E_{\mathrm{u}}\gamma^2$, with $E_{\mathrm{u}}T_{\mathrm{u}} = \hbar$ for the common clock. These conversion formulas do not select the units or fix an absolute rest-energy offset. Binding energy is not total rest mass. For the cube, the following rounded displays have exact rational energy and weight intervals in the receipt, with the original contact unit $U = 1$:

H energy	$E - 3$	Probe weight
1.816498920	−1.183501080	0.002465366
1.860537501	−1.139462499	0.005644624
1.955137632	−1.044862368	0.005881219
2.108692846	−0.891307154	0.007137664
2.167639734	−0.832360266	0.004528123
3.015861346	+0.015861346	0.421577939
3.030769653	+0.030769653	0.415789141
3.044862368	+0.044862368	0.136975924

Weights refer to the normalized centered density probe; the phase-kicked target has 15/16 times these weights plus an elastic weight 1/16 at energy 3. The lowest displayed line is not the $N = 6$ ground: $-3\sqrt{3}/8 \le E_{0,N=6} \le -\sqrt{3}/8$, by the free one-particle bound and a polarized zero-contact variational state. These are finite energy levels and signed transitions, not eight rest masses. The physical species assignment count remains zero. On the cube the density probe, its phase-kicked reference and all spectral components remain in $N = 6$: the transition operator is number-neutral and even under $\Gamma = (-1)^N$. An isometric dictionary intertwining this grading cannot map that

response to a single odd fermion, since an odd creation operator changes parity. This does not exclude other odd model excitations or a fermion–antifermion response; even parity alone does not establish Bose exchange or Lorentz spin. Hidden spin and model number have not been identified with physical spin and electric charge. The physical target conventions are the PDG electron and quark quantum numbers and relativistic kinematics.[6]

The supplied cube reference also fails the vacuum ground-state spectral condition for its own H. Its positive density-probe spectral measure has five negative and three positive frequencies $\omega = z/16 - 1$; all eight weights are strictly positive by the exact Krylov certificate. In particular $z_- = 8 - 2\sqrt{19}$ has weight $(1 - 4/\sqrt{19})/14 > 0$ and gives an $N = 6$ eigenenergy $5/2 - \sqrt{19}/8 < 2$, below the reference energy 3. A number-preserving unitary swapping these two states lowers the energy, so the reference is not passive for the full field unitary algebra. This is a mathematical witness, not an executed extraction protocol: the actual local phase preserves the mean energy. These are frequencies of a positive greater correlation, not the two-sided frequencies of a retarded commutator. A scalar or common number chemical potential leaves the gaps unchanged, and time reversal still leaves mixed signs.

The moving-pair limit $\epsilon(P) = |P|^2/4 - \gamma^2$ is a genuine controlled nonrelativistic band. For any fixed affine energy map $E = \alpha\epsilon + \beta$, $\alpha \neq 0$, and fixed invertible linear momentum map $p = LP$, the residual $E^2 - c^2|p|^2 - m^2c^4$ has nonzero quartic coefficient $\alpha^2/16$ along a coordinate line. It cannot vanish on an open momentum set. Thus these maps do not turn this limit into an exact massive or massless Lorentz shell. A low-momentum approximation remains possible: after matching constant and quadratic terms, the difference between $E_0 + Ar^2$ and $\sqrt{E_0^2 + 2E_0Ar^2}$ is $A^2r^4/(E_0 + Ar^2 + \sqrt{E_0^2 + 2E_0Ar^2})$, for rest energy $E_0 > 0$ and $A > 0$.

`model/particle_identification` binds these scoped tests and the remaining vacuum, gauge-charge, spin, clock, detector and mass-shell requirements. The earlier dimer electron mass was a calibration input, not an independent mass prediction; its retrospective equality comparison does not identify the new cube or pair modes. No mass fitting, new eligible physical comparison, or whole-program exclusion is added.

The preceding identity audit leaves a relativistic construction open. The following extension constructs a massless finite-particle limit on the growing lattice. Its carrier and reference remain distinct from the fixed protected cube, and its limiting interaction vanishes.

A controlled massless relativistic sector of the unchanged action. The same Weyl/contact family admits a simultaneous eight-valley continuum comparison on every fixed finite-particle momentum domain. Set

$$H_a = d\Gamma(h_a) + U(a)D, \qquad h_a(k) = -\frac{v}{a}\sum_{j=1}^{3}\sigma_j \sin k_j, \qquad D = \sum_x n_{x\uparrow}n_{x\downarrow},$$

with fixed $v > 0$ and finite $U(a) \geq 0$. This declares the scaling $c = v/a$; it adds no mass term, energy subtraction or time reversal. For $\nu = \pi b$, $b \in \{0,1\}^3$, let $C = [-\Lambda, \Lambda]^3$ and $a\Lambda < \pi/2$. With both Fourier measures normalized by $(2\pi)^{-3}$,

$$(W_a f)(\nu + ap) = a^{-3/2} f_b(p), \qquad W_a : \bigoplus_b L^2(C, \mathbb{C}^2) \longrightarrow L^2(\mathbb{T}^3, \mathbb{C}^2)$$

[6] https://pdg.lbl.gov/2025/listings/rpp2025-list-electron.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-model.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-kinematics.pdf.

is an isometry, with zero output outside the eight disjoint patches. There is no additional factor $1/\sqrt{8}$. Its exterior powers give an isometry $W_a^{\le N}$ on all particle numbers through a fixed N, with the vacuum and every valley retained.

Writing $s_j = (-1)^{b_j}$ and $\chi_b = -s_1 s_2 s_3$, the limiting blocks are

$$h_b(p) = -v \sum_j s_j \sigma_j p_j, \qquad E_\pm(p) = \pm v|p|, \qquad E^2 - v^2|p|^2 = 0.$$

All eight continuum masses are zero. Four blocks have each chirality. The proper frame $R_b = -\chi_b \operatorname{diag}(s)$ has a constant $SU(2)$ lift and spin $S_{b,j} = R_{b,jj}\sigma_j/2$, so $h_b = 2\chi_b v S_b \cdot p$. The positive-energy helicity is $\chi_b/2$; the negative-energy helicity is $-\chi_b/2$. At $p = 0$ the two spin states are degenerate and helicity is undefined. In these frames an opposite pair has $\alpha_j = \operatorname{diag}(-\sigma_j, \sigma_j)$. With off-diagonal identity β, the matrices $\gamma = (\beta, \beta\alpha_1, \beta\alpha_2, \beta\alpha_3)$ satisfy the Clifford relations for signature $(1,-1,-1,-1)$, checked exactly. Thus the limiting differential equation is massless Weyl/Dirac in coordinate $x^0 = vt$, not merely a fitted linear dispersion.[7] The fixed cutoff and the lattice regulator are not Lorentz invariant; neither the valley pairing nor this spin frame selects physical species.

The full interacting evolution has an explicit comparison bound. The sine remainder gives, uniformly on C,

$$\|h_a(\nu + ap) - h_b(p)\| \le \frac{va^2}{6} \sum_j |p_j|^3 \le \frac{va^2\Lambda^3}{2}.$$

For $a\Lambda \le 1$ and $p \ne 0$, both spectral-projector errors are at most $a^2\Lambda^2/6$. The squared-energy defect lies between $-v^2a^2 \sum_j p_j^4/3$ and zero. No projector at the degenerate node is assigned.

Let Ω_a project onto the initial Fock band and $d = \lfloor N/2 \rfloor$. On the infinite lattice the site-diagonal band weight is exactly $f_a = 8(a\Lambda/\pi)^3$. The positive countable partial sums defining D are bounded by d, hence converge strongly and satisfy $D^2 \le dD$. On band states each annihilation has norm at most $\sqrt{f_a}$ and preserves the band for the remaining particles. Summing its squared norms gives $\Omega_a D \Omega_a \le f_a d$, separately on each up/down number sector, and therefore $\|D\Omega_a\| \le d\sqrt{f_a}$. This proves the infinite-volume attachment of the finite dilute-band estimate without substituting a continuum fraction for a finite count.

Duhamel's formula now yields

$$\|e^{-itH_a} W_a^{\le N} - W_a^{\le N} e^{-itd\Gamma(h_{\mathrm{W}})}\| \le \min\{2, |t|\epsilon_a\},$$
$$\epsilon_a = \frac{Nva^2\Lambda^3}{2} + |U(a)|d\sqrt{8(a\Lambda/\pi)^3}.$$

The left evolution is the full source Hamiltonian: no band projection is inserted and every outgoing contact state remains present. The estimate also holds with arbitrary untouched references; it bounds half trace distance and individual probability errors, capped by one. For fixed N, Λ, v and bounded times it tends to zero whenever $U(a) = o(a^{-3/2})$. At $N \le 1$ the contact is exactly zero for every $U(a)$. Thus the proved limit is free; the critical pair scaling $U \sim a^{-2}$ and finite-density limits are outside this conclusion. For $N = 2$, $\Lambda = v = U = t = 1$ and $a = 1/64$, the complete-vector bound is less than 0.001237 and the projector bound is $1/24576$.

[7] For the standard Weyl equation and helicity convention, see D. Tong, *Quantum Field Theory*, Sections 4.4 and 4.7, https://www.damtp.cam.ac.uk/user/tong/qft/qfthtml/S4.html.

The finite source check retains all 128 one-particle Fourier columns on a periodic 4^3 cube. An eight-valley two-particle state there has exact $\|D\psi\|^2 = 1/8$ and $\|(1-\Omega)D\psi\|^2 = 7/64$, so the contact escape is real. Propagation on all 8,128 two-particle states at $t = 1/8$ gives a numerical vector difference about 0.0441641, below the rigorous $\sqrt{1/8}/8$. These matrix diagnostics and exact source identities do not constitute an additional native preparation or detector execution.

Resulting energies and the physical boundary. At $v = 1$, $a = 1/64$, every valley has the following energies; both signs are retained, and only their common positive magnitude is displayed. The receipt gives rational enclosures for the rounded lattice values.

p	lattice $\lvert E_a\rvert$	continuum $\lvert E\rvert$	continuum mass
$(0,0,0)$	0	0	0
$(1,0,0)$	0.999959310393	1	0
$(1,1,0)$	1.414156018578	$\sqrt{2}$	0
$(1,1,1)$	1.731980331101	$\sqrt{3}$	0

These are original-generator model energies, without an SI calibration. The finite-spacing mass-shell defect is a regulator error, not a tachyonic mass. This growing-lattice branch is distinct from the protected eight-site cube: neither its repair invariance nor the identity of its eight response lines is transferred to the valleys. The empty Fock state is not a ground state: already at finite spacing, a one-particle mode at $(\pi/2, 0, 0)$ has energy $-c$, with zero contact for every U. This compares different number sectors; it is not a number-preserving extraction protocol from the empty sector. A filled vacuum, its interacting stability and a physical antiparticle dictionary remain unconstructed. A single 2×2 constant Hermitian mass anticommuting with all three kinetic Pauli matrices is exactly zero; this limited obstruction does not exclude valley coupling or many-body mass generation. The package `model/relativistic_sector_audit` therefore adds a controlled massless relativistic dynamical sector, not massive physical species, a photon assignment, an interacting relativistic vacuum theory or Einstein dynamics.

The preceding continuum construction does not itself execute a detector. The following finite periodic construction attaches model flavor and energy labels to a source-compiled occupation readout, and compares the resulting massless sector with observed particle data.

A detector identifies the model's propagating Weyl flavors. The massless sector now has a concrete model-particle readout. On the periodic 8^3 lattice at $c = 1/8$, retain all 1,024 physical CAR modes of the original Weyl/contact action. For each $b \in \{0,1\}^3$, choose momentum labels $\ell_b = (4b_x, 4b_y, 4b_z + 1)$, with $k_b = \pi\ell_b/4$. For component eigenvalue $r = \pm1$ of σ_z, the normalized mode is

$$\psi_{b,r}(x,s) = \frac{e^{ik_b\cdot x}}{\sqrt{512}}\delta_{s,(1-r)/2}, \qquad H\psi_{b,r} = E_{b,r}\psi_{b,r}, \qquad E_{b,r} = -r(-1)^{b_z}\frac{\sqrt{2}}{16}.$$

The contact term is exactly zero on the entire one-particle sector for every supplied $U \geq 0$. These sixteen modes are orthonormal, checked in Gaussian integers with amplitudes $(A+\sqrt{2}B)/32$. Their rank-one projectors satisfy $H\Pi_{b,r} = E_{b,r}\Pi_{b,r}$; the remaining one-particle projector commutes with H. Thus a label gives an actual source-generator energy, not a fitted interpretation of a peak. All modes have $N = 1$, odd CAR parity and fermionic exchange statistics from the supplied CAR algebra.

The operational species are eight Weyl flavors indexed by measured valley momentum. Their common energy magnitude is $\sqrt{2}/16 \simeq 0.0883883476483$ in original model units; it alone does not

distinguish the flavors. With $\chi_b = -(-1)^{b_x+b_y+b_z}$, the positive-energy branch has cone helicity $\chi_b/2$ and the negative branch has the opposite helicity.

flavor b	momentum labels ℓ_b	positive-branch component	positive cone helicity
000	$(0,0,1)$	1	$-1/2$
001	$(0,0,5)$	0	$+1/2$
010	$(0,4,1)$	1	$+1/2$
011	$(0,4,5)$	0	$-1/2$
100	$(4,0,1)$	1	$+1/2$
101	$(4,0,5)$	0	$-1/2$
110	$(4,4,1)$	1	$-1/2$
111	$(4,4,5)$	0	$+1/2$

Here helicity uses the explicit proper valley frame $R_b = -\chi_b \operatorname{diag}((-1)^{b_j})$; it is $(-1)^{b_x+b_y} r/2$ on this family. The raw detector measures momentum and CAR component, so bare $\sigma_z/2$ is not silently identified with this cone spin or with physical particle spin. The total directional current $J_z(k) = -c \cos k_z \, \sigma_z$ has eigenvalue $E_{b,r}$ here, with nonzero current $\pm\sqrt{2}/8192$ on every forward z bond. Current is stationary on these selected modes; $[H, J_z] \neq 0$ at generic momenta. The witness is propagating flux, even for a uniform-density plane wave.

The Fourier readout is compiled with complete native errors. The detector applies the fermionic extension of $F[k,x] = e^{-2\pi i k\cdot x/8}/\sqrt{512}$ separately to each component, then copies all 1,024 occupations to fresh pointers. Its output includes every binary occupation string, the vacuum, all other one-particle addresses, multiple occupations and every export abort. On the chosen sixteen effects it measures the energies above; the other 1,008 one-particle records remain present without that energy assignment. On higher-number inputs these same records measure mode occupations, not the total interacting energy.

An explicit three-axis FFT uses 4,608 two-mode butterflies and 768 bit-reversal transpositions. Each butterfly is a mode phase followed by the native hopping word with coefficient $-i$ and duration $\pi/4$. All nonadjacent operations use graded FSWAP routes, including their double-occupation sign. Exact arithmetic in $\mathbb{Z}[\zeta_8]/(\zeta_8^4 + 1)$ proves the 64 entries and 64 Gram sums of the one-axis transform; tensoring gives the full mode map. One transform requests 1,260,800 adjacent FSWAPs and 32,909,824 native pulses. Native excitation from the vacuum, inverse-transform preparation, forward-transform readout and all occupation pointers total 65,822,721 formal pulses. Original-H propagation is a separate specified model-law service, not an uncounted synthesized field word.

The complete literal-source error for preparation and detection is

$$\epsilon_{\text{source}} < 8.31 \times 10^{-9}.$$

The estimate retains all first-export failures and is valid with untouched references. It does not assume that literal native maps are exactly Gaussian or number preserving. Unitary original-H propagation on the full Fock space does not amplify preparation error. Therefore, at every model time, a prepared catalog member produces its correct native label with unconditional probability at least $1 - \epsilon_{\text{source}}$; all wrong, outside, vacuum, multiple-occupation and abort outputs together have probability at most ϵ_{source}.

The replay executes the ideal Gaussian comparison on all 1,024 one-particle rows and all sixteen inputs. At time eight its complete output-vector discrepancy from the exact spectral prediction is about 1.94×10^{-15}, separate from the rigorous native bound. A seeded 65,536-record illustration retains all sixteen label counts and every zero outside count for the exact uniform input mixture.

These are model-simulated counts, not a literal 2^{1024}-row native execution or laboratory data. No survival normalization is used.

Mass readout and comparison with measured particles. For the periodic family $L = 8\cdot 2^m$, $a = 2^{-m}$ and $c = (1/8)/a$, the model period stays eight and the momentum offset stays $p_z = \pi/4$. The exact branch energies satisfy

$$E_\pm(a) = \pm\frac{\sin(a\pi/4)}{8a} \longrightarrow \pm\frac{\pi}{32}, \qquad m_{\text{continuum}} = 0.$$

This is a direct finite-torus consequence of the original spectrum. The finite-spacing mass-shell defect is a regulator correction, not a tachyonic mass. No SI scale is assigned.

A separate retrospective comparison uses fixed PDG 2025 targets, which never enter the source or detector calculation:

physical target	measured mass in MeV/c^2	present continuum mass
electron	$0.51099895000 \pm 0.00000000015$	0
muon	$105.6583755 \pm 0.0000023$	0
tau	1776.93 ± 0.09	0

No common positive unit conversion maps zero to these positive masses; equal finite-momentum energies are not a rest-mass hierarchy.[8] The photon has spin one, so direct spin-preserving identification with these cone-helicity-1/2 modes also fails. Three such massless flavors give zero mass-squared differences, outside the nonzero ranges of the specified three-neutrino fit in PDG Table 14.7.[9] A single massless neutrino or a composite photon in another sector is not excluded. The model's Weyl flavors are now operationally identified; direct identifications with measured physical species remain zero. The empty reference is not a ground state, and the negative branch is retained rather than renamed as a physical antiparticle. Interacting vacuum stability, mass generation, physical charges and the empirical dictionary remain additional work.

The missing ground-reference and mass-generation steps admit further constructions with explicit domains. The original cube has a global interacting ground, and its one-dimensional restriction has an interaction-generated canonical charge gap. A specified refinement of that restriction also permits a controlled relativistic mass calculation. The following results retain the reference and coupling choices rather than combining these distinct domains into an assumed physical sector.

A global interacting ground on the original cube. The unchanged open $2 \times 2 \times 2$ Weyl/contact source, with $c = 1/8$ and $U = 1$, now admits a certified comparison across its entire 65 536-dimensional Fock space:

$$E_G = -\frac{\sqrt{3}}{4}, \qquad N_G = 4, \qquad \dim\ker(H - E_G) = 5.$$

The five states form one hidden-spin $S = 2$ multiplet. No chemical potential, energy recentering, mass term, or replacement Hamiltonian is introduced. This is a finite interacting ground reference, not a claim of a unique or Lorentz-invariant infinite-volume vacuum.

[8] PDG 2025 listings: https://pdg.lbl.gov/2025/listings/rpp2025-list-electron.pdf; https://pdg.lbl.gov/2025/listings/rpp2025-list-muon.pdf; https://pdg.lbl.gov/2025/listings/rpp2025-list-tau.pdf.

[9] The photon listing is https://pdg.lbl.gov/2025/listings/rpp2025-list-photon.pdf. The selected normal-ordering fit gives $\Delta m^2_{21} \in [6.92, 8.05] \times 10^{-5}\,\text{eV}^2/c^4$ and $\Delta m^2_{32} \in [2.388, 2.531] \times 10^{-3}\,\text{eV}^2/c^4$; these are reported 3σ fit ranges, not exact bounds: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf.

The existing onsite unitary frame sends the original hopping to the spin-independent signed cube adjacency $A/16$, where

$$A = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}, \quad BB^T = 3I_4, \quad K = \begin{pmatrix} \sqrt{3}I_4 \\ -B^T \end{pmatrix}, \quad K^*K = 6I_4.$$

Filling the four negative orbitals $K/\sqrt{6}$ with hidden spin up gives an explicit normalized state G. Its occupation coefficients belong to $\mathbb{Q}(\sqrt{3})$, it has no doublons, and the original physical-frame eigenvalue equation is checked exactly. With $\lambda = \sqrt{3}/16$, the $N = 4$ bound $H \geq -4\lambda$ is saturated precisely by negative-band states annihilated by every local pair annihilator. On the full 70-dimensional four-particle negative-band space, the stacked contact map has spin-weight block dimensions $1, 16, 36, 16, 1$ and ranks $0, 15, 35, 15, 0$. Nonzero integer minors modulo101 certify the rank lower bounds; five explicit spin-lowered vectors certify the matching upper bounds. This proves the complete five-dimensional ground kernel in that number sector.

For $N \leq 3$, positivity gives $H \geq -N\lambda > -4\lambda$. For $N \geq 9$, the occupancy bound $D \geq N - 8$ gives $H \geq -N\lambda + N - 8 > 0$. For $N = 5, 6, 7, 8$, hidden $SU(2)$ symmetry ensures that every eigenvalue occurs in the balanced spin weight. Those block dimensions are $1568, 3136, 3920, 4900$. A numerical factorization proposes an integer matrix Q; only the following subsequently checked integer identity enters the proof:

$$H = 2^{-40}QQ^T - \frac{27}{64}I + R, \qquad \|R\|_2 \leq \|R\|_\infty < \frac{1}{512}.$$

All integer product and row-sum overflow bounds are checked before evaluation. Since QQ^T is positive semidefinite for every proposal, the argument requires no trusted floating eigenvalue, factorization error estimate, or nonzero-pivot premise. The exact comparison

$$\frac{\sqrt{3}}{4} > \frac{441}{1024} > \frac{217}{512}$$

then proves that every $N \neq 4$ lies more than $7/1024$ above E_G. This last bound concerns charged sectors; no numerical lower bound for neutral excitations inside $N = 4$ is asserted. Any density matrix supported on this finite ground space is passive for the stated H.

Positive charged spectral data above this ground. At the origin the onsite frame is the identity, so the probes below are the original physical CAR modes. Their exact local spectral data are

probe or component	N	weight	$\omega = E - E_G$
$c_\uparrow G$	3	1/2	λ
$c_\uparrow^\dagger G$	5	1/2	λ
$c_\downarrow G$	3	0	absent
$S = 5/2$ component of $c_\downarrow^\dagger G$	5	1/10	λ
$\eta = 5/2$ component of $c_\downarrow^\dagger G$	5	1/10	$1 + \lambda$

Thus the resolved model gaps are approximately 0.108253175473 and 1.108253175473 in source energy units. They are finite-volume gaps, not relativistic rest masses. The remaining down-addition weight is $4/5$, with strictly positive spectral support and nonzero variance.

More explicitly, set $v = c_\downarrow^\dagger G$, with $\|v\| = 1$. The exact vectors

$$v_s = \frac{1}{5}S^-S^+v, \qquad v_\eta = \frac{1}{5}\eta^+\eta^-v, \qquad v_r = v - v_s - v_\eta$$

are mutually orthogonal and have squared norms $1/10, 1/10, 4/5$. The first two obey the original Hamiltonian eigenvalue equations at -3λ and $1-3\lambda$. Consequently the complete greater addition measure satisfies

$$\mu_{\downarrow} = \frac{1}{10}\delta_{\lambda} + \frac{1}{10}\delta_{1+\lambda} + \frac{4}{5}\mu_r, \qquad \operatorname{supp}\mu_r \subset (7/1024, \infty).$$

The displayed $1/10$ values are exact symmetry-resolved contributions; without further resolution of μ_r, coincident residual atoms are not excluded. All moments below are checked over $\mathbb{Q}(\sqrt{3})$:

$$\mathbb{E}\omega = \frac{1}{2}, \qquad \mathbb{E}\omega^2 = \frac{131}{256}, \qquad \operatorname{Var}(\omega) = \frac{67}{256},$$

$$\mathbb{E}_r\omega = \frac{1}{2} - \frac{\sqrt{3}}{64}, \qquad \mathbb{E}_r\omega^2 = \frac{131}{256} - \frac{\sqrt{3}}{64}, \qquad \operatorname{Var}_r(\omega) = \frac{1069}{4096} > 0.$$

The contact expectation in v is $1/2$; deleting that interaction changes its mean gap from $1/2$ to zero. This channel therefore probes the actual contact interaction. Addition and removal carry model number changes $+1$ and -1 and odd CAR grading relative to G; electric charge and physical Lorentz spin do not follow from these labels.

Attachment to the declared repair process. The existing four-dimer repair completion has24 active columns, all entirely in $N = 6$. Its complete130-outcome dictionary therefore acts on $N = 3 \oplus N = 4 \oplus N = 5$ as zero for every active Kraus outcome and as identity for the outside projector. This6748-dimensional statement retains cross-number coherences and arbitrary reference systems. Since the original H preserves number, it also survives every finite interleaving with the ordinary field evolution. This is passive outside-service protection, not active correction or autonomous preparation of the ground.

The new finite ground removes the negative-frequency objection that applied to the earlier excited reference. It does not retroactively turn that earlier preparation into ground-state preparation. A native source and detector for G, the unresolved charged spectrum, a controlled growing-volume interacting vacuum, relativistic massive poles, physical dictionaries and units, and the geometric Einstein premises remain separate obligations. The exact certificate and negative controls are retained in `model/interacting_vacuum`.

An interaction-generated charge scale in the original one-dimensional law. There is a positive many-body charge threshold without inserting a Dirac mass. On the unchanged periodic Weyl/contact geometry $(L, 1, 1)$, $L = 4m$, the onsite frame $G_x = (-1)^x (i\sigma_x)^x$ satisfies $G_x(i\sigma_x)G_{x+1}^{\dagger} = -I$ and has period four. Its CAR lift fixes local doublons, so the original action is exactly the repulsive Hubbard chain with $t = c/2 = 1/16$ and $U = 1$. Four exact local frame identities, source-matrix checks through $L = 16$, and an independent 256-state full-Fock witness bind this join. Unlike the earlier logarithmic-reference alternative Hamiltonian, no coupling is reassigned.

For the *half-filled canonical ground sector*, the imported Lieb–Wu charge-gap theorem gives

$$\Delta_c = \lim_{L\to\infty}[E_L(L+1) + E_L(L-1) - 2E_L(L)] = U - 4t + 8t\sum_{k\geq 1}\binom{1/2}{k}(2t/U)^{2k-1}\eta(2k-1).$$

Here the series is used only for $2t/U < 1$. Twelve binomial terms and 96-term rational Euler bounds for each Dirichlet eta value enclose $\Delta_c = 0.7715517157380375\ldots$ with width below $1.67\,10^{-25}$. The binomial tail is bounded by $8t|\binom{1/2}{K+1}|(2t/U)^{2K+1}/[1-(2t/U)^2]$; the nonnegative Euler coefficients give tail 2^{-96}. Floating quadrature is an independent check only.

Particle–hole symmetry fixes the charge costs for the original H:

$$E_+ = \frac{U + \Delta_c}{2} = 0.885775857869\ldots, \qquad E_- = \frac{\Delta_c - U}{2} = -0.114224142131\ldots.$$

Thus half filling is a specified canonical reference, not the original $\mu = 0$ grand-canonical vacuum. The symmetric threshold $\Delta_c/2 = 0.3857758578690188\ldots$ uses the explicitly centered $K = H - (U/2)N$; Δ_c itself is invariant under any common $-\mu N$. The spin sector remains gapless, contact still vanishes in the one-particle sector, and no preparation or complete-repair preservation of this many-body reference is established by the gap theorem.

The fixed-interaction continuum boundary. The equivalent positive integral for the charge gap and the elementary K_1 integral bounds imply, for every $t, U > 0$,

$$\Delta_0 \le \Delta_c \le \Delta_0 \frac{1 + 3/(8A)}{1 - e^{-2A}}, \qquad A = \frac{2\pi t}{U}, \qquad \Delta_0 = \frac{8}{\pi}\sqrt{Ut}\, e^{-A}.$$

Consequently fixed U with $t = v/(2a)$ has $\Delta_c \to 0$ as $a \to 0$: this lattice gap does not itself supply a nonzero continuum mass. A sufficient additional running family is

$$a(A) = \frac{4ve^{-A}}{M\sqrt{2\pi A}}, \qquad t(A) = \frac{v}{2a(A)}, \qquad U(A) = \frac{\pi v}{a(A)A},$$

for a *chosen* $M, v > 0$. It gives $\Delta_0/2 = M$ exactly and $1 \le \Delta_c/(2M) \le [1 + 3/(8A)]/[1 - e^{-2A}] \to 1$. This is a quantitative charge-threshold limit, not a derivation of M from the fixed source, a complete relativistic scaling theory, or a $3+1$-dimensional massive species. The receipt `model/mass_generation_audit/results.json` keeps these reference, coupling-family, and dimensional premises explicit.

The preceding charge-threshold result alone does not prove a mass shell. The next construction supplies energy and momentum together, with a controlled relativistic limit along an explicitly chosen coupling family.

A controlled relativistic charge mass on an explicit running family. The preceding charge gap can be promoted to a relativistic dispersion *in one spatial dimension on a specified coupling trajectory*. The source-bound half-filled Hubbard reference and centered generator $K = H - (U/2)N$ remain explicit. The staggered particle–hole map $d_{x,\downarrow} \mapsto (-1)^x d^\dagger_{x,\downarrow}$, with $d_{x,\uparrow}$ unchanged, preserves hopping and reverses the centered interaction. A complete 256-state CAR witness checks this duality and the translation identity $PTP^\dagger = -(-1)^{N_\downarrow}T$. Consequently the attractive spin dispersion becomes the repulsive charge dispersion, with the appropriate momentum-carrier shift. Here p is relative to that low-energy branch; the onsite source frame also carries its specified uniform internal rotation. The even-ring SO(4) selection rules are retained: an elementary thermodynamic quasiparticle is not automatically an isolated finite-ring state or a local-CAR pole.

Writing $A = 2\pi t/U$, $\theta = A\sin k$, $|k| < \pi/2$, and $v = 2ta$, the imported exact dressed dispersion has the source-normalized form

$$E = \frac{8t}{\pi} \sum_{\substack{m \ge 1 \\ m \ \mathrm{modd}}} \frac{K_1(mA)}{m} \cosh(m\theta), \qquad vp = \frac{8t}{\pi} \sum_{\substack{m \ge 1 \\ m \ \mathrm{modd}}} \frac{K_0(mA)}{m} \sinh(m\theta).$$

On $a(A) = 4ve^{-A}/[M\sqrt{2\pi A}]$, $t = v/(2a)$, $U = \pi v/(aA)$, the leading Bessel coefficient is exactly M. For $A \ge 1$ and $|\theta| \le \Theta < A$, put

$$q = e^{-2(A-\Theta)}, \qquad r_E = \frac{1 + 3/(8A)}{1 - q} - 1, \qquad r_p = \max\{1/(8A), q/(1-q)\}.$$

The Gamma-integral tangent bounds $1-1/(8z) \le K_0(z)/b(z) \le 1$ and $1 \le K_1(z)/b(z) \le 1+3/(8z)$, with $b(z) = \sqrt{\pi/(2z)}e^{-z}$, and the positive odd-mode tail prove

$$|E/M - \cosh\theta| \le r_E \cosh\Theta, \qquad |vp/M - \sinh\theta| \le r_p \sinh\Theta,$$

$$\left|(E^2 - v^2p^2)/M^2 - 1\right| \le \cosh^2\Theta(2r_E + r_E^2) + \sinh^2\Theta(2r_p + r_p^2) \;\longrightarrow\; 0.$$

At $\Theta = 1$, the last bound is below 0.034 for $A = 64$ and 0.017 for $A = 128$. Thus the limiting charge-doublet dispersion has $E = M\cosh\theta$, $p = (M/v)\sinh\theta$, and the calculated rest mass is $m = M/v^2$. Exact rational finite-parameter enclosures and independent Bessel/integral controls are retained in `model/hubbard_relativistic_sector/results.json`.

For a concrete trajectory through the original point, take $A_0 = \pi/8$, $v = 1/8$ and $M_0 = e^{-\pi/8}/\pi$. Then $a(A_0) = 1$, $t(A_0) = 1/16$, $U(A_0) = 1$, and

$$M_0 = 0.2149329913552644\ldots, \qquad m_0 = \frac{64e^{-\pi/8}}{\pi} = 13.75571144673692\ldots \quad \text{in model mass units.}$$

This is the mass of the *chosen running trajectory*, not a unique mass selected by the original bare point. Indeed, for any $M > 0$, multiplying $a_M(A)$ by $1 + (M/M_0 - 1)e^{-(A-A_0)}$ gives the same starting point and limiting rest energy M. The initial lattice half-gap is different, and the fixed-U continuum family has zero limiting charge mass. No native ground preparation, repair-invariant charged pole, $3+1$-dimensional massive species, or empirical particle-mass identification follows from this one-dimensional result.

The original three-dimensional action also permits a growing-volume ground construction at fixed spacing. This closes ground-state existence and positive excitation energy without selecting the one-dimensional canonical reference or its coupling continuation. Its local response is then connected to the existing geometry theorem interfaces.

An original-action infinite-volume ground representation at fixed spacing. The full three-dimensional Weyl/contact action now admits a translation-invariant, gauge-invariant ground state at its original zero chemical potential, without choosing a particle-number sector. This is a compactness existence theorem, not a computed or natively prepared state. Keep fixed finite $c > 0$, $U > 0$ and the complete quasi-local CAR algebra with two modes per site. The previously proved full-Fock locality estimate supplies the strongly continuous dynamics τ_t. On each even-sided cube Λ_L of side $2L$, take the normalized projector onto the absolute minimum of H_{Λ_L}, extend this even state by the exterior vacuum, and average its translates over all sites of Λ_L. A diagonal subsequence of the bounded local matrix entries defines a state ω; positivity, normalization and CAR compatibility pass to the limit. The vanishing boundary-to-volume ratio makes ω translation invariant.

For a local A, including an odd charged probe, the local derivation $[H, A]$ contains only interaction terms touching its support. Whenever the support and its nearest neighbors lie inside the box,

$$\omega_L(A^*[H, A]) = \operatorname{Tr}\rho_L A^*(H_{\Lambda_L} - E_L)A \ge 0.$$

Bad translates occupy a vanishing fraction. Their real-part lower defect and imaginary magnitude are each bounded by $2\|A\|^2|\operatorname{supp} A|(U + 6c)$, so $\omega(A^*[H, A]) \ge 0$ in the limit. This uses the full ground minimum and would fail for a merely canonical ground with an unrestricted charged probe.

For a radius-R buffer, let $B_R(t)$ denote the existing volume-uniform local-dynamics boundary bound. Comparing both finite and infinite dynamics to this intermediate cube gives, for $\|A\| \le 1$,

$$|\nu_L(\tau_t(A)) - \nu_L(A)| \le \min\Big\{2,\, 2B_R(t) + 2\left[1 - (1 - R/L)^3\right]\Big\}.$$

First $L \to \infty$, then $R \to \infty$, establishes stationarity. The same comparison applies to two-point correlations. Every finite full-ground correlation has a positive spectral measure supported on $[0, \infty)$; compact-time convergence and bounded correlations pass all Schwartz Fourier tests supported on negative energies to the limit. The GNS implementation therefore satisfies

$$U_t \pi(A)\Omega = \pi(\tau_t(A))\Omega, \qquad U_t = e^{itL_\omega}, \qquad L_\omega \geq 0, \quad L_\omega \Omega = 0.$$

The generator measures excitation energy in this representation; no chemical potential has been inserted into the source action.

The state can be chosen at nonzero density. Tiling each even box by the already certified four-particle cube vector gives contact zero, energy $-2\sqrt{3}c$ per eight-site tile, and zero expectation for every cross-tile hopping. Hence $E_L/|\Lambda_L| \leq -\sqrt{3}c/4$. Cartesian difference operators commute and the Pauli mixed terms cancel, giving $h^2 \leq 3c^2 I$ and $H \geq -\sqrt{3}cN$. Thus every such full ground has density at least $1/4$. The exact real-bipartite particle–hole identity $E(2V-N) = E(N) + U(V-N)$ excludes $N > V$ from an absolute minimum when $U > 0$. Spatial averaging before the subsequence passes both bounds:

$$\frac{1}{4} \leq \omega(n_{0\uparrow} + n_{0\downarrow}) \leq 1.$$

Half-filling and a particular density are not assumed or concluded. The original ground energy per site also has a limit e_∞. For every even cube side ℓ, tiling larger boxes gives the computable bracket, conditional on its finite full-ground energy,

$$\frac{E_\ell}{\ell^3} - \frac{3c}{\ell} \leq e_\infty \leq \frac{E_\ell}{\ell^3}, \qquad -\sqrt{3}c \leq e_\infty \leq -\frac{\sqrt{3}c}{4}.$$

The lower surface term counts intertile hopping norms; a product of gauge-invariant tile grounds gives the upper bound. Untiled boundary terms vanish per volume, and the bracket width tends to zero. For the constructed averaged ground, $\omega(h_0) = e_\infty$, where h_0 assigns half of each incident bond to the origin. This supplies an energy-density observable, not a state-convergence rate or rest mass.

The contact interaction remains represented:

$$[H_U - H_0, a_0^* a_2] = U a_0^* a_2 (n_1 - n_3) \neq 0.$$

Every unital GNS representation is injective on each full local matrix algebra, so this quartic operator cannot disappear in representation. This faithfulness of the representation does not assert faithfulness of the ground-state functional. The claim is a nonzero represented interaction derivation, not a scattering or connected-vacuum-correlator theorem.

An exact source countercontrol prevents simply substituting the free filled sea. For a pure Slater projector with $P_{xx} = I_2/2$,

$$\mathrm{Var}(D) = \frac{V}{16} + \sum_{x \neq y} |\det P_{xy}|^2 \geq \frac{V}{16}.$$

If it is also a kinetic eigenstate, its interacting energy variance is $U^2 \mathrm{Var}(D) > 0$. On the original cube, the complete 3844-term negative-band filling has $N = 8$, $\langle D \rangle = 2$ and $\mathrm{Var}(H) = 2/3$ at $U = 1$. The exact checks and proof are retained in `model/thermodynamic_ground/results.json`. For local CAR probes, the same subsequence also supplies the analytic retarded response

$$G_{AB}(z) = -i \int_0^\infty e^{izt} \omega(\{\tau_t(A), B^*\})\, dt, \qquad \mathrm{Im}\, z > 0.$$

Each time tail after T is at most $2\|A\|\|B\|e^{-\eta T}/\eta$ for $\operatorname{Im} z \geq \eta > 0$. Compact-time locality and weak-star convergence, followed by this tail bound, give uniform convergence on compact subsets of the upper half-plane. The unknown finite-state convergence error is retained; no real-axis pole, local inverse or metric is inferred. The result closes fixed-spacing ground-state existence and the positive-energy representation for the actual interacting lattice. It supplies no selected ground functional, convergence rate, native preparation, repair stationarity, uniform gap, Lorentz continuum, physical particle assignment or Einstein dynamics.

Exact matter response before a geometric identification. The unchanged open $2 \times 2 \times 2$ Weyl/contact cube at $c = 1/8$, $U = 1$ now supplies a state-dependent retarded response, including its exact eliminated-state memory. In the existing onsite hidden-spin frame, write $h = A/16$, where A is the real signed cube adjacency and $A^2 = 3I$. The retained full-Fock ground theorem gives the normalized highest-weight reference G, with four occupied negative-band up orbitals, $E_G = -\sqrt{3}/4$, and a strict gap $\delta = 7/1024$ to every different particle-number sector. The second exact stationary reference is $G_+ = \sqrt{2}\, c_{0\uparrow}^\dagger G$, of energy $-3\sqrt{3}/16$. No chemical potential, fitted mass, or new geometry law is inserted. Physical CAR responses are obtained by the existing onsite unitary frame; hidden spin is not assigned a physical Lorentz or electric-charge meaning.

For either reference ψ of energy E, the complete eight-site fermionic response, for $\operatorname{Im} z > 0$, is

$$\begin{aligned}\mathcal{G}_{xy}(z) &= -i \int_0^\infty e^{izt} \langle \{c_x(t), c_y^\dagger\} \rangle_\psi \, dt \\ &= \langle c_x [z - (H - E)]^{-1} c_y^\dagger \rangle_\psi + \langle c_y^\dagger [z + (H - E)]^{-1} c_x \rangle_\psi .\end{aligned}$$

There is an exact isometric dilation

$$\begin{aligned} V_y &= c_y^\dagger \psi \oplus \overline{c_y \psi}, & V^\dagger V &= I, \\ L &= (H_{N+1} - E) \oplus -\overline{(H_{N-1} - E)}, & \mathcal{G}(z) &= V^\dagger (z - L)^{-1} V. \end{aligned}$$

The conjugate removal space fixes the matrix indices; the present source arrays happen to be real. With $P = VV^\dagger$, $Q = I - P$, define $M_1 = V^\dagger L V$, $W = QLV$ and $B = QLQ|_{\operatorname{ran} Q}$. Exact block elimination gives

$$\mathcal{G}(z) = \left[zI - M_1 - \Sigma(z) \right]^{-1}, \qquad \Sigma(z) = W^\dagger (z - B)^{-1} W, \qquad W^\dagger W = M_2 - M_1^2.$$

Both L and B are self-adjoint; consequently $-\operatorname{Im} \mathcal{G}(z) > 0$ and $-\operatorname{Im} \Sigma(z) \geq 0$ in the upper half-plane. The eight observed columns do not replace the source Hilbert space: their exact complement remains in the memory kernel.

A computed change of response at fixed microscopic geometry. For the up channel of both references, the full cyclic identity $LV = Vh$ gives $\mathcal{G}_\uparrow(z) = (z - h)^{-1}$ and $W = 0$. For G, the up dilation retains all 56 addition and 56 removal states. The ground down channel has no removal component and retains all 560 states with $N_\uparrow = 4$, $N_\downarrow = 1$. Exact arithmetic over $\mathbb{Q}(\sqrt{3})$ gives

$$M_{1,\downarrow} = h + \tfrac{1}{2} I, \qquad W_\downarrow^\dagger W_\downarrow = \tfrac{1}{4} I.$$

Thus a static Hartree term alone does not reproduce this interacting response. Indeed $W_x = (n_{x\uparrow} - 1/2) c_{x\downarrow}^\dagger G$; orthogonal down locations and up occupation variance $1/4$ give the displayed Gram matrix directly.

For G_+, the down channel retains its entire 448-state invariant block. In lexicographic order of the eight cube sites, its exact matrices are

$$M^+_{1,\downarrow} - h = \mathrm{diag}(1, \tfrac{2}{3}, \tfrac{2}{3}, \tfrac{1}{2}, \tfrac{2}{3}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}),$$
$$(W^+_\downarrow)^\dagger W^+_\downarrow = \mathrm{diag}(0, \tfrac{2}{9}, \tfrac{2}{9}, \tfrac{1}{4}, \tfrac{2}{9}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}).$$

The first-moment change is exactly the added orbital density, $(1/2, 1/6, 1/6, 0, 1/6, 0, 0, 0)$. Removing the contact term in the explicit $U = 0$ control removes both this Hartree term and the ground down memory, giving $(z - h)^{-1}$. That comparison does not declare G a full free-Fock ground. The excited reference receives no positive-gap assertion.

The complete local charged spectral measure. For the original physical origin probe $v = c^\dagger_{0\downarrow} G$, the retained calculation now resolves all 78 distinct positive spectral lines, including the previously unresolved weight 4/5. A monic degree-78 polynomial p over $\mathbb{Q}(\sqrt{3})$ is admitted by the exact source equation $p(16H)v = 0$. A nonzero 78×78 Hankel determinant modulo 97, with $\sqrt{3} \mapsto 10$, proves the matching cyclic rank lower bound. Exact moments determine a numerator q such that

$$\langle v, (z - 16H)^{-1} v \rangle = \frac{q(z)}{1296\, p(z)}.$$

The denominator 1296 is the squared norm of the inherited integer state numerator. Modular reconstruction and a numerical eigensolver only propose coefficients and brackets. Integer annihilation and rank checks, 78 disjoint strict sign changes, and outward 1024-bit interval arithmetic certify every root and its strictly positive residue $q(x)/(1296p'(x))$. Each charged gap is $(x + 4\sqrt{3})/16$.

The smallest gap is approximately 0.029950938342053226, with weight 0.3130113772155167; the largest is approximately 1.8360744654423855, with weight $2.711810625844545 \times 10^{-10}$. The receipt retains every gap and weight in rational intervals rounded outward to 24 decimal places. Polynomial substitution proves that the two lines $\sqrt{3}/16$ and $1 + \sqrt{3}/16$ each have total weight exactly 1/10. The remaining 76 lines have total weight 4/5, leaving zero unresolved weight for this local probe. This is the complete local cyclic measure, not the entire 560-state spectrum, a native detector execution, an infinite-volume pole theorem, or a relativistic mass assignment.

Controlled low-frequency coefficients. For the interacting ground down channel, put $K = H_{4,1} - E_G$. The full-ground certificate gives $K > \delta I$. Moreover, $D \le 1$ in this block, the kinetic norm is at most 5λ, and $-E_G = 4\lambda$, with $\lambda = \sqrt{3}/16$; hence $K \le 1 + 9\lambda < 2$. The same bounds hold for its compression B. For $|z| \le r < \delta$ the exact memory has the convergent expansion

$$\Sigma(z) = -\sum_{n=0}^{p} z^n C_n + R_p(z), \qquad C_n = W^\dagger B^{-n-1} W \ge 0,$$
$$\|R_p(z)\| \le \frac{r^{p+1}}{4\delta^{p+1}(\delta - r)}.$$

These are finite gapped frequency coefficients, not a spatial derivative expansion or a relativistic cone. All retarded moments $M_0, \ldots, M_4$ are also computed exactly; the corresponding large-frequency remainder is $2^{p+1}/[R^{p+1}(R-2)]$ for $|z| \ge R > 2$.

The static response is computed with an exact residual certificate. For $C = V^\dagger K^{-1} V = -\mathcal{G}_\downarrow(0)$,

$$\|C - (12.9058381883 I - 4.9026412320 A)\| < 10^{-7},$$
$$\|\Sigma(0) - (-0.3633623064 I - 0.0105943755 A)\| < 10^{-6}.$$

A numerical solve only proposes a matrix with denominator 2^{40}. Integer arithmetic encloses its full $\mathbb{Q}(\sqrt{3})$ residual; division by the proved δ bounds the solution error. The exact isometry then bounds C, and $C > I/2$ bounds its inverse. An independent rational inverse residual certifies $\Sigma(0) = C^{-1} - M_1$. No numerical tolerance admits the theorem, and exact equality to an affine adjacency polynomial is not inferred.

The existing geometry interface and its remaining join. The same-action thermodynamic-ground construction separately supplies analytic local responses along its ground subsequence on compact subsets of $\operatorname{Im} z > 0$. The cube calculation does not supply a rate of convergence to that state, and its finite gap is not transferred to infinite volume. The existing `wave_density_metric` constructor takes a complete local action density $\mathcal{C} = \mu \operatorname{diag}(-w, a)$, with $\mu, w > 0$ and $a > 0$, and recovers $g = \sqrt{-\det \mathcal{C}}\, \mathcal{C}^{-1}$. The existing `source_tetrad` interface in the common matter join then constructs the common coframe and the scalar, Weyl, Dirac and Maxwell principal-symbol identities for its supplied density. The present result supplies an actual matter-dependent inverse response from which such coefficients must be derived. It does not substitute a Hartree diagonal for a spatial principal symbol, identify the cube with the distinct screen source, or complete the common action-normalization join. Fixed microscopic geometry is compatible with this emergent-response route; a new fundamental metric register is not required. Common local limits, modular and area response, full stress, and the same-family Einstein balance premises remain to be proved. No new physical species, native response detector, or completed G1–G6 premise is claimed.

Other finite interfaces. The full-CAR locality bounds used above control dynamics at fixed lattice spacing; a chiral spatial continuum remains a further limit. For partition expectations E_c and positive rates,

$$L = \sum_c \gamma_c (E_c - I), \qquad \operatorname{Tr}(X^* L X) = -\sum_c \gamma_c \|X - E_c X\|_{\mathrm{HS}}^2$$

identifies $\ker L$ with the intersection of the partition commutants, without assuming commuting pinchings or independent no-cancellation. Applying this theorem requires the stated generator. Equivariant rational chain maps and homotopies on the supplied finite support tower are homology maps, not quantum channels or a dynamical refinement limit.

What the negative controls exclude. Several finite benchmarks constrain possible identifications. A supplied Weyl/contact action has an isolated moving bound-pair band, complete contact residues and an optical identity, but its positive contact section makes the band topologically trivial. Its certified curvature fails the rest/kinetic mass relation with the same speed parameter. Contact interactions also couple proper Weyl momentum pockets to their complements. The lattice scattering theorem, finite thermal positivity and KMS identities do not remove those obstructions or supply physical asymptotic detectors, a chiral continuum or calibrated cross sections.

The centered repulsive dimer gives a sharper family-wide exclusion. For $t = c/2 > 0$, $U \geq 0$, $\mu = U/2$ and $d = \sqrt{U^2 + 16t^2}$, its two charge-one doublets have gaps $g_\pm = d/2 \pm t$, hence

$$1 < g_+/g_- \leq 3.$$

No common positive linear conversion yields the charged-lepton hierarchy or three distinct same-charge families there. Allowing another chemical potential can evade the ratio bound but does not create a third doublet. The fixed complete charged-frequency catalogue also fails its broader lepton-ratio comparison. The periodic four-site extension fails its declared $E(k)^2 = m^2 + c^2 \sin^2 k$

benchmark at $c = 1/8$; a two-parameter fit to two distinct momentum values does not rescue that test. These exclude stated finite dictionaries, not every larger action or a renormalized continuum limit. Electron/positron masses set by one common measured calibration, and retrospective parameter comparisons, provide no independent mass prediction.

Finite current response also has a precise scope. An open-bond phase is pure gauge and measures no magnetic moment; the contact kernel is not a Coulomb kernel. On the complete periodic ring, a genuine loop phase has gauge-invariant current and total curvature,

$$E_0' = \langle \partial_\phi H \rangle, \qquad E_0'' = \langle \partial_\phi^2 H \rangle - 2 \sum_{n \neq 0} \frac{|\langle n | \partial_\phi H | 0 \rangle|^2}{E_n - E_0},$$

under the isolated-ground hypothesis. Degenerate free and atomic controls lie outside that formula. The calculation retains all 256 Fock states but does not identify a photon or physical Lorentz spin.

Clock and memory exclusions are similarly conditional. A frozen real-operation grammar has a normalized trace-distance lower bound $1/2$ for a specified nontrivial modular action of its nontracial reference. Finite independent copies of the original native reference also fail the required field modular-gap ratios. These do not exclude the explicitly enlarged clock and control laws above. The earlier pair/dark-reference encoding covers a fixed corner, not the whole spatial field. Finite stationary observer references concern source/buffer marginals at the stated repair rate and on their recurrent support; delayed variants need their own construction, and growing exported histories are not made stationary by these marginal references. Actively corrected coherent memories can have flat logical energy and a fixed core, so memory lifetime alone does not establish a moving massive particle. Properties proved on distinct carriers require an explicit intertwiner before they can be combined.

Physical target and remaining obligations. The target is an interacting quantum field theory with gapless photons and experimentally matched massive sectors, not a positive global mass gap. A gapless photon rules out such a global gap. An exact nontrivial positive-energy conformal theory in a flat-space vacuum representation with unbroken unitary dilations also cannot have a positive flat-space vacuum mass gap: for a spectral value $E > 0$ and proposed gap $g > 0$, scale covariance also puts $g/2$ in the spectrum. This does not exclude a gap in scaling dimensions or discrete finite-radius cylinder levels; the cylinder radius supplies an energy scale. Conformal behavior may describe a critical or short-distance regime, but a deformation or symmetry-breaking mechanism responsible for massive flat-space sectors must then be specified and implemented.

The finite results leave concrete physical joins. A localized defect must carry the protected charge and the same measured field observables. Spatial refinement must control the additional lattice modes and yield the required Lorentz propagation, while keeping the interactions and reference state. The electromagnetic field and gauge–Higgs interactions must be dynamical, with charge constraints, detector coupling and their infrared behavior treated on that same theory. Massive excitations need a common energy and clock normalization, including the relation $E''(0) = c^2/E_0$ for a relativistic branch with $E_0 > 0$, together with the appropriate stress and proper-time relations. The neutrino sector and other measured particle properties require their own matching tests. Neither a successful finite source compilation nor a calibrated spectrum alone discharges these obligations.

This paper's hypothesis and place in the tower

This paper is layer 1 of the tower: the mathematics of local agreement repair. Its hypothesis is that local agreement repair between overlapping patches reaches global consistency, as a schedule-independent quotient normal form that preserves its protected boundary data and orders its commits by provenance, and that the residue local repair cannot remove is cycle holonomy, or frustration. Hidden local representation is gauge: implementation detail that the public normal form does not see.

The finite theorems are proved outright for every structure meeting their stated hypotheses: confluence of an admissible accepted repair law (Theorem 3.25), the adaptive and capacity horizons, the holonomy criterion and the frustration bound (Theorem 4.1, Corollary 4.3), gauge descent (Theorem 5.2), refinement-limit classes, the record-algebra and finite observable-net results, distributed invariance and the generated provenance order. The noisy fair-block bound and the protocol and code extensions are implications from certificates that an implementation must supply.

Two results pass to higher layers. Nonzero cycle holonomy with fixed edge offsets forces a strictly positive residual mismatch; this is the mathematical seed of the matter hypothesis of layer 5 (Section 4.1). Gauge as implementation hiding is the quotient on which the gauge layer reconstructs transport (Section 5.2). The normal form and its records feed the observer layer. The provenance order is informational and reaches the spacetime layer only through further premises. The flagship paper composes the layers [14].

These theorems describe the executable model only after its native law is presented as an admissible quotient repair law. The integer pair-repair and transactional contracts satisfy the descent premise. The native seam rule waits or swaps at mismatch one, so its binary shell keeps moving and its accepted moves need not lower a measure [13]. A quotient presentation of the native law that meets the admissibility premises, or a separate theorem for its continuing regime in the sense of self-stabilization [17], is an open join. The matter bridge needs native transition data that carry a nonzero holonomy class preserved by the admitted repairs, together with a localized core, its transport and a derived energy; none of these is constructed.

An admissible presentation in which two accepted schedules from one initial state reach different quotient normal forms, with no holonomy obstruction and no mere gauge difference, refutes the consensus hypothesis for that law. So do two inequivalent normal forms from the same boundary data where the fiber-uniqueness hypothesis of Theorem 5.4 holds. An exhaustive proof that native transport admits no nonzero holonomy class preserved by the admitted repairs, over every sector and refinement of a declared defect class, stops the defect route to matter.

Carrier signature, screens and observers

The tower panel above orders the layers. Three conventions are shared by every layer that uses the carrier.

Three meanings of screen

The word "screen" is used for three related objects that are identified only through an explicit map.

1. The *local carrier boundary* is the twelve-port oriented interface of one icosahedral carrier. Its incidence has $(V, E, F) = (12, 30, 20)$.

2. The *federation screen* is the routed system of interfaces, records, repairs and checkpoints of many carriers at finite cutoff.

3. The *support screen* is the observer-facing geometric chart. On the spherical branch it is the refined conformal S^2 used for caps, collars, modular flow and Lorentz reconstruction.

Local icosahedral incidence does not determine the topology of the federation nerve. Identical local carriers can be routed as a path, a cycle, a higher-genus complex or a spherical complex. The map from routed carriers to a support-visible spherical nerve is a physical bridge with its own premises.

Presentation invariance and architecture sensitivity

OPH is invariant under changes of presentation that preserve the complete observer-visible carrier signature. On the icosahedral branch that signature is

$$\mathcal{C}_{i,r} = \big(\mathcal{A}_{i,r}, \rho_{i,r}, P_{i,r}, I^{\mathrm{or}}_{i,r}, \mathcal{R}_{i,r}, \mathcal{U}_{i,r}, \mathsf{Chk}_{i,r}, \mathsf{Resp}_{i,r}, c_{sr}\big),$$

with accessible algebra, state, port set, oriented incidence, record algebra, repair or feedback interface, checkpoints, visible response law and refinement lineage. Hidden coordinates, port names, worker partitions, materials and wiring presentations are silent when an isomorphism preserves this tuple and its error model. A change of port number, incidence, orientation, accessible algebra, response, repair law, clock or refinement lineage can change the admitted physics. A cube and an icosahedron are different carrier contracts even when built from the same material.

When a carrier is an observer

A carrier body realizes an observer when it supplies bounded access, self-readback, durable records, record-conditioned feedback, boundary prediction against controls and checkpoint continuation. One carrier may pass that test, or a connected subfederation may pass it instead; no theorem fixes the primitive observer size by counting carrier bodies. Equal-frequency phase locking can synchronize an interface; it makes the interface an observer only together with these record and feedback clauses.

How the layers compose

Consensus, geometry and gravity are composable branches of one carrier tower: accepted repair produces the quotient-visible normal form, the exact port response supplies the rank-three slow-response carrier, and the support screen feeds the modular and Lorentz constructions of the spacetime layer. The gauge layer uses the same carrier architecture, where the complete twelve-port response and endogenous holonomy force the abstract local Lie algebra $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. The matrix current and matter action are conditional realizations; a common source construction joining them to the geometric tower is the matter layer's open join. Matching dimensions or symmetry labels does not supply that join.

1 Introduction

A world is represented here as a finite graph of observer patches. Each patch carries local state, neighboring patches compare their data on overlaps, and local repair moves reconcile mismatch. The questions are whether this repair reaches one shared world, and how the obstructions are encoded when it cannot.

"Dynamics" is meant in the rewriting-system sense. The repair schedule selects terminal normal forms, and schedule independence is proved under stated hypotheses. The selected observer-facing structure can carry an internal record order, which is read as history. Identifying the normal form with intrinsic computation of a complete history requires an observation- and intervention-preserving realization map, and a physical-clock reading additionally requires an observer-readable transition process, event correspondence and clock calibration. The patch-net algorithm is a theorem device for fixed-point selection; the description contains no global timeline on which spacetime contents are rendered.

The mathematics is standard where it should be. Constraint satisfaction, rewriting, well-founded descent, local-diamond confluence [19], inverse limits and recovery channels enter with their usual meanings. OPH adds the reading: each variable is an observer patch, each constraint an overlap-visible record check, and each accepted rewrite a local repair move that descends to the quotient seen by neighboring observers. Two precedents separate the available convergence tools. Dijkstra's self-stabilizing ring reaches a legitimate regime under its central daemon while permitted activity continues [17], which differs from termination of a repair relation. For finite-dimensional inner-product stalks and linear restriction maps on a finite graph, sheaf diffusion converges to the orthogonal projection onto compatible sections [18]. Each result requires its own state, map and scheduling hypotheses, and neither selects a physical law from its equilibria. The finite rewrite construction below is a separate instantiation.

The results are as follows.

1. **Constraint codes.** A finite overlap net is a finite constraint code with $C = \Phi^{-1}(0)$ (Proposition 2.3). It is not by itself a quantum error-correcting code, and the bare graph does not determine a code distance (Theorem C.5).

2. **Asynchronous confluence.** Primitive recovery proposals commit only through transactional snapshot validation, protected-data preservation and exact descent. The accepted relation induces a total, idempotent, boundary-preserving quotient repair map $\mathrm{Rep}_\lambda : Q \to Q$. Semantic-complete read sets and one coherent aggregate per conflict component prove the quotient local diamond, and with repair completeness every initial state has a unique schedule-independent quotient normal form (Theorems 3.9 and 3.25). Attempt streams, fairness and cumulative capacity are classified exactly (Theorems 3.20 and 3.21).

3. **Cycle obstruction and frustration.** For affine abelian overlap constraints, global consistency holds exactly when every cycle holonomy vanishes (Theorem 4.1); fixed nonzero holonomy forces positive residual mismatch (Corollary 4.3); the crossed-module class extends the obstruction to the noncentral branch (Theorem 4.4).

4. **Gauge quotient and boundary data.** The normal form descends to the quotient by implementation hiding, and preserved boundary data with a unique consistent extension fix the quotient bulk (Theorems 5.2, 5.4, 5.7, 5.10 and 5.16).

5. **Refinement.** Natural restriction systems assemble normal forms and holonomy classes into inverse-limit classes, and controlled coarse-graining commutes with reconciliation up to declared errors (Theorems 6.2 and 6.9).

6. **Records and finite observable nets.** A supplied density state and a declared Lüders instrument give Born–Lüders formulas on central record projectors, with explicit stability bounds for approximate records (Theorem 7.2). Authenticated provenance generates a finite informational order, and the finite consensus tower carries a normed completion, selected-state GNS representations and exact finite local dynamics.

7. **Implementations.** Admissible distributed presentations of one global carrier have the monolithic normal form (Theorem 9.5), and fair-block contraction certificates control noisy repair (Theorem 10.1).

The repair step is a concrete recovery move. On the fixed-cutoff collar branch a local update comes from exact Markov splice or from a declared Petz/Fawzi–Renner recovery channel, read on overlap-invariant data. The union-collar payload is parenthesization independent only with an exact aligned Markov construction, a coherent canonical recovery construction or a primitive aggregate payload (Proposition B.7). Repair completeness, the quotient local diamond and, on the Petz branch, the support clause of Proposition C.4 are separate conditions. The rooted-tree packet domain of Theorem 3.14 proves all of them on a nontrivial exported net.

Synchronization and consensus are distinct. A phase-lock process may establish a stable relative phase between two exposed ports. It enters these theorems only after its phase records are tied to the accepted transaction ledger, an independent clock, the record surface and the confluence premises; frequency entrainment can coexist with conflicting records.

2 Patch Nets, Overlaps, and Global Consistency

Definition 2.1 (Patch net)**.** *Let $G = (V, E)$ be a finite connected graph. Each vertex $i \in V$ is an* ***observer patch*** *with finite local state space S_i. The global state space is*

$$\Sigma := \prod_{i \in V} S_i.$$

For each edge $e = \{i, j\} \in E$, let I_e be an interface alphabet and let

$$\pi_{i,e} : S_i \to I_e, \qquad \pi_{j,e} : S_j \to I_e$$

be the interface projection maps. A global state $s = (s_i)_{i \in V} \in \Sigma$ is ***consistent on edge*** *$e = \{i, j\}$ iff*

$$\pi_{i,e}(s_i) = \pi_{j,e}(s_j).$$

The global consistency set is

$$C := \{s \in \Sigma : \forall\, e = \{i, j\} \in E,\ \pi_{i,e}(s_i) = \pi_{j,e}(s_j)\}.$$

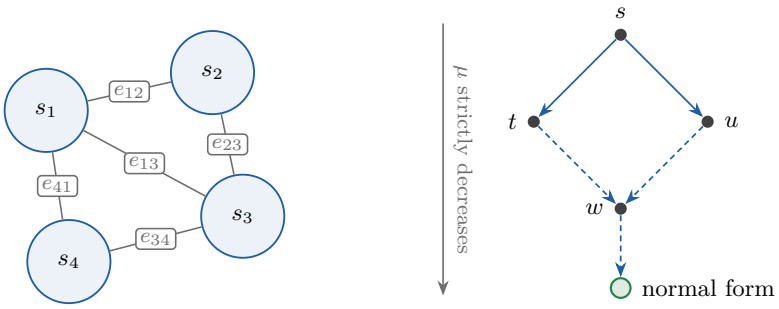
For exposition we use a finite pairwise-overlap graph. The hypergraph version is straightforward: replace edges by hyperedges and pairwise equality by a common interface label on each hyperedge. Nothing in the proofs depends on the pairwise restriction.

The picture: each observer holds a local state, and neighboring observers share an interface through which they can compare notes. A universe-state is physically admissible exactly when all neighbors agree on their shared data. This is a constraint satisfaction problem (CSP), and the consistent states are the codewords.

Definition 2.2 (Inconsistency potential)**.** *For each edge e, choose a weight $w_e > 0$ and a function $d_e : I_e \times I_e \to \mathbb{R}_{\geq 0}$ with $d_e(a, b) = 0 \iff a = b$. On the declared fixed-cutoff branch, d_e is the overlap score used by the local acceptance contract on that interface. Define*

$$\Phi(s) := \sum_{e=\{i,j\} \in E} w_e\, d_e\big(\pi_{i,e}(s_i), \pi_{j,e}(s_j)\big).$$

Then $s \in C \iff \Phi(s) = 0$.



(a) Patches i with local states s_i; on each overlap $e = \{i, j\}$ the interface projections must agree, $\pi_{i,e}(s_i) = \pi_{j,e}(s_j)$.

(b) Every one-step peak $t \leftarrow s \rightarrow u$ closes at a common w (local diamond) and every accepted repair lowers the measure μ; Newman's lemma then gives one normal form, whatever the order of repairs.

Figure 1: (a) A patch net: a global state is consistent when the interface projections agree on every overlap (Definition 2.1). (b) The finite consensus mechanism: accepted repairs strictly decrease the exact measure, every one-step peak closes in a local diamond, and Newman's lemma gives one normal form whatever the order of repairs (Theorem 3.9).

Proposition 2.3 (Bare overlap nets are finite constraint codes)**.** *For every finite patch net of Definition 2.1, the consistency set $C \subseteq \Sigma$ is a finite constraint code whose codewords are exactly the globally overlap-consistent states. With the mismatch potential of Definition 2.2,*

$$C = \Phi^{-1}(0).$$

This proposition is the theorem-grade content of the unqualified phrase "the overlap network is a code." It supplies a finite constraint code, not a quantum error-correcting code, not a topological code, not a graph-theoretic formula for code distance, and not a Lorentzian or Einstein-geometry theorem.

Proof. Finiteness follows from $\Sigma = \prod_i S_i$ with finite S_i. The definition of C is a finite family of interface-equality constraints, so C is the set of satisfying assignments. Since each $w_e > 0$ and $d_e(a, b) = 0$ exactly when $a = b$, every term in Φ is nonnegative and vanishes exactly on a satisfied edge constraint. Therefore all edge constraints hold if and only if $\Phi(s) = 0$. □

Definition 2.4 (Bare finite consensus reduct)**.** *A bare finite consensus reduct at regulator r is*

$$\mathsf{Cons}_r = (\Sigma_r, \Gamma_r, Q_r, \Phi_r, \rightarrow_r, n_r, C_r, B_r),$$

where Σ_r is the finite presentation space, Γ_r is the presentation-redundancy groupoid, $Q_r = \Sigma_r/\Gamma_r$ is the physical quotient, Φ_r is the mismatch functional, $\rightarrow_r$ is the accepted repair relation, n_r is the quotient normal-form map, and

$$C_r = \Phi_r^{-1}(0)$$

is the globally overlap-consistent set, and B_r is its boundary-data readout.

Theorem 2.5 (Bare finite consensus is not Einstein-complete)**.** *The bare finite consensus reduct Cons_r does not determine a Lorentzian metric g_{ab}, stress tensor T_{ab}, Newton coupling G, area/edge operator L_C, generalized entropy S_{gen}, modular geometric flow, or the equation*

$$G_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab}.$$

Consequently, Einstein geometry is not a theorem of the bare finite consensus language.

Proof. The symbols of Cons_r describe finite states, quotienting, mismatch, accepted repair, normal forms, and consistency. They do not include a metric, curvature tensor, stress tensor, cap modular automorphism, area operator, or entropy-area normalization. Hence two model extensions can share the same Cons_r while assigning different geometry/stress data. One extension may attach Minkowski metric, $T_{ab} = 0$, and $\Lambda = 0$, so Einstein holds. Another may attach a Lorentzian metric g'_{ab} and stress tensor T'_{ab} for which

$$G'_{ab} + \Lambda' g'_{ab} \neq 8\pi G' T'_{ab}$$

somewhere. All bare consensus statements have the same truth value in the two extensions, while the Einstein equation has different truth values. Therefore the Einstein equation is not entailed by the bare reduct. □

The theorem excludes definability of the Einstein equation from free decorations of Cons_r: the two extensions attach geometry with no arrow from the quotient. Its machine-checked form covers the integer scalar Einstein identity on a constant demonstration tower [12]; the metric, stress, coupling, area, entropy and modular-flow clauses rest on the model extension above. Extensions whose geometry is a computable function of the quotient normal forms are the subject of the producer and composition theorems and lie outside this exclusion. Bare consensus supplies quotient normal forms, boundary data and obstruction classes. It supplies no future null cone, round-cap support chart, conformal transport, time orientation or observer-frame homogeneous space; the spacetime layer adds these with its own geometric-readout, modular, stress, entropy and calibration premises [32]. The functional Φ measures mismatch and is no energy: repair weights and counts acquire an energy meaning only through a derived maintenance-to-energy map in a calibrated clock.

3 Asynchronous Reconciliation and the Main Fixed-Point Theorem

Definition 3.1 (Recovery-derived local repair law)**.** *Fix for each patch i a finite collar chart A_i–B_i–D_i around the overlaps touched by i, together with a fixed local decoder from repaired collar data back to the finite patch label at i. A* ***law*** *λ is a family of local repair maps*

$$T_i^{\lambda} : \Sigma \to \Sigma \qquad (i \in V)$$

such that T_i^{λ} changes only the state of patch i (or, more generally, only a bounded neighborhood of i), and the local update is induced by one of the declared OPH recovery moves on that collar, where $\omega_{A_iB_i}(s)$ denotes the $A_i \cup B_i$ marginal of the input collar state encoded by s:

(i) exact Markov splice on the collar, using Theorem A.1 when $I(A_i : D_i \mid B_i) = 0$; or

(ii) a declared recoverability channel

$$(\mathrm{id}_{A_i} \otimes \mathcal{R}_i)(\omega_{A_iB_i}(s)), \qquad \mathcal{R}_i = \mathcal{R}_{\sigma_i, \mathcal{N}_i},$$

with $\mathcal{R}_i$ in the Petz/Fawzi–Renner class of Definition C.3 and Theorem A.1.

The decoder back to S_i is bookkeeping for the finite patch presentation; the physical content is the repaired collar state on the declared fixed-cutoff branch. Write $s \rightsquigarrow_i t$ iff $t = T_i^{\lambda}(s) \neq s$. These are primitive proposals, not accepted physical rewrite steps. A proposal is committed only through the transactional acceptance layer below.

Definition 3.2 (Touched-overlap potential and accepted local-fit contract)**.** *For each repair site i, let*

$$E_i^{\mathrm{touch}} := \{e \in E : \text{the interface data on } e \text{ may change under } T_i^{\lambda}\}.$$

Define the touched-overlap potential

$$\Phi_i(s) := \sum_{e=\{u,v\}\in E_i^{\mathrm{touch}}} w_e\, d_e\big(\pi_{u,e}(s_u), \pi_{v,e}(s_v)\big).$$

On the declared fixed-cutoff branch, a recovery-derived primitive candidate is eligible for transactional commitment only if it strictly lowers this touched-overlap score:

$$s \rightsquigarrow_i t \implies \Phi_i(t) < \Phi_i(s).$$

This is the patch-net form of the regulator-side monotone local-fit contract carried by the declared repair package.

Definition 3.3 (Overlap-associative union-collar gluing)**.** *Fix $s \in \Sigma$ and two enabled primitive proposals $s \rightsquigarrow_i t$, $s \rightsquigarrow_j u$. Write*

$$E_{ij}^{\mathrm{touch}} := E_i^{\mathrm{touch}} \cup E_j^{\mathrm{touch}}.$$

*The declared repair branch is **overlap-associative** if the following hold.*

(i) If $E_i^{\mathrm{touch}} \cap E_j^{\mathrm{touch}} = \varnothing$, the two local proposals have disjoint support on the declared branch and therefore commute if they are committed as separate transactions.

(ii) If $E_i^{\mathrm{touch}} \cap E_j^{\mathrm{touch}} \neq \varnothing$, there is a finite union collar U_{ij} covering the interfaces in E_{ij}^{touch} such that the physical glued state on U_{ij} is parenthesization-independent on the quotient, in the sense of Proposition B.7, and the local decoders/lifts of Definition 3.1 are restriction-compatible on nested collars.

This is the concrete compatibility package used below to build canonical aggregate payloads for conflicting repair components.

Definition 3.4 (Transactional acceptance layer)**.** *Let X denote the finite physical presentation on which a repair step is being checked: before quotienting one may take $X = \Sigma$, and on the physical branch X is the quotient by hidden representatives. Registers are the finite patch, interface, sector, and record coordinates. Fix a boundary/sector/holonomy record map*

$$B : X \to \mathcal{B}$$

and a well-founded exact measure

$$\mu : X \to (W, \prec),$$

for example a lexicographic integer vector $(N_{\mathrm{hard}}, \Phi, N_{\mathrm{unresolved}})$.

A prepared transaction is a tuple

$$\tau = (R_\tau, W_\tau, \sigma_\tau, p_\tau)$$

with read set, write set, read snapshot, and payload. It commits at $x \in X$ only if all of the following hold:

(i) the snapshot is current: $x|_{R_\tau} = \sigma_\tau$;

(ii) the payload changes no register outside W_τ;

(iii) boundary, sector, holonomy, and protected record data are preserved: $B(\mathrm{Apply}_\tau(x)) = B(x)$;

(iv) exact descent holds:

$$\mu(\mathrm{Apply}_\tau(x)) \prec \mu(x).$$

A stale, aborted, ambiguous, or obstructed transaction is not a rewrite step. The read set is ***semantic-dependency-complete*** *as follows. Let $\mathcal{F}$ contain every finite-support functional whose value enters acceptance: the supported terms of μ, the protected boundary/sector/holonomy functions, enablement predicates, semantic-history and event-parent functions, observer-registry updates, and checkpoint-continuation functions. For a write set W, define*

$$D_{\mathcal{F}}(W) := \bigcup_{\substack{f \in \mathcal{F} \\ \mathrm{supp}(f) \cap W \neq \varnothing}} \mathrm{supp}(f).$$

Every prepared transaction satisfies $R_\tau \supseteq D_{\mathcal{F}}(W_\tau)$; its enablement and payload are functions of the read snapshot, and every acceptance functional affected by its write is revalidated at commit. For seam potentials this reads both endpoints of every seam whose score can change. Protected-support completeness and protected-conflict completeness are the restrictions of this condition to protected functionals; they are not substitutes for the full semantic closure.

At a state x, form the conflict graph of enabled primitive repair proposals, with

$$\tau \# \sigma \quad \Longleftrightarrow \quad W_\tau \cap (R_\sigma \cup W_\sigma) \neq \varnothing \text{ or } W_\sigma \cap (R_\tau \cup W_\tau) \neq \varnothing.$$

Each connected conflict component K is replaced by exactly one canonical aggregate transaction τ_K, computed on the union-collar or conflict-component support, with final declared supports R_K, W_K satisfying $R_K \supseteq D_{\mathcal{F}}(W_K)$. If aggregation or collar completion expands either support, conflicts are recomputed on the final aggregate supports and components are merged again until a fixed point is reached; distinct fixed-point aggregates are pairwise nonconflicting. These fixed-point components form a prepared source batch. After one component commits, another source-batch aggregate whose snapshot remains current and whose acceptance functionals revalidate remains admissible before later-enabled proposals are batched. Equivalently, an implementation may recompute immediately only if it certifies this surviving-component property. Primitive members of K never commit separately. Write $x \to y$ for a successful aggregate transaction commit, and let $\to^*$ be its reflexive-transitive closure. A state is a ***normal form*** *when no aggregate transaction is enabled.*

3.1 The quotient repair operator

The transactional layer above defines the physically accepted one-step relation. The object used as law is the induced finite normal-form map on the physical quotient, not a hidden-representative rewrite.

Definition 3.5 (Finite quotient repair presentation)**.** *A* ***finite quotient repair presentation*** *is a tuple*

$$\mathcal{P} = (\Sigma, \Gamma, q, Q, C_Q, B, \mu, \mathsf{A}, \prec_{\mathsf{A}})$$

where

$$Q = \Sigma/\Gamma, \qquad q : \Sigma \to Q$$

is the physical quotient by hidden representative data, $C_Q \subseteq Q$ is the quotient-level consistency set,

$$B : Q \to \mathcal{B}$$

is the protected boundary, sector, root-packet, charge, or holonomy-sector map, and

$$\mu : Q \to (W, \prec)$$

is a well-founded exact descent measure whose image on Q is finite. The finite set A consists of accepted aggregate repair transactions, equipped with a fixed total order $\prec_{\mathsf{A}}$. Each $a \in \mathsf{A}$ has a domain $D_a \subseteq Q$ and a map

$$a : D_a \to Q.$$

The accepted one-step relation is

$$x \to_{\mathcal{P}} y \quad \Longleftrightarrow \quad \exists a \in \mathsf{A},\ x \in D_a,\ y = a(x).$$

*The presentation is **OPH-admissible** when:*

$$(H_B) \qquad x \to_{\mathcal{P}} y \implies B(y) = B(x),$$

$$(H_{\downarrow}) \qquad x \to_{\mathcal{P}} y \implies \mu(y) \prec \mu(x),$$

$$(H_{\diamond}) \qquad \to_{\mathcal{P}} \text{ is locally confluent on } Q,$$

and

$$(H_{\mathrm{comp}}) \qquad x \in C_Q \quad \Longleftrightarrow \quad \text{no accepted aggregate transaction is enabled at } x.$$

Definition 3.6 (Local quotient repair operator)**.** *For $x \in Q$, let*

$$\mathsf{A}(x) := \{\, a \in \mathsf{A} : x \in D_a \,\}$$

be the enabled aggregate transaction set. Define

$$\mathrm{locRep}_{\lambda}(x) := \begin{cases} a_{\min}(x), & \mathsf{A}(x) \neq \varnothing, \\ x, & \mathsf{A}(x) = \varnothing, \end{cases}$$

where $a_{\min}$ is the $\prec_{\mathsf{A}}$-least enabled aggregate transaction.

Proposition 3.7 (Local quotient repair is total, protected, and descending)**.** *For every OPH-admissible finite quotient repair presentation,*

$$\mathrm{locRep}_{\lambda} : Q \to Q$$

is a total map. For every $x \in Q$,

$$B(\mathrm{locRep}_{\lambda}(x)) = B(x),$$

and either $\mathrm{locRep}_{\lambda}(x) = x$ or

$$\mu(\mathrm{locRep}_{\lambda}(x)) \prec \mu(x).$$

Finally,

$$\mathrm{locRep}_{\lambda}(x) = x \quad \Longleftrightarrow \quad x \in C_Q.$$

Proof. The set A is finite and totally ordered, so a least enabled transaction exists whenever $\mathsf{A}(x) \neq \varnothing$. If no transaction is enabled, the definition returns x. Hence locRep_λ is total.

If no transaction is enabled, boundary preservation is immediate. If $a_{\min}$ is enabled, (H_B) gives $B(a_{\min}(x)) = B(x)$. The descent statement is immediate from $(H_\downarrow)$. Since strict descent excludes $a_{\min}(x) = x$, the local operator fixes exactly those states with no enabled aggregate transaction. By (H_{comp}), these are exactly the elements of C_Q. □

Definition 3.8 (Global quotient repair operator)**.** *For $x \in Q$, define the canonical local-repair iterates by*

$$x_0 = x, \qquad x_{n+1} = \mathrm{locRep}_\lambda(x_n).$$

By Proposition 3.7, the sequence either stops or strictly descends inside the finite value set $\mu(Q)$. Hence there is a least $N(x)$ such that

$$x_{N(x)+1} = x_{N(x)}.$$

*The **global quotient repair operator** is*

$$\mathrm{Rep}_\lambda(x) := x_{N(x)}.$$

Theorem 3.9 (Global Repair is the finite quotient normal-form map)**.** *For every OPH-admissible finite quotient repair presentation,*

$$\mathrm{Rep}_\lambda : Q \to Q$$

is total and satisfies

$$\mathrm{Rep}_\lambda(x) \in C_Q, \qquad B(\mathrm{Rep}_\lambda(x)) = B(x),$$
$$\mathrm{Rep}_\lambda(\mathrm{Rep}_\lambda(x)) = \mathrm{Rep}_\lambda(x),$$

and

$$\mathrm{Rep}_\lambda(x) = x \quad \Longleftrightarrow \quad x \in C_Q.$$

If $x \to_{\mathcal{P}}^{*} y$ and y is terminal, then

$$y = \mathrm{Rep}_\lambda(x).$$

Thus

$$\mathrm{Rep}_\lambda = \overline{\mathrm{nf}}_\lambda : Q \to Q$$

is independent of the accepted asynchronous repair schedule.

Proof. Totality follows from finite descent. If $x_{n+1} \neq x_n$, then Proposition 3.7 gives

$$\mu(x_{n+1}) \prec \mu(x_n).$$

Since $\mu(Q)$ is finite and well founded, no infinite strictly descending sequence exists. Thus the canonical iteration reaches a fixed point x_N. By Proposition 3.7,

$$x_N = \mathrm{locRep}_\lambda(x_N) \quad \Longleftrightarrow \quad x_N \in C_Q,$$

so $\mathrm{Rep}_\lambda(x) \in C_Q$. Boundary preservation follows by induction from (H_B), hence $B(\mathrm{Rep}_\lambda(x)) = B(x)$. Idempotence follows because every point of C_Q is fixed by Proposition 3.7.

For schedule independence, $(H_\downarrow)$ gives termination of $\to_{\mathcal{P}}$, and $(H_\diamond)$ gives local confluence. Newman's lemma gives confluence. A terminating confluent rewrite system has a unique terminal normal form reachable from each initial state. The canonical iteration defining Rep_λ is one accepted repair execution, so it reaches that unique terminal normal form. Therefore every other maximal accepted repair execution from x reaches the same value. □

Proposition 3.10 (Validation support for local mismatch measures)**.** *Suppose the exact repair measure has finite local supports,*

$$\mu(x) = \sum_{a \in A} \mu_a(x|_{S_a}).$$

If a transaction writes W, then only terms with $S_a \cap W \neq \varnothing$ can change. Consequently, the descent validation is snapshot-local once the read set contains

$$R^{\mu}(W) := \bigcup_{a : S_a \cap W \neq \varnothing} S_a.$$

For the OPH overlap potential

$$\Phi(x) = \sum_{e=\{i,j\}} w_e \, d_e(\pi_{i,e}(x_i), \pi_{j,e}(x_j)),$$

this means reading both endpoints of every overlap whose score may change when a written register changes.

Proof. If $S_a \cap W = \varnothing$, the transaction leaves every register read by μ_a unchanged, so that term cancels between the pre- and post-state. Every remaining term is determined by the payload together with the restriction to $R^{\mu}(W)$. The displayed edge-potential formula has one two-endpoint support for each overlap term, giving the stated seam rule. □

Remark 3.11 (Inputs and branch conditions). The repair step is therefore not an abstract rewrite primitive. Its declared inputs are the fixed-cutoff collar chart, either exact Markov splice or a chosen Petz/Fawzi–Renner recovery channel, a local decoder/lift back to the finite patch presentation, and the touched-overlap local-fit contract of Definition 3.2, together with the support-local disjoint-commutation clause and the restriction-compatible union-collar package of Definition 3.3. The parenthesization-independent quotient-local glue used there is supplied by Proposition B.7 from the fixed-cutoff center-sector / higher-gauge gluing package. The actual accepted step is the validated aggregate transaction of Definition 3.4. The theorem package takes repair completeness as an explicit branch condition. On the Petz branch, full CPTP action on all inputs also requires the support clause recorded in Proposition C.4. On broader branches one must also prove that the declared union-collar compatibility is preserved under refinement or branch change.

The theorem package separates the imported repair-law data from the theorem-local inputs cleanly:

Assumption 3.12 (Repair completeness)**.** *For the accepted transactional relation $\to$, $s \in C$ if and only if no aggregate repair transaction is enabled at s.*

Normal forms are exactly the globally consistent states. The dynamics is neither too weak (missing some inconsistencies) nor too strong (repairing things that were fine).

Definition 3.13 (Verified rooted-tree packet-net domain)**.** *Fix a finite rooted tree $T = (V, E, r)$. For each non-root vertex i, write $p(i)$ for its parent and write $w_i > 0$ for the weight of the edge $\{p(i), i\}$. Let A be a finite packet alphabet with $|A| \geq 2$, and let K_i be a finite hidden-label set. The patch state space is*

$$S_i = A \times K_i, \qquad s_i = (x_i, k_i).$$

For every edge $e = \{i, j\}$, the interface alphabet is $I_e = A$, and both endpoint projections read the packet component:

$$\pi_{i,e}(x_i, k_i) = x_i, \qquad \pi_{j,e}(x_j, k_j) = x_j.$$

Thus e is consistent exactly when $x_i = x_j$. Choose the weights so that

$$w_i > \sum_{j:p(j)=i} w_j \qquad \text{for every non-root } i,$$

with an empty sum equal to 0. Define

$$\Phi(s) = \sum_{\{p(i),i\}\in E} w_i \, \mathbf{1}[x_i \neq x_{p(i)}].$$

The repair map at a non-root vertex i is

$$T_i(s)_i = (x_{p(i)}, k_i),$$

with all other vertices unchanged; if $x_i = x_{p(i)}$, the map is a no-op. The root repair map is the identity. The hidden labels k_i are acted on by arbitrary finite gauge relabelings and are not read by any interface projection.

Theorem 3.14 (Rooted-tree packet repair completeness and quotient closure)**.** *On the domain of Definition 3.13, Assumption 3.12 is a theorem. Every enabled repair strictly decreases Φ. Every maximal asynchronous repair run terminates at the unique state*

$$x_i = x_r \quad \text{for all } i \in V,$$

with all hidden labels k_i unchanged. The normal-form map descends to the quotient by hidden gauge relabeling. If several tree repairs lie in one conflict component, the canonical aggregate transaction applies them in increasing tree depth and is the same as the corresponding serial tree repair path. The four-vertex instance with edges $r-a$, $a-b$, $a-c$, alphabet $A = \mathbb{Z}_3$, hidden labels $K_i = \mathbb{Z}_2$, and weights $5, 1, 1$ is exported as a verified domain record with the cited consensus code.

Proof. First, repair completeness is immediate from the rooted tree. If $s \in C$, every edge has $x_i = x_{p(i)}$, so every non-root repair is a no-op. Conversely, if every repair is a no-op, then $x_i = x_{p(i)}$ for every non-root vertex, hence every edge is consistent and $s \in C$.

Let $i \neq r$ be enabled. Only the parent edge $\{p(i), i\}$ and child edges $\{i, j\}$ with $p(j) = i$ can change their contribution to Φ. The parent edge changes from inconsistent to consistent, contributing $-w_i$. Each child edge can increase by at most its weight. Therefore

$$\Phi(T_i(s)) - \Phi(s) \leq -w_i + \sum_{j:p(j)=i} w_j < 0.$$

So every enabled repair is accepted by the Lyapunov contract.

Since Φ takes finitely many values, every maximal repair run terminates. At a terminal state no non-root vertex differs from its parent, so the terminal packet label is x_r on every vertex. The root label and every hidden label are invariant under all repairs, so the terminal state is unique and independent of update order. Because interface projections, enabledness, Φ, and the repair maps depend only on x_i, arbitrary relabelings of the hidden K_i commute with quotienting:

$$q \circ T_i = \overline{T}_i \circ q.$$

Thus the normal-form map descends to the hidden-label quotient. □

Corollary 3.15 (Physical-law map on the verified packet domain)**.** *On the rooted-tree packet domain, every gauge-invariant observable*

$$M : \prod_i (A \times K_i) \to Y$$

has a schedule-independent repaired value

$$M(\mathrm{nf}(s)),$$

and this value depends only on the quotient class of s. Hence the normal-form map is usable as physical law on this verified packet branch without adding a representative-level gauge-covariance assumption.

Proof. Theorem 3.14 gives a unique terminal state for every asynchronous repair schedule and shows that the normal-form map descends to the quotient by hidden-label relabeling. A gauge-invariant observable factors through that quotient, so its value on the terminal state is independent of both the repair schedule and the hidden representative. □

Proposition 3.16 (Classical full-support Petz packet domain)**.** *Let B and D be finite packet alphabets, let the collar algebras be diagonal, and let $\mathcal{N} : \mathbb{C}^{B\times D} \to \mathbb{C}^B$ be the marginal channel $(\mathcal{N}p)(b) = \sum_d p(b,d)$. Fix a reference state σ_{BD} with $\sigma_B(b) > 0$ for every b. Let*

$$\gamma_\sigma := \min_{b\in B} \sigma_B(b) > 0.$$

Define the Petz recovery channel

$$\mathcal{R}_{\sigma,\mathcal{N}}(\mu)(b,d) = \mu(b)\,\sigma(d \mid b), \qquad \sigma(d \mid b) := \frac{\sigma_{BD}(b,d)}{\sigma_B(b)}.$$

Then $\mathcal{R}_{\sigma,\mathcal{N}}$ is stochastic, completely positive and trace preserving on the diagonal algebra, and ℓ^1-contractive:

$$\|\mathcal{R}_{\sigma,\mathcal{N}}(\mu) - \mathcal{R}_{\sigma,\mathcal{N}}(\nu)\|_1 \le \|\mu - \nu\|_1.$$

The support inverse is uniformly bounded on this domain by

$$\|\sigma_B^{-1/2}\| \le \gamma_\sigma^{-1/2}.$$

If a collar state has the exact classical Markov form

$$\omega_{ABD}(a,b,d) = \omega_{AB}(a,b)\sigma(d \mid b),$$

then $(\mathrm{id}_A \otimes \mathcal{R}_{\sigma,\mathcal{N}})(\omega_{AB}) = \omega_{ABD}$. The support obstruction is exact: if $\sigma_B(b) = 0$ and an input assigns mass to b, the Petz inverse on that sector is undefined unless the channel domain is restricted or a separate trace-preserving completion is declared.

Proof. The displayed formula is the finite diagonal specialization of the Petz map

$$\sigma_{BD}^{1/2}\left(\sigma_B^{-1/2}\mu\,\sigma_B^{-1/2} \otimes \mathbf{1}_D\right)\sigma_{BD}^{1/2}.$$

Full support of σ_B makes the inverse well defined, with support gap $\gamma_\sigma > 0$, hence $\|\sigma_B^{-1/2}\| \le \gamma_\sigma^{-1/2}$. Nonnegativity is immediate, and

$$\sum_{b,d} \mathcal{R}_{\sigma,\mathcal{N}}(\mu)(b,d) = \sum_b \mu(b) \sum_d \sigma(d \mid b) = \sum_b \mu(b),$$

so the map is trace preserving. Diagonal positive trace-preserving maps are completely positive on the diagonal algebra. For signed diagonal inputs,

$$\|\mathcal{R}_{\sigma,\mathcal{N}}(\mu) - \mathcal{R}_{\sigma,\mathcal{N}}(\nu)\|_1 = \sum_{b,d} |\mu(b) - \nu(b)|\, \sigma(d \mid b) = \sum_{b} |\mu(b) - \nu(b)|,$$

which gives the stated contraction. The exact Markov recovery identity follows by substitution. If $\sigma_B(b) = 0$, the factor $\sigma_B^{-1/2}$ has no value on that sector; mass placed there by an input is outside the Petz support domain. That is the claimed obstruction. □

Proposition 3.17 (Accepted repair moves are decreasing)**.** *For the accepted transactional repair law of Definitions 3.1–3.4, every enabled repair strictly decreases the declared exact measure:*

$$s \to t \implies \mu(t) \prec \mu(s).$$

On the scalar Φ-branch, this specializes to $\Phi(t) < \Phi(s)$.

Proof. This is part of the commit validation in Definition 3.4. For a single-site scalar-Φ transaction it reduces to the touched-overlap contract of Definition 3.2; for an aggregate conflict component it is checked on the aggregate payload before the component can commit. □

Proposition 3.18 (Termination from the OPH Lyapunov functional)**.** *Under Proposition 3.17, every repair sequence is finite; equivalently, the repair relation $\to$ is terminating.*

Proof. Along any nontrivial repair step $s \to t$, Proposition 3.17 gives $\mu(t) \prec \mu(s)$. The measure order is well founded, so an infinite strictly descending chain is impossible. On the scalar-Φ branch this is the finite-value argument on $\Phi(\Sigma)$. □

Proposition 3.19 (Finite repair step bound)**.** *Let $s_0 \in \Sigma$. Under Proposition 3.17, every accepted repair run starting at s_0 has length at most the number of distinct μ-values reachable below $\mu(s_0)$ minus one. On the scalar-Φ branch this gives*

$$\left|\{\, \Phi(s) : s \in \Sigma,\ \Phi(s) \le \Phi(s_0) \,\}\right| - 1 \le |\Phi(\Sigma)| - 1.$$

If, additionally, Φ is integer-valued, or more generally every accepted move lowers Φ by at least a fixed $\eta > 0$, then the run length satisfies

$$T(s_0) \le \left\lceil \frac{\Phi(s_0)}{\eta} \right\rceil.$$

This is a finite descent bound in repair steps only. It is not a wall-clock bound, not a probability-one scheduling statement, and not spectral or exponential convergence.

Proof. The values of μ strictly decrease along the run, so no μ-value can occur twice. This gives the finite reachable-value bound. On the scalar-Φ branch, the values of Φ strictly decrease along the run, so no value in $\{\, \Phi(s) : s \in \Sigma,\ \Phi(s) \le \Phi(s_0) \,\}$ can occur twice. This gives the finite value-set bound. If each move lowers Φ by at least η, after T moves the value has dropped by at least $T\eta$. Since $\Phi \ge 0$, $T\eta \le \Phi(s_0)$, giving the displayed ceiling bound. □

Finite settlement of the integer pair-repair law. The integer pair-repair implementation is an empirical check on finite observer patches joined at ports. At each of levels eight through ten of the declared icosahedral cell tower, sixteen schedules from one initial load reach the balanced component multiset. The largest rung has 20,971,520 twelve-port carriers and settles in 84 to 100 sweeps. This is a sampled statement about the specified law and gluing; the fitted logarithmic sweep count across tested rungs does not follow from the finite strict-descent bound. Potential-preserving swaps are excluded from the strict-descent counter and continue within the balanced class. The balanced multiset is fixed by the conserved component totals, so the check tests termination at the balanced class; it is no evidence for the local diamond, which Proposition 3.23 supplies under its own premises. The quotient multiset fixes no spatial texture: at level ten, ancestral groups of 4,096 cells retain amplitude response 0.95232 of the common initial configuration with schedule-noise fraction 0.594%, from an unbiased split of 48 terminal states into schedule variance and cross-schedule common power. These equal-count groups carry no derived physical area, and repeated schedules are not independent initial worlds.

Theorem 3.20 (Adaptive repair attempts: stabilization, fairness, and work conservation)**.** *Let C be a finite OPH carrier, let x_0 be an initial record, and let σ choose one site from the current record at each attempt. Write $x_n = \mathrm{adaptiveRun}_C(n, \sigma, x_0)$. Then:*

(i) for every σ, the sequence x_n is eventually constant;

(ii) if σ is pathwise weakly fair, so that a site which remains enabled along a tail is eventually selected on that tail, the stable record is a normal form;

(iii) if σ is work conserving, so that it selects an effective repair whenever one exists, there is a normal and stable index $N \le \mathrm{mismatchCount}(x_0)$.

Proof. Every attempt either stutters or is an accepted repair step. The mismatch count never increases and strictly decreases at every nonstuttering attempt, so only finitely many genuine changes are possible. If a pathwise weak-fair run were eventually constant at a reducible record, an enabled site would remain enabled along that tail and would eventually be selected, contradicting constancy. Work conservation removes stuttering before normality, so the initial natural-number rank bounds the normal horizon. □

Theorem 3.21 (Cumulative attempt-capacity classification)**.** *Charge one unit for every canonical adaptive scheduler invocation, including an equality stutter. Then:*

(i) no function of the initial mismatch alone bounds the first normal attempt over all normalizing adaptive schedulers;

(ii) if, from every scheduler index and every reducible record, one genuine change occurs among the next $q+1$ attempts, there is a stable normal index

$$N \le (q+1)\,\mathrm{mismatchCount}(x_0);$$

(iii) the coefficient is sharp at $q = 0$: for every $m \ge 1$, a canonical independent-defect carrier has initial mismatch m and a work-conserving run whose first normal index is exactly m. The committed TwoCell source is a second sharp instance with a different patch cardinality.

Proof. For the first clause, delay the enabled TwoCell probe by selecting its edge-free site for any prescribed finite number of attempts, then select the probe. The run normalizes one attempt later

while the initial mismatch stays one. For the second clause, every block of $q+1$ attempts starting from a reducible record contains a genuine change, and each such change strictly lowers the natural mismatch rank. Strong induction gives the product bound. For the sharp family, each self-loop mismatch belongs to one site and one canonical local repair removes exactly that mismatch. A state-dependent scheduler always chooses a broken site, so its mismatch count is $m-n$ through attempt m. The threshold is therefore both necessary and sufficient. The TwoCell calculation repeats the threshold-one conclusion on a nonisomorphic source. □

The premise in (ii) is a bounded stutter allowance. It is stronger than eventual fairness and weaker than work conservation when $q > 0$. The cumulative unit is only a count of chosen-site attempts; it is not a physical clock, rate, energy, bandwidth allocation, hardware quota, or fee.

Remark 3.22 (Termination is not confluence). Proposition 3.18 proves only that accepted repair runs stop. It does not prove that two update orders stop at the same physical state. Schedule-independence enters only after the local-diamond property of Proposition 3.23 is combined with termination via Newman's lemma and with repair completeness in Assumption 3.12. If two accepted repair schedules from the same initial state reach different observer-facing quotient normal forms, with no declared holonomy or higher-gauge obstruction and no mere gauge-representative difference, then the proposed repair law is not OPH-admissible as a consensus mechanism.

Proposition 3.23 (Semantic-complete transactional local diamond)**.** *For the accepted repair law of Definitions 3.1–3.4, assume semantic-dependency-complete read sets and revalidation, payload determination from the read snapshot, and a coherent canonical aggregate union-collar payload from Proposition B.7. Assume additionally the aggregate-support fixed-point closure and surviving prepared-component rule of Definition 3.4. Then every one-step quotient peak*

$$t \longleftarrow s \longrightarrow u$$

admits a quotient state v with $t \to v \leftarrow u$. Hence the transactional repair relation is locally confluent. The finite implementation receipt records the functional supports, dependency closures, read/write sets, support-closed conflict components, aggregate payload hashes, finite state transitions, and recomputed protected-record and descent checks needed to check that its concrete reference engine realizes these premises.

Proof. A one-step peak cannot choose two different payloads from one conflict component because that component has one canonical aggregate transaction. Its two steps therefore come from distinct support-closed components, say τ and σ. By the fixed-point conflict definition,

$$W_\tau \cap (R_\sigma \cup W_\sigma) = \varnothing, \qquad W_\sigma \cap (R_\tau \cup W_\tau) = \varnothing.$$

The writes are disjoint and neither changes the other's read snapshot. If an acceptance functional can change under τ, semantic dependency closure puts its support in R_τ; hence σ leaves that functional's input unchanged. Thus τ's enablement, descent comparison, protected-data check, semantic parents, checkpoint continuation, and payload remain valid after σ, and conversely. The surviving prepared-component rule prevents a later-enabled proposal from absorbing either old component before its second commit. Both second commits are legal. Since their writes are disjoint and their payloads are determined by unchanged snapshots,

$$\mathrm{Apply}_\sigma \,\mathrm{Apply}_\tau(s) = \mathrm{Apply}_\tau \,\mathrm{Apply}_\sigma(s) = v.$$

This is the local diamond on the physical quotient. □

Finite premise receipt. The executable check enumerates all states and one-step peaks of a finite conflict-component reference engine and recomputes separate countermodels for semantic closure, atomic component commits, canonical aggregate coherence, aggregate-support reclosure, surviving prepared components, snapshot-determined payloads, strict descent, repair completeness, and protected-record preservation. It verifies a concrete realization of the proposition's premises; it does not infer those premises for an arbitrary engine.

Remark 3.24 (Atomic commits do not prove the local diamond). Let $Q = \{0,1\}^2$, $s = (0,0)$, and protect $B(x_1, x_2) = x_1 x_2$. The transactions

$$\tau_1 : (0, x_2) \mapsto (1, x_2), \qquad \tau_2 : (x_1, 0) \mapsto (x_1, 1)$$

have disjoint write sets and each preserves B at s. Both lower $\mu(x_1, x_2) = 2 - x_1 - x_2$. Their peak has endpoints $(1,0)$ and $(0,1)$, while either second write would reach $(1,1)$ and change B. Neither continuation is admissible. The missing dependency is the nonlinear protected observable shared by the two writes. Protected-support and conflict completeness expose that dependency; an implementation must check the resulting quotient peak.

Theorem 3.25 (Asynchronous confluence / fixed-point law)**.** *For the accepted repair law of Definitions 3.1–3.4, assume that the induced physical relation on $Q = \Sigma/\Gamma$ is the OPH-admissible finite quotient presentation of Definition 3.5; in particular use Proposition 3.23 and Assumption 3.12 on that quotient. Each fixed initial state $s \in \Sigma$ then has a unique* quotient *normal form*

$$\overline{\mathrm{nf}}_\lambda(q(s)) = \mathrm{Rep}_\lambda(q(s)) \in q(C),$$

and every maximal asynchronous repair execution from s terminates at a representative $t \in C$ with $q(t) = \overline{\mathrm{nf}}_\lambda(q(s))$. Microscopic terminal representatives need not be equal. When one such representative must be named below, write $\mathrm{nf}_\lambda(s)$ for a choice; only its quotient is canonical. The quotient endpoint is independent of update order because descent, local confluence on the physical quotient, and repair completeness all hold; termination alone is not enough.

Proof. Proposition 3.18 gives termination, while Proposition 3.23 gives local confluence on Q. Newman's lemma [19] therefore makes the quotient relation confluent. By Assumption 3.12, its normal forms are exactly $q(C)$. Theorem 3.9 identifies the unique quotient endpoint with $\mathrm{Rep}_\lambda(q(s))$. A lifted maximal execution terminates at some microscopic representative of that orbit; no representative equality follows from quotient confluence. □

Corollary 3.26 (Objective law is schedule-independent)**.** *Let $\overline{M} : Q \to Y$ be a quotient observable, equivalently let $M : \Sigma \to Y$ factor as $M = \overline{M} \circ q$. Under the hypotheses of Theorem 3.25, $\overline{M}(\overline{\mathrm{nf}}_\lambda(q(s)))$ is independent of the asynchronous update schedule. If physical law is identified with this quotient-observable map, then physical law is objective.*

Proof. All schedules from the same initial s terminate in the same quotient normal form, so a quotient observable has the same value on every terminal representative. □

Here objectivity is identified with schedule-independent convergence on the physical quotient; Theorem 5.16 lifts this to the physical observable algebra while explicitly allowing distinct microscopic terminal representatives.

Corollary 3.27 (Canonical endpoint for weak-fair adaptive repair)**.** *Assume the completeness and confluence hypotheses of Theorem 3.25. Every pathwise weak-fair canonical adaptive run from one initial record reaches a stable consistent record equal to the canonical repair endpoint. Its image is therefore the same public fixed-point object for every such scheduler.*

Proof. Theorem 3.20 gives a reachable stable normal form. Completeness makes that normal form consistent. Confluence identifies it with the canonical repair normal form from the same start, and the public projection preserves the equality. □

Corollary 3.28 (Capacity-bounded canonical endpoint)**.** *Assume the completeness and confluence hypotheses of Theorem 3.25. Let the scheduler have waste bound q, and let the cumulative attempt budget satisfy*

$$B \geq (q+1)\,\mathrm{mismatchCount}(x_0).$$

Then the run reaches a stable consistent record at some attempt $N \leq B$. That record equals the canonical repair endpoint and maps to the same existing public fixed-point object for every scheduler satisfying the same contract.

Proof. Theorem 3.21 gives the normal index within budget. Completeness makes the record consistent. Its accepted-step reachability and confluence identify it with canonical repair, and the public projection preserves the equality. □

Remark 3.29 (Quantifier on normal-form uniqueness)*.* Theorem 3.25 proves quotient-normal-form uniqueness from a fixed initial state; Theorem 5.2 below records representative-lift independence for a fixed physical quotient state. It does not say that all possible initial data settle to one universal quotient state. Different boundary conditions, conserved charges, root packets, holonomy sectors, or external records can legitimately determine different normal forms.

3.2 Repair as computation and its complexity

On a fixed finite patch net the accepted reconciliation dynamics is a finite-state asynchronous rewrite system. Every accepted run has at most the number of distinct reachable μ-values below its start value minus one (Proposition 3.19), so exact normal-form computation is decidable by direct iteration of accepted aggregate transactions. This bound counts accepted nontrivial repairs; scheduler attempts may stutter and are classified by Theorems 3.20 and 3.21. No uniform polynomial-time bound, complexity-class placement for growing families of nets or hardness lower bound is proved here.

Two finite computation results have distinct scopes. An input-specialized federation with one ordered sweep produces the value of any Boolean formula and a unique consensus output; since that federation is compiled from the input and the weak `RepairStep` relation permits a stuttering move that leaves a failing observer unfixed, this is straight-line satisfiability, and that relation admits a one-member fair-stuttering execution that never repairs.

The fixed computation federation is the stronger device. Its generated formula nodes are independent of the input, and it reads immutable input ports from the initial state. Its canonical accepted step performs the selected failing node's declared write. A superincreasing dependency rank strictly descends on every genuine step; arbitrary attempts stabilize, and pathwise weak fairness promotes the stable state to consensus. Same-input normal endpoints have one output. Tail-relative pathwise weak fairness is equivalent to eventual stable consensus on a fixed run; the reverse implication is vacuous after stable consensus and supplies no recurrence. Member and register-site recurrence coincide at the generated-node granularity because output registers are unique, and recurrence is strictly stronger than tail fairness. A state-blind round-robin function is recurrent, pathwise weak-fair and bounded-waste. Its first n attempts are exactly the emission-order compiler sweep, so it reaches stable consensus and the correct output within n attempts for n emitted nodes.

The compiler gives every generated register at most one downstream consumer. Linear defect weights therefore bound every canonical accepted path by $n(n+1)/2$ for n emitted nodes. A

recursively nested NAND family from that compiler has a consensus-reaching path with $8m + 1 = n^2 + 8n$, making the same-metric classification order-sharp $\Theta(n^2)$, not constant-sharp. Well-formedness alone admits fewer than 2^n accepted steps, and an explicit fanout chain attains $2^n - 1$, so the single-consumer structure is load-bearing for the quadratic bound. Arbitrarily long finite stuttering prefixes remain pathwise weak-fair, so weak fairness gives no uniform attempt horizon. A separately supplied fixed-node bounded-waste value q gives a stable horizon of $(q+1)n(n+1)/2$, and round robin supplies $q = n - 1$; the one-cycle theorem is the sharper linear bound for emission-order round robin. The bounded-waste counting proof is shared with the adaptive route through a neutral ranked-attempt theorem.

Separately, a width-five permutation word program is fixed independently of the input and evaluates the formula under the stated encoding. Its formal carrier is the ambient permutation group on five points; membership of every instruction in the alternating subgroup is not a typed theorem. These results give a fixed finite repair device with accepted nontrivial repairs and fair asynchronous convergence, and a mathematical round-robin scheduler with conditional attempt counts. They supply no physical scheduler source, wall-clock horizon, rate, energy, bandwidth, fee, hardware resource, noise stability, encoding-overhead bound or physical device realization, and no identity with the width-five engine [12].

3.3 Finite support propagation and algebraic no-signalling

The concrete single-site record repair has a finite support theorem that is independent of termination and confluence. For a patch i, let $\mathcal{N}[i]$ be i together with every endpoint of an edge incident to i, and define

$$\mathrm{Grow}(S) = S \cup \bigcup_{i \in S} \mathcal{N}[i], \qquad B_0(S) = S, \qquad B_{n+1}(S) = \mathrm{Grow}(B_n(S)).$$

Theorem 3.30 (Finite repair support bound)**.** *Let L_i be the concrete* `localRepair` *map: it reads only $\mathcal{N}[i]$ and writes only the record at i. For any schedule word $w = (i_1, \dots, i_n)$, region S, and records x, y,*

$$x|_{B_n(S)} = y|_{B_n(S)} \quad \Longrightarrow \quad L_w(x)|_S = L_w(y)|_S,$$

where $L_w = L_{i_n} \circ \cdots \circ L_{i_1}$. Consequently, changing one initial record at a site outside $B_n(S)$ cannot change the final readout on S after any fixed exogenous schedule word of length n.

Proof. One repair move preserves agreement on T whenever its two inputs agree on $\mathrm{Grow}(T)$: away from the firing site it writes nothing, and at the firing site its trigger, solvability predicate, and selected repair value depend only on the closed incident-edge neighborhood. Induction on the schedule word gives the displayed implication. A one-site update outside $B_n(S)$ leaves the two initial records equal throughout that set. The statement is machine checked for the committed repair map [12]. □

This fixed-word bound has an exact conditional adaptive extension. If the next-site scheduler is supplied together with a region R on which its choice depends, then agreement on $B_n(S \cup R)$ forces the two n-step runs to agree on $S \cup R$; equivalently, a change outside both $B_n(S)$ and $B_n(R)$ cannot alter the probe readout. Fixed words embed as record-blind schedulers, while a two-cell countermodel proves that the consultation-region term cannot be dropped. Naturality also follows under the laws of a separately declared refinement structure. These results do not produce the scheduler, consultation region, channel, or refinement maps from the source. Neither theorem supplies a converse, minimal cone, physical graph metric, propagation speed, or claim that influence reaches the cone boundary.

Proposition 3.31 (Bipartite marginal invariance)**.** *Let A, B be finite sets.*

(i) For an arbitrary real array $p : A \times B \to \mathbb{R}$ and real matrix $K : A \times A \to \mathbb{R}$ satisfying $\sum_{a'} K(a, a') = 1$, define

$$p'(a', b) = \sum_{a} p(a, b) K(a, a').$$

Then $\sum_{a'} p'(a', b) = \sum_{a} p(a, b)$ for every b.

(ii) For an arbitrary matrix M on $\mathbb{C}^A \otimes \mathbb{C}^B$ and a finite family K_r on $\mathbb{C}^A$ satisfying $\sum_r K_r^\dagger K_r = I_A$, define

$$\Phi(M) = \sum_{r} (K_r \otimes I_B) M (K_r^\dagger \otimes I_B).$$

Then $\operatorname{Tr}_A \Phi(M) = \operatorname{Tr}_A M$.

Proof. For the classical identity, interchange the two finite sums and use the row normalization of K. For the matrix identity, expand the partial trace in indices and contract the acted-factor indices with $\sum_r K_r^\dagger K_r = I_A$. The one-point choice $K = 2$ is a negative control: without row normalization the remote marginal is doubled. Both identities are machine checked [12]. □

These marginal identities are standard finite algebra. Their bipartite split and local lift are supplied inputs. Positivity and unit mass are additional premises when the classical arrays are read as probabilities. No physical spacelike separation, observer-region factorization, density-state realization, or laboratory operation follows. A local observable-net construction must attach disjoint observer regions to the declared factors.

4 Why Local Agreement Is Not Enough: Cycle Holonomy and Frustration

Pairwise neighbor agreement does not by itself imply global consistency.

Theorem 4.1 (Cycle-obstruction / holonomy criterion)**.** *Let A be an abelian group, and let $G = (V, E)$ be a connected graph with an arbitrary orientation on each edge. For each oriented edge $e : u \to v$, assign a label $b_e \in A$. Consider the affine consistency equations*

$$x_v - x_u = b_e \qquad \text{for every oriented edge } e : u \to v,$$

where $x_v \in A$ are unknown patch labels. A global solution $x : V \to A$ exists if and only if for every cycle $C \subseteq G$,

$$\sum_{e \in C} \varepsilon_C(e)\, b_e = 0,$$

where $\varepsilon_C(e) = +1$ if the cycle traverses e in the chosen orientation and -1 otherwise.

Proof. **Necessity.** Suppose x is a solution. Summing the edge equations around any cycle C,

$$\sum_{e : u \to v \in C} \varepsilon_C(e)\, (x_v - x_u) = \sum_{e \in C} \varepsilon_C(e)\, b_e.$$

The left side telescopes to 0 because every vertex appears once with + sign and once with − sign.

Sufficiency. Fix a root $r \in V$. For any vertex v, choose a path $P_{r \to v}$ and define

$$x_v := \sum_{e \in P_{r \to v}} \varepsilon_{P_{r \to v}}(e)\, b_e, \qquad x_r := 0.$$

If P and P' are two paths from r to v, traversing P followed by the reverse of P' yields a cycle. By the vanishing-holonomy assumption, the total signed sum is zero, so x_v is well-defined. For any edge $e : u \to v$, extending a path to u by that edge gives $x_v = x_u + b_e$. □

Corollary 4.2 (Parity triangle: pairwise consistency is not enough)**.** *Take $A = \mathbb{Z}_2$ on the triangle A-B-C-A with edge labels $b_{AB} = 0$, $b_{BC} = 0$, $b_{CA} = 1$. Each individual edge equation is satisfiable. But the global system is not: the cycle sum is $0 \oplus 0 \oplus 1 = 1 \neq 0$.*

This example shows that pairwise consistency does not imply a global solution. The obstruction is carried by the cycle.

Corollary 4.3 (Persistent frustration at fixed holonomy)**.** *On a finite connected graph with fixed edge labels b_e and strictly positive weights $w_e > 0$, define the mismatch cost*

$$\Phi_b(x) := \sum_{e:u\to v\in E} w_e \, \mathbf{1}[x_v - x_u \neq b_e] \, .$$

If the cycle-holonomy condition fails, then $\min_{x:V\to A} \Phi_b(x) > 0$. Every minimizer contains irreducible residual inconsistency.

Proof. The cost takes values in the finite set of subset sums of the positive weights, so its minimum is attained. If that minimum were zero, an assignment would satisfy every edge equation, contradicting Theorem 4.1. □

No sequence of local changes of x, holding b fixed, removes this residual; Section 4.1 records what it supplies to the matter layer.

Theorem 4.4 (Higher-gauge defect hierarchy)**.** *Let a finite overlap nerve carry crossed-module defect data (g_{ij}, h_{ijk}) for a compact crossed module*

$$H \xrightarrow{\partial} G.$$

Under local rechartings by

$$C^1(N, H) \rtimes C^0(N, G),$$

the nonabelian Čech class

$$q = [(g, h)] \in \check{H}^2(N, H \to G)$$

is invariant and classifies the full crossed-module orbit. Its neutral element is equivalent to full gauge trivialization of both levels. The higher associator alone is removable precisely when q lies in the image of the natural strict-locus map

$$\check{H}^1(N, G) \longrightarrow \check{H}^2(N, H \to G), \qquad [g] \longmapsto [(g, 1)].$$

After such a strictification, endpoint-only ordinary reconciliation additionally requires an allowed strict representative with trivial represented loop holonomy. The strict-locus map need not be injective, so q need not determine one ordinary H^1 class.

Proof. The allowed rechartings are exactly the crossed-module coboundaries, so they preserve the full orbit. The neutral orbit admits the completely trivial representative. More generally, the higher associator is removed when the orbit meets the strict locus $(g, 1)$; different points of that locus can be related by H-valued edge changes. Ordinary loop coherence is therefore the separate existential holonomy test stated above. □

4.1 Frustration as the seed of the matter hypothesis

The OPH matter hypothesis reads stable massive matter as localized topological defects of native overlap transport: a bounded core whose boundary transport carries a holonomy that admitted local repairs cannot erase, with rest energy derived from its excess maintenance in the same physical clock and energy dictionary. Theorem 4.1 and Corollary 4.3 are the finite mathematics behind the first clause of that statement, and this subsection records exactly what they supply.

What the theorem gives. On a fixed finite graph with positive mismatch weights and fixed edge offsets b, nonzero cycle holonomy forces a strictly positive minimum mismatch $\min_x \Phi_b(x) > 0$. Repairs of the vertex assignments cannot remove it. The violated edges of every assignment meet every cycle of nonzero holonomy, because a cycle whose edge equations all hold has telescoping holonomy zero. The obstruction persists under changes of b that keep its cycle class. On the noncentral branch the invariant is the crossed-module class of Theorem 4.4, and the refinement-limit holonomy class of Theorem 6.2 carries it across a natural refinement system.

What it does not give. The offsets b are fixed data, so the theorem supplies no update law for them. An admitted update that changes b outside its cycle class removes the obstruction, and gauge invariance alone does not prove that a physical update preserves the class. The theorem localizes nothing and moves nothing: it constrains the violated edges only to meet every frustrated cycle, and it selects no core, no reference-like exterior and no transport of a defect. The residual $\min_x \Phi_b(x)$ is a mismatch of the whole compatibility problem, not an excitation above that problem's own ground state. Mismatch weights, repair counts and host work are distinct from rest energy; identifying the residual with rest energy requires a derived maintenance-to-energy map in a calibrated clock, with one common signal speed and no conversion fitted per species.

The native carrier supplies no free holonomy. Native load contrasts are differences of port values. They are exact, so their holonomy vanishes on every cycle and their face curl is zero; the oriented-face curvature of the native carrier has kernel exactly the port gradients (Section 8). The full two-dimensional carrier complex has vanishing first homology, so every cycle of its one-skeleton bounds faces and a face-curl-free offset has zero holonomy on every cycle. The seam graph alone has a nonzero cycle space, and a frustrated offset there must carry nonzero face curl. A native defect therefore needs transition data that the native law itself supplies and protects. Such data are unconstructed. Declared $\mathbb{Z}_6$ links and punctures support a separate flux experiment, but they are additional state and update premises, and unrestricted local link changes can erase a declared pair [13].

The tested local candidate, a doubled port load in the native total-load-two C/E process, is transient. Under independent uniform choice of the 30 seams it survives n seam attempts with probability $(5/6)^n$, with mean lifetime six seam-attempt counts, and splits into two unit loads; both complete C/E instruments give the same survival law on its quantum collision subspace. No physical duration is assigned to a seam attempt, and quantum and history sectors beyond that subspace are untested [13]. The matter hypothesis therefore requires an actual native sector and boundary invariant preserved by the full admitted law, a localized core with a reference-like exterior, transport of the invariant, and a derived energy. An exhaustive invariant and localization no-go over every sector and admitted refinement of a declared defect class would stop that mechanism.

5 Gauge Symmetry as Implementation Hiding

Definition 5.1 (Gauge action)**.** *Let $\Gamma = \prod_{i \in V} \Gamma_i$ act on Σ componentwise. The action is a* ***gauge action*** *if for every $e = \{i, j\} \in E$,*

$$\pi_{i,e}(\gamma_i \cdot x) = \pi_{i,e}(x) \qquad \forall\, x \in S_i,\ \forall\, \gamma_i \in \Gamma_i.$$

Gauge changes alter hidden local representations and leave overlap data fixed.

Write

$$q : \Sigma \to \Sigma/\Gamma, \qquad q(s) = [s],$$

for the gauge-orbit map. A **physical repair law** is the family of quotient-local maps

$$\overline{T}_i^{\lambda} : \Sigma/\Gamma \to \Sigma/\Gamma$$

induced by the recovery-derived collar updates of Definition 3.1. A **representative repair family** is any choice of lifts

$$T_i^{\lambda} : \Sigma \to \Sigma$$

such that

$$q \circ T_i^{\lambda} = \overline{T}_i^{\lambda} \circ q.$$

This is the finite patch-net form of defining the repair step first on overlap-invariant physical data and only then lifting it to hidden representatives. Once repair is formulated on the quotient, no separate gauge-covariance premise is needed.

Theorem 5.2 (Gauge quotient theorem)**.** *Under the gauge action of Definition 5.1, any representative lift of a physical repair law as just defined, and the hypotheses of Theorem 3.25,*

$$q(\mathrm{nf}_{\lambda}(\gamma \cdot s)) = q(\mathrm{nf}_{\lambda}(s)) \qquad \forall\, \gamma \in \Gamma,\ \forall\, s \in \Sigma.$$

Equivalently, the canonical normal-form map is the quotient operator constructed above:

$$\overline{\mathrm{nf}}_{\lambda} : \Sigma/\Gamma \to \Sigma/\Gamma, \qquad [s] \mapsto \mathrm{Rep}_{\lambda}([s]) = q(\mathrm{nf}_{\lambda}(s)).$$

Proof. Let $s \to t$ be an accepted aggregate transaction, and let $\overline{\tau}$ be its quotient payload. If $s' = \gamma \cdot s$, the representative-lift hypothesis gives an accepted lift $s' \to t'$ of the same quotient transaction, and

$$q(t') = \overline{\tau}(q(s')) = \overline{\tau}(q(s)) = q(t).$$

Thus gauge-equivalent inputs induce the same repaired orbit, and by induction the orbit reached after any repair sequence depends only on the initial orbit. Theorem 3.25 says that every maximal repair sequence from s ends at some terminal representative whose orbit is $\mathrm{Rep}_{\lambda}(q(s))$. Hence $q(\mathrm{nf}_{\lambda}(s))$ depends only on $q(s) = [s]$. This makes

$$[s] \longmapsto \mathrm{Rep}_{\lambda}([s])$$

well-defined on Σ/Γ. □

Corollary 5.3 (Repair respects gauge)**.** *Let $Q = \Sigma/\Gamma$, let $q : \Sigma \to Q$ be the quotient map, and let*

$$\mathrm{Rep}^{\Sigma}_{\lambda} := \mathrm{Rep}_{\lambda} \circ q : \Sigma \to Q$$

be the quotient-valued physical repair of a representative. Then for every $\gamma \in \Gamma$ and $s \in \Sigma$,

$$\mathrm{Rep}^{\Sigma}_{\lambda}(\gamma \cdot s) = \mathrm{Rep}^{\Sigma}_{\lambda}(s).$$

Equivalently,

$$\mathrm{Rep}_{\lambda}(q(\gamma \cdot s)) = \mathrm{Rep}_{\lambda}(q(s)).$$

If $M : Q \to Y$ is any physical observable, then

$$M(\mathrm{Rep}^{\Sigma}_{\lambda}(\gamma \cdot s)) = M(\mathrm{Rep}^{\Sigma}_{\lambda}(s)).$$

Proof. Since q is the quotient map, $q(\gamma \cdot s) = q(s)$. Therefore

$$\mathrm{Rep}^{\Sigma}_{\lambda}(\gamma \cdot s) = \mathrm{Rep}_{\lambda}(q(\gamma \cdot s)) = \mathrm{Rep}_{\lambda}(q(s)) = \mathrm{Rep}^{\Sigma}_{\lambda}(s).$$

Applying M gives the observable statement. □

Theorem 5.4 (Boundary-conditioned quotient uniqueness)**.** *Let*

$$B : \Sigma/\Gamma \to \mathcal{B}$$

record fixed external boundary data, conserved charge, root packet, holonomy sector, or task input. Assume accepted quotient repairs preserve B:

$$[s] \to [t] \implies B([s]) = B([t]).$$

For $b \in \mathcal{B}$, define the consistent quotient fiber

$$C_b := \{\, x \in q(C) : B(x) = b \,\}.$$

If every C_b has at most one element, then all initial states with the same boundary value settle to the same observer-facing quotient normal form:

$$B([s]) = B([s']) \implies \overline{\mathrm{nf}}_{\lambda}([s]) = \overline{\mathrm{nf}}_{\lambda}([s']).$$

Proof. Theorem 5.2 gives unique quotient normal forms $\overline{\mathrm{nf}}_{\lambda}([s])$ and $\overline{\mathrm{nf}}_{\lambda}([s'])$ in $q(C)$. Boundary preservation along accepted repair sequences gives

$$B(\overline{\mathrm{nf}}_{\lambda}([s])) = B([s]), \qquad B(\overline{\mathrm{nf}}_{\lambda}([s'])) = B([s']).$$

If $B([s]) = B([s']) = b$, both quotient normal forms lie in C_b. By the unique consistent extension assumption, C_b has at most one element, so the quotient normal forms are equal. □

Remark 5.5 (Same-source and cross-source quantifiers). Theorem 3.25 compares repair schedules from one fixed initial source. The stronger comparison of endpoints reached from different sources with the same protected boundary is controlled by the consistent boundary fibers. Under boundary preservation and NF $= q(C)$, Ref. [10] proves that cross-source endpoint agreement, modulo any declared silent equivalence, is equivalent to injectivity of the induced boundary map on the consistent quotient. Weak normalization and all-schedule liveness are separate premises. The layered and functional carriers below establish the injectivity premise on their declared domains. On the verified rooted-tree packet-net domain of Definition 3.13, the declared boundary map is the root-packet readback, and its injectivity modulo hidden-label gauge on the consistent set is established class-wide, together with the resulting cross-source endpoint form for the tree repair. Same-source confluence alone does not establish the injectivity premise for an arbitrary physical boundary map, and outside the declared domains it is a named per-net hypothesis with explicit failure witnesses.

5.1 A multi-edge finite carrier for boundary reconstruction

Definition 5.6 (Layered functional boundary carrier)**.** *A **layered functional boundary carrier** is a finite directed layered graph*

$$G = (V, E), \qquad V = L_0 \sqcup L_1 \sqcup \cdots \sqcup L_D,$$

with $D \geq 2$. The boundary layer is L_0. For each $v \in L_d$, $d \geq 1$, choose a nonempty parent set

$$P(v) \subseteq L_0 \sqcup \cdots \sqcup L_{d-1}$$

and a finite alphabet A_v. Each interior vertex has a deterministic local rule

$$F_v : \prod_{u \in P(v)} A_u \to A_v.$$

The carrier is genuinely multi-edge when at least two dependency edges occur and at least one layer contains more than one dependency or more than one repaired vertex.

The quotient state space is

$$Q = \prod_{v \in V} A_v,$$

and the boundary map is

$$B : Q \to \prod_{v \in L_0} A_v, \qquad B(a) = a|_{L_0}.$$

Given boundary data $b \in \prod_{v \in L_0} A_v$, define its functional extension $E(b) \in Q$ recursively by

$$E(b)_v = b_v \quad (v \in L_0),$$

and, for $v \in L_d$, $d \geq 1$,

$$E(b)_v = F_v\big((E(b)_u)_{u \in P(v)}\big).$$

Optional cross-check edges may be included by choosing finite predicates

$$\chi_e(a_u, a_v) = 0.$$

A state $a \in Q$ is consistent when every interior functional equation

$$a_v = F_v\big((a_u)_{u \in P(v)}\big)$$

holds and all cross-check predicates pass. Let C_Q be the set of consistent states. A boundary b is admissible when $E(b) \in C_Q$; equivalently,

$$C_Q = \{\, E(b) : b \text{ admissible} \,\}.$$

For each layer $d = 1, \ldots, D$, define a layer repair map $R_d : Q \to Q$ by

$$(R_d(a))_v = \begin{cases} F_v((a_u)_{u \in P(v)}), & v \in L_d, \\ a_v, & v \notin L_d. \end{cases}$$

The full staged repair sweep is

$$R_{\text{sweep}} = R_D \circ R_{D-1} \circ \cdots \circ R_1.$$

Gauge representatives may be added by replacing A_v with $A_v \times H_v$, letting Γ_v act only on H_v, and defining all boundary and consistency maps through the A_v-coordinate. The quotient is then the Q above.

Theorem 5.7 (Layered carrier proves $H_B \wedge H_{\mathrm{fib}}$)**.** *Let G be a layered functional boundary carrier. For every initial quotient state $a \in Q$, set*

$$a^{(0)} = a, \qquad a^{(d)} = R_d(a^{(d-1)}), \quad d = 1, \dots, D.$$

Let $b = B(a)$. Then

$$B(a^{(d)}) = b \qquad \text{for all } d = 0, \dots, D.$$

If b is admissible, then

$$a^{(D)} = E(b) \in C_Q.$$

The consistent boundary fiber is a singleton:

$$C_Q \cap B^{-1}(b) = \{E(b)\}.$$

Thus preserved boundary data plus consistency reconstruct the observer-facing quotient bulk on this carrier.

Proof. Boundary preservation is immediate because each R_d writes only layer L_d, with $d \geq 1$. No R_d writes L_0, so $B(a^{(d)}) = B(a) = b$ for every stage.

We prove by induction on d that after stage d,

$$a_v^{(d)} = E(b)_v \qquad \text{for every } v \in L_0 \sqcup \cdots \sqcup L_d.$$

For $d = 0$, this is the definition of b. Assume the claim holds through layer $d-1$, and let $v \in L_d$. Every parent $u \in P(v)$ lies in an earlier layer, so $a_u^{(d-1)} = E(b)_u$. The layer repair therefore gives

$$a_v^{(d)} = F_v\big((a_u^{(d-1)})_{u \in P(v)}\big) = F_v\big((E(b)_u)_{u \in P(v)}\big) = E(b)_v.$$

Earlier layers are not modified by R_d, so the induction closes. At $d = D$, all layers agree with $E(b)$. If b is admissible, $E(b) \in C_Q$.

For boundary-fiber uniqueness, let $c \in C_Q \cap B^{-1}(b)$. Since c has boundary b, it agrees with $E(b)$ on L_0. Since $c \in C_Q$, every interior vertex satisfies

$$c_v = F_v((c_u)_{u \in P(v)}).$$

The same induction on layers gives $c_v = E(b)_v$ for every vertex. Hence $c = E(b)$. □

Corollary 5.8 (Boundary reconstruction by global Repair)**.** *Assume the layered carrier's layer updates are included among the accepted aggregate transactions of an OPH-admissible finite quotient repair presentation, and let $b = B(x)$ be admissible. Then*

$$\mathrm{Rep}_\lambda(x) = E(b).$$

Consequently, for every physical readout

$$\mathrm{Read} : Q \to \mathcal{Y}$$

that factors through the quotient normal form,

$$\mathrm{Read}(\mathrm{Rep}_\lambda(x)) = \mathrm{Read}(E(B(x))).$$

Proof. By Theorem 3.9,

$$\mathrm{Rep}_\lambda(x) \in C_Q \qquad \text{and} \qquad B(\mathrm{Rep}_\lambda(x)) = B(x) = b.$$

Therefore $\mathrm{Rep}_\lambda(x) \in C_Q \cap B^{-1}(b)$. Theorem 5.7 identifies this fiber with $\{E(b)\}$, so $\mathrm{Rep}_\lambda(x) = E(b)$. Applying Read gives the readout statement. □

Remark 5.9 (A sharp linear reconstruction fixture). The layered carrier is feed-forward. A linear binary update on a finite cylinder tests the same logic without that structure. A row is $x_t : \mathbb{Z}_n \to \mathbb{F}_2$ with $x_{t+1}(j) = x_t(j-1) + x_t(j+1) \pmod 2$; this is a finite test object, not a proposed physics law. On the slab $0, \dots, t$, the two-column tube $\{j_0, j_0 + g\} \times \{0, \dots, t\}$ determines the whole finite record exactly when

$$\gcd(g, n) = 1 \quad \text{and} \quad n \le 2(t+1).$$

Adjacent columns reach the capacity threshold, and a non-coprime stride loses information at every horizon. A local transactional decoder that writes one non-tube cell per transaction through a bounded edge-local window, under the declared rank schedule, terminates in one pass and preserves the tube reading; at the threshold any two records with the same tube reading settle to the same normal form, which is consistent exactly when the tube fiber is realizable. On the same cylinder no single-patch frustration-free local repair satisfies the strongest $H_1 \wedge H_2 \wedge H_3$ binder form, and the canonical single-patch repair stalls at $(n, t) = (3, 2)$ on a record whose tube fiber is unrealizable. Boundary reconstruction is therefore a theorem only after the preserved boundary, the realizable-fiber condition and the repair roster are specified.

Theorem 5.10 (Functional selected-fiber uniqueness)**.** *Let $Q_b := B^{-1}(b) \subseteq \Sigma/\Gamma$ be a same-boundary quotient fiber presented on a rooted finite overlap graph with spanning tree T, root r, and patch state sets X_v. Assume the boundary value fixes the root value*

$$u_r = \beta(b),$$

and every non-root vertex has a deterministic extension map

$$f_v : X_{p(v)} \times \mathcal{B} \to X_v.$$

Define u_b recursively by

$$u_v = f_v(u_{p(v)}, b).$$

Evaluate all non-tree overlap equations, sector constraints, and holonomy checks on this candidate. If they all pass, then $C_b = \{u_b\}$. If any check fails, then $C_b = \varnothing$.

Proof. Every consistent state in Q_b must have root value $\beta(b)$. Along each tree edge, the deterministic extension equation forces the child value to be $f_v(u_{p(v)}, b)$. Induction over tree depth therefore forces every consistent state in the fiber to equal the single recursively constructed candidate u_b. The remaining constraints are exactly the non-tree overlaps, sector equations, and holonomy checks. If they pass, u_b is a consistent state and no other state can be. If one fails, the only tree-compatible candidate is inconsistent, so no element of C_b exists. □

The layered functional carrier of Theorem 5.7 is the finite multi-edge, multi-step witness that the boundary-fiber hypothesis is non-vacuous beyond one-dimensional propagation intuition.

Proposition 5.11 (Tree repair realizes the selected extension)**.** *Under the hypotheses of Theorem 5.10, choose positive weights*

$$w_v > \sum_{c:p(c)=v} w_c$$

and set

$$\Phi_T(x) = \sum_{v \neq r} w_v \, \mathbf{1}[x_v \neq f_v(x_{p(v)}, b)].$$

The repair that sets a non-root vertex to $f_v(x_{p(v)}, b)$ strictly decreases Φ_T, and the aggregate transaction for a connected tree-repair conflict component computes the same result in increasing tree depth.

Proof. Repairing v removes the v-term of weight w_v. Only children of v can become unmatched, contributing at most $\sum_{c:p(c)=v} w_c$, which is strictly smaller than w_v. Thus the tree potential decreases. For a connected conflict component, applying parent repairs before child repairs is forced by the same extension equations, and any different parenthesization has the same quotient result because each child value is a deterministic function of the repaired parent and b. □

Corollary 5.12 (Nontrivial branch elimination)**.** *Assume transactional confluence, boundary preservation, repair completeness, and Theorem 5.10. If a selected boundary fiber has $|Q_b| \geq 2$ and $C_b = \{u_b\}$, then*

$$\overline{\mathrm{nf}}_\lambda(Q_b) = \{u_b\},$$

while $Q_b \setminus C_b \neq \varnothing$. Thus multiple candidate interiors exist, inconsistent candidates are eliminated by repair, and all surviving candidates share one quotient normal form. If $C_b = \varnothing$, the branch is obstructed. If a union solver returns two physically distinct consistent quotient endpoints for the same selected data, the result is an ambiguous union repair and requires an explicit continuation gate, not selection by hash.

Remark 5.13 (Receipts do not replace the theorem)*.* Implementation receipts for seam descent, atomic commit, the distributed local diamond, repair completeness, same-boundary multistart confluence and the quotient normal-form hash are evidence contracts for the finite theorem hypotheses above, never proof substitutes. A normal-form hash certifies that two emitted quotient normal forms agree after the declared quotienting; it may not choose between two physically distinct minimizing states. Declared booleans, labels, hashes or plots cannot override failed residuals, missing objects or undeclared quotient maps, and a receipt must expose the raw objects needed to recompute it.

Corollary 5.14 (Gauge-invariant law)**.** *If $M : \Sigma \to Y$ is gauge-invariant ($M(\gamma \cdot s) = M(s)$ for all γ), then $M(\mathrm{nf}_\lambda(s))$ depends only on the gauge orbit $[s]$, not the representative.*

Proof. Because M is gauge-invariant, it factors through the orbit map: $M = \overline{M} \circ q$ for some $\overline{M} : \Sigma/\Gamma \to Y$. Theorem 5.2 gives

$$q(\mathrm{nf}_\lambda(\gamma \cdot s)) = q(\mathrm{nf}_\lambda(s)),$$

hence

$$M(\mathrm{nf}_\lambda(\gamma \cdot s)) = \overline{M}(q(\mathrm{nf}_\lambda(\gamma \cdot s))) = \overline{M}(q(\mathrm{nf}_\lambda(s))) = M(\mathrm{nf}_\lambda(s)).$$

□

Definition 5.15 (Physical observable algebra on the quantum lift)**.** *Fix a finite patch region or union collar R on the declared fixed-cutoff quantum lift of Appendix B. Its* ***physical observable algebra*** *$\mathcal{A}_{\mathrm{phys}}(R)$ is the fixed-point collar algebra under the compact boundary redundancy action on the ordinary or central-defect branch, or the corresponding quotient-local algebra on the genuinely noncentral branch. The central sector projectors carried by the collar decomposition belong to $Z(\mathcal{A}_{\mathrm{phys}}(R))$. A* ***physical observable*** *is any $X \in \mathcal{A}_{\mathrm{phys}}(R)$. For a microscopic representative $s \in \Sigma$, let ω_R^s denote the induced state on $\mathcal{A}_{\mathrm{phys}}(R)$.*

Theorem 5.16 (Observable-level confluence on the quantum lift)**.** *Under the hypotheses of Theorems 3.25 and 5.2, every initial orbit $[s] \in \Sigma/\Gamma$ has a unique quotient normal form*

$$\overline{\mathrm{nf}}_\lambda([s]) \in q(C).$$

Fix a declared region R. Let $t, u \in \Sigma$ be terminal microscopic representatives reached by representative lifts of maximal repair sequences from initial states in the same orbit $[s]$. If the induced R-collar states of t and u are representatives of the same quotient-local glued state in the sense of Proposition B.7, then for every physical observable $X \in \mathcal{A}_{\mathrm{phys}}(R)$,

$$\omega_R^t(X) = \omega_R^u(X).$$

Hence all physical observables converge to the same overlap-consistent values even when the microscopic terminal representatives differ by gauge relabelings globally or by sector/higher-gauge relabelings inside one declared quotient-local glued state.

Proof. Theorems 3.25 and 5.2 give the unique quotient normal form $\overline{\mathrm{nf}}_\lambda([s]) = [\mathrm{nf}_\lambda(s)] \in q(C)$. Let t and u be as stated. By Corollary B.9, representatives of the same quotient-local glued state induce the same state on the physical observable algebra $\mathcal{A}_{\mathrm{phys}}(R)$, including the central sector projectors carried by that algebra. Therefore $\omega_R^t(X) = \omega_R^u(X)$ for every $X \in \mathcal{A}_{\mathrm{phys}}(R)$. □

Remark 5.17 (Inputs and boundary for observable-level confluence). Theorem 5.16 does not add a new repair hypothesis beyond the confluence and fixed-cutoff gluing package. Its inputs are exactly the quotient-level confluence theorem, quotient descent of the repair law, Definition 5.15, and Corollary B.9 from the fixed-cutoff union-collar gluing package. The boundary item is extension of the same statement to broader refinement-stable branches where the declared union-collar compatibility is only approximate or where the chosen physical observable algebra itself changes under refinement.

Corollary 5.18 (Inert ancillary refinement does not change physical law)**.** *Let $K = \prod_i K_i$ be a finite ancillary state space and define $\Sigma^\eta := \Sigma \times K$. Lift the repair maps by*

$$T_i^{\lambda,\eta}(s,k) := (T_i^\lambda(s), k).$$

If $M^\eta : \Sigma^\eta \to Y$ ignores the ancillary factor,

$$M^\eta(s,k) = M(s),$$

with M gauge-invariant on Σ, then

$$M^\eta(\mathrm{nf}_\lambda^\eta(s,k)) = M(\mathrm{nf}_\lambda(s))$$

for all $(s,k) \in \Sigma^\eta$.

Proof. Because the ancillary factor is inert,

$$\mathrm{nf}^{\eta}_{\lambda}(s,k) = (\mathrm{nf}_{\lambda}(s), k).$$

Therefore $M^{\eta}(\mathrm{nf}^{\eta}_{\lambda}(s,k)) = M(\mathrm{nf}_{\lambda}(s))$, and Corollary 5.14 supplies gauge-orbit independence. □

Physical uniqueness therefore holds on the quotient by gauge or implementation hiding. The same statement is unchanged under inert ancillary stabilization.

5.2 Implementation hiding and the gauge layer

Gauge here means implementation hiding: a change of hidden local representation that leaves every overlap readout fixed (Definition 5.1). This subsection supplies the gauge layer with three exact inputs. Physical law factors through the quotient Σ/Γ, and repair defined first on overlap data and then lifted needs no separate gauge-covariance premise (Theorem 5.2, Corollaries 5.3 and 5.14). Holonomy classes, abelian or crossed-module, are invariants of that quotient (Theorems 4.1 and 4.4); they are the part of the overlap transport that implementation hiding cannot remove. At fixed cutoff, overlap rechartings generate a compact boundary redundancy group whose invariant collar algebra has the central sector decomposition of Appendix B; these central sectors are the edge-sector seed that the gauge layer transports.

The gauge layer adds what the quotient does not determine [15]. On a branch carrying an explicit compact-gauge refinement receipt, coherent pullback functors reconstruct a compact group from the tensor-generated zero-obstruction sectors; zero obstruction there means central or higher-associator strictifiability together with an allowed strict representative of trivial represented holonomy. Separately, complete reversible response and endogenous overlap transport force the local Lie algebra $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ under that paper's response premises. The consensus theorems fix no gauge group, Lie algebra, connection dynamics, coupling or matter representation: they hold verbatim for every redundancy group that preserves overlap data. A transformation that changes overlap data changes the compatibility problem; it hides no implementation.

Gauge and matter meet at the holonomy. A defect in the sense of Section 4.1 is a transport class that implementation hiding cannot remove, and the matter hypothesis requires that class to be carried and protected by native transport.

5.3 Finite packet closure and the simulation criterion

Definition 5.19 (Finite packet closure simplex)**.** *Assume the finite fixed-cutoff consensus branch of Theorems 3.25 and 5.2, so the physical quotient $Q := \Sigma/\Gamma$ is finite and carries the schedule-independent quotient normal-form map*

$$\overline{\mathrm{nf}}_{\lambda} : Q \to Q.$$

Let

$$N_{\lambda} := \overline{\mathrm{nf}}_{\lambda}(Q) = q(C)$$

*be the quotient normal-form set. The **finite packet simplex** on Q is*

$$\Delta(Q) := \left\{ \mu : Q \to \mathbb{R}_{\geq 0} \;\middle|\; \sum_{x \in Q} \mu(x) = 1 \right\},$$

*and the **finite packet closure map** is the pushforward*

$$\mathcal{C}_{\lambda} : \Delta(Q) \to \Delta(Q), \qquad \mathcal{C}_{\lambda}(\mu) := (\overline{\mathrm{nf}}_{\lambda})_{*}\mu,$$

equivalently

$$\mathcal{C}_\lambda(\mu)(y) = \sum_{\substack{x \in Q \\ \overline{\mathrm{nf}}_\lambda(x) = y}} \mu(x).$$

Theorem 5.20 (Finite packet-quotient closure map)**.** *On the finite fixed-cutoff consensus branch of Definition 5.19, the map $\mathcal{C}_\lambda$ is an affine continuous idempotent self-map of $\Delta(Q)$. Its image is exactly the normal-form simplex*

$$\Delta(N_\lambda) = \{\mu \in \Delta(Q) \mid \mathrm{supp}(\mu) \subseteq N_\lambda\}.$$

Hence the fixed points of $\mathcal{C}_\lambda$ are exactly the packets supported on quotient normal forms. For every initial quotient state $[s] \in Q$,

$$\mathcal{C}_\lambda(\delta_{[s]}) = \delta_{\overline{\mathrm{nf}}_\lambda([s])}.$$

Proof. Because $\overline{\mathrm{nf}}_\lambda : Q \to Q$ is a set map on a finite set, its pushforward on probability packets is affine and continuous in the finite-dimensional simplex topology.

By Theorems 3.25 and 5.2, the quotient normal form is schedule-independent and terminal, so

$$\overline{\mathrm{nf}}_\lambda \circ \overline{\mathrm{nf}}_\lambda = \overline{\mathrm{nf}}_\lambda.$$

Pushforward therefore gives

$$\mathcal{C}_\lambda^2 = (\overline{\mathrm{nf}}_\lambda)_*(\overline{\mathrm{nf}}_\lambda)_* = (\overline{\mathrm{nf}}_\lambda \circ \overline{\mathrm{nf}}_\lambda)_* = \mathcal{C}_\lambda,$$

so $\mathcal{C}_\lambda$ is idempotent.

If $\nu = \mathcal{C}_\lambda(\mu)$, then every point in $\mathrm{supp}(\nu)$ lies in the image of $\overline{\mathrm{nf}}_\lambda$, namely N_λ. Thus $\mathrm{im}(\mathcal{C}_\lambda) \subseteq \Delta(N_\lambda)$. Conversely, if $\nu \in \Delta(N_\lambda)$, then $\overline{\mathrm{nf}}_\lambda(y) = y$ for every $y \in N_\lambda$, so

$$\mathcal{C}_\lambda(\nu) = \nu.$$

Hence $\Delta(N_\lambda) \subseteq \mathrm{im}(\mathcal{C}_\lambda)$, proving $\mathrm{im}(\mathcal{C}_\lambda) = \Delta(N_\lambda)$. The same identity shows that ν is a fixed point if and only if it is supported on N_λ. The Dirac-mass formula is the pushforward of a point mass under $\overline{\mathrm{nf}}_\lambda$. □

Theorem 5.20 concerns one fixed quotient carrier whose repair law is terminating, confluent and quotient-descended. It gives no closure map for arbitrary state-and-law data and no uniqueness or stability estimate beyond this finite branch.

Fixed points are not simulations. A fixed-point equation holds for a selected finite OPH packet, but it does not define one. The content lies in the further clauses below.

Definition 5.21 (OPH simulation certificate)**.** *On the branch of Definition 5.19 write $n := \overline{\mathrm{nf}}_\lambda$, let $B : Q \to \mathcal{B}$ be a boundary/sector map preserved by accepted quotient repairs, fix $b \in \mathcal{B}$, and set $Q_b := B^{-1}(b)$, $C_b := N_\lambda \cap Q_b$. A selected pair (b, u) with $u \in C_b$, equivalently the packet $\mathfrak{U}_u := \delta_u$, has an **OPH simulation certificate** when:*

(S1) ***Endogenous update.*** *The local maps are the recovery-derived collar maps of Definition 3.1, descend to the quotient and read only declared patch, overlap, recovery, decoder and acceptance data.*

(S2) ***Observer-readable records.*** *A declared observer-accessible region carries an exact record presentation (Definition 7.1) with at least two nonzero orthogonal event projectors, collected in a map* $\mathrm{Read} : Q \to \mathcal{Y}_{\mathrm{rec}}$ *whose terminal value depends only on the quotient normal form.*

(S3) ***Overlap repair.*** *Accepted moves strictly lower the declared touched-overlap score, normal forms are exactly the consistent states, and* n *is terminating and schedule independent.*

(S4) ***Nontrivial branch elimination.*** $C_b = \{u\}$, $Q_b \setminus C_b \neq \varnothing$ *and* $n(Q_b) = \{u\}$*; a candidate with nonzero holonomy or higher-gauge obstruction counts as obstructed, never as an alternative world.*

(S5) ***Implementation and clock closure.*** *For auxiliary implementation and clock sets with physical projection* $\widetilde{p : Q \times \mathcal{E}_{\mathrm{ext}} \times \Theta_{\mathrm{ext}} \to Q}$*, every admissible lifted update and readout satisfy* $p \circ \widetilde{n} = n \circ p$ *and* $\widetilde{\mathrm{Read}} = \mathrm{Read} \circ p$*. Schedules and repair counters are proof data, and any history is read from records in the terminal quotient state.*

Proposition 5.22 (Selected packet and fixed-point firewall)**.** *Assume Definitions 3.1, 3.2 and 3.3, Assumption 3.12, and the hypotheses of Theorems 5.2, 5.16 and 7.2. If a preserved value* b *has* $C_b = \{u\}$ *and* $Q_b \setminus C_b \neq \varnothing$*, and the selected observer surface has a nontrivial exact record presentation whose order data are quotient-observable, then* δ_u *has an OPH simulation certificate, and* $n(u) = u$*,* $\mathcal{C}_\lambda(\delta_u) = \delta_u$*. The converse fails: for any set* X *with* $|X| > 1$*, the identity map fixes every point and every point minimizes the constant functional, while no proper candidate basin collapses to any point, so clause (S4) fails.*

Proof. (S1) is Definition 3.1 with quotient descent from Theorem 5.2; (S2) follows from Theorems 7.2 and 5.16; (S3) from Definition 3.2 and Theorem 3.25. For $x \in Q_b$, boundary preservation and completeness give $n(x) \in C_b = \{u\}$, which is (S4); the rooted-tree packet domain of Theorem 3.14 shows the clause is nonvacuous. The update and readout are defined on the quotient alone, schedule independence removes the scheduler, and Corollary 5.18 removes inert carrier enlargement, giving (S5). Theorem 5.20 gives $\mathcal{C}_\lambda(\delta_u) = \delta_{n(u)} = \delta_u$. The identity-map example proves the non-implication. □

A certificate fails if an accepted update reads undeclared oracle, carrier or clock data; if the record algebra is trivial or not quotient-readable; if a move fails strict descent; if terminal states differ from consistent states; if two schedules or two candidates in the selected fiber give different physical normal forms; if the fiber has nothing to eliminate; or if a carrier or clock lift changes the projected update or readout.

6 Refinement-Limit Consensus Classes

The finite consensus theorem is the operational object used by the continuum branches. To pass from one finite patch net to a refining family, the paper uses an explicit inverse-limit package with named compatibility clauses.

Definition 6.1 (Separated cofinal refinement consensus system)**.** *Let* $(R, \preceq)$ *be a directed refinement set. For each* $r \in R$*, let*

$$Q_r := \Sigma_r / \Gamma_r$$

be the finite physical quotient state space of a patch presentation, let

$$n_r : Q_r \to Q_r$$

be the finite-stage quotient normal-form map supplied by Theorems 3.25 and 5.2, and let

$$h_r : Q_r \to \mathcal{H}_r$$

be the finite-stage holonomy or higher-gauge obstruction map supplied by Theorems 4.1 and 4.4. For $r \preceq s$, assume restriction maps

$$\rho_{sr} : Q_s \to Q_r, \qquad \chi_{sr} : \mathcal{H}_s \to \mathcal{H}_r,$$

with $\rho_{rr} = \mathrm{id}$, $\chi_{rr} = \mathrm{id}$, and the cocycle identities

$$\rho_{tr} = \rho_{sr} \circ \rho_{ts}, \qquad \chi_{tr} = \chi_{sr} \circ \chi_{ts} \qquad (r \preceq s \preceq t).$$

The system is a separated cofinal refinement consensus system when:

1. ***Normal-form naturality:***

$$\rho_{sr} \circ n_s = n_r \circ \rho_{sr} \qquad (r \preceq s).$$

2. ***Holonomy naturality:***

$$\chi_{sr} \circ h_s = h_r \circ \rho_{sr} \qquad (r \preceq s).$$

3. ***Visible separation:*** *two compatible families in $\varprojlim Q_r$, or in $\varprojlim \mathcal{H}_r$, are equal whenever their projections agree on a cofinal subset of stages.*

Theorem 6.2 (Refinement-limit consensus and holonomy classes)**.** *For any separated cofinal refinement consensus system, the formulas*

$$n_\infty((x_r)_r) := (n_r(x_r))_r, \qquad h_\infty((x_r)_r) := (h_r(x_r))_r$$

define maps

$$n_\infty : \varprojlim Q_r \to \varprojlim Q_r, \qquad h_\infty : \varprojlim Q_r \to \varprojlim \mathcal{H}_r.$$

The class $n_\infty(x)$ is the unique schedule-independent refinement-limit normal form of x. The class $h_\infty(x)$ is the refinement-limit holonomy obstruction. The pair $(n_\infty(x), h_\infty(x))$ is the refinement-limit consensus class. In the inverse-limit topology, finite-stage normal forms and holonomy classes converge to these two classes: for every stage r, all refinements $s \succeq r$ restrict to the fixed values

$$\rho_{sr}(n_s(x_s)) = n_r(x_r), \qquad \chi_{sr}(h_s(x_s)) = h_r(x_r).$$

The relation $h_\infty(x) = 0$ holds if and only if every finite projection has zero finite-stage obstruction. If $h_\infty(x) \neq 0$, visible separation gives a finite stage that witnesses the nonzero obstruction. If two refinement-limit candidates have the same normal-form and holonomy projections on a cofinal tail, then they determine the same refinement-limit consensus class.

Proof. Let $x = (x_r)_r \in \varprojlim Q_r$, so $\rho_{sr}(x_s) = x_r$ for $r \preceq s$. Normal-form naturality gives

$$\rho_{sr}(n_s(x_s)) = n_r(\rho_{sr}(x_s)) = n_r(x_r),$$

so $(n_r(x_r))_r$ is a compatible family and n_∞ is well defined. The same argument with holonomy naturality gives

$$\chi_{sr}(h_s(x_s)) = h_r(\rho_{sr}(x_s)) = h_r(x_r),$$

so h_∞ is well defined.

Each finite $n_r(x_r)$ is independent of update schedule by Theorems 3.25 and 5.2. Therefore every finite projection of $n_\infty(x)$ is schedule-independent, and visible separation makes the inverse-limit class unique. The displayed restriction identities are exactly the cylinder convergence statement in the inverse-limit topology.

The zero-obstruction statement follows from the definition of the inverse-limit zero class. The identity $h_\infty(x) = 0$ holds exactly when every projection $h_r(x_r)$ is zero. If $h_\infty(x) \neq 0$, at least one finite projection is nonzero, and any cofinal tail containing a refinement of that stage carries the same nonzero projected obstruction by holonomy naturality. The last claim is visible separation applied to the compatible normal-form and holonomy families. □

Remark 6.3 (Scope of the refinement theorem). Theorem 6.2 is a controlled inverse-limit bridge. It proves persistence, exhaustion, and collapse of the consensus data once the OPH refinement system supplies the restriction maps and the two naturality clauses in Definition 6.1. It leaves uniform complexity bounds, automatic fair-block noisy-consensus certificates, and automatic normal-form naturality for arbitrary changing repair laws as separate branch conditions. It does not identify this inverse limit with a physical continuum or spacetime.

Definition 6.4 (Repair morphism). *Let $(Q_s, \to_s, C_s, n_s)$ and $(Q_r, \to_r, C_r, n_r)$ be two finite quotient repair systems covered by Theorems 3.25 and 5.2. A map*

$$\rho_{sr} : Q_s \to Q_r$$

*is a **repair morphism** when:*

(i) every fine repair step $x \to_s y$ maps to a coarse repair path or a stutter,

$$\rho_{sr}(x) \to_r^* \rho_{sr}(y);$$

(ii) fine consistency maps into coarse consistency,

$$\rho_{sr}(C_s) \subseteq C_r.$$

For a concrete local repair law this is checked by giving, for every fine repair generator, a coarse witness word in the accepted coarse repair generators or the empty word, and by verifying that the restriction maps carry every fine consistency equation to a coarse consistency equation.

Theorem 6.5 (Normal-form naturality from repair morphisms). *If $\rho_{sr} : Q_s \to Q_r$ is a repair morphism, then*

$$\rho_{sr} \circ n_s = n_r \circ \rho_{sr}.$$

Proof. Fix $x \in Q_s$. Since $x \to_s^* n_s(x)$, repeated use of the step-simulation clause gives

$$\rho_{sr}(x) \to_r^* \rho_{sr}(n_s(x)).$$

The consistency-preservation clause gives $\rho_{sr}(n_s(x)) \in C_r$, because $n_s(x) \in C_s$. Thus $\rho_{sr}(n_s(x))$ is a coarse normal form reachable from $\rho_{sr}(x)$. The coarse system has a unique normal form reachable from each initial quotient state, so $\rho_{sr}(n_s(x)) = n_r(\rho_{sr}(x))$. □

Definition 6.6 (Holonomy cochain morphism). *For obstruction maps $h_s : Q_s \to \mathcal{H}_s$ and $h_r : Q_r \to \mathcal{H}_r$, a map $\chi_{sr} : \mathcal{H}_s \to \mathcal{H}_r$ is a **holonomy cochain morphism** when the edge, cycle, or crossed-module cochains that compute h_s restrict to the cochains that compute h_r, modulo the declared quotient identifications. Equivalently,*

$$\chi_{sr} \circ h_s = h_r \circ \rho_{sr}.$$

Remark 6.7 (How exact refinement naturality is discharged). Definition 6.1 may be used as an interface contract, but on an implemented exact branch the normal-form part is proved by Theorem 6.5 from explicit coarse witness words for fine repairs. The holonomy part is proved by the cochain morphism check of Definition 6.6. Approximate RG branches use the controlled defects of Definition 6.8 instead.

Definition 6.8 (Controlled coarse-graining / reconciliation square)**.** *Fix refinement stages $r \preceq s$. Let Q_r, Q_s be the physical quotient state spaces, n_r, n_s their finite-stage normal-form maps, and h_r, h_s their holonomy or higher-gauge obstruction maps. Let*

$$\rho_{sr} : Q_s \to Q_r, \qquad \chi_{sr} : \mathcal{H}_s \to \mathcal{H}_r$$

be the coarse-graining maps on quotient states and obstruction data. Equip Q_r and $\mathcal{H}_r$ with pseudometrics d_r^Q and $d_r^{\mathcal{H}}$ that define the macroscopic readout scale at stage r.

The square is $(\varepsilon_{sr}^n, \varepsilon_{sr}^h)$-controlled when, for every $x \in Q_s$,

$$d_r^Q(\rho_{sr}(n_s(x)),\, n_r(\rho_{sr}(x))) \le \varepsilon_{sr}^n,$$

and

$$d_r^{\mathcal{H}}(\chi_{sr}(h_s(x)),\, h_r(\rho_{sr}(x))) \le \varepsilon_{sr}^h.$$

*The errors are **cofinally vanishing** when, for every fixed macroscopic stage r and every $\delta > 0$, there is a refinement stage $s_0 \succeq r$ such that $\varepsilon_{sr}^n, \varepsilon_{sr}^h < \delta$ for all $s \succeq s_0$.*

Theorem 6.9 (Coarse-graining commutes with reconciliation up to controlled error)**.** *Suppose the refinement square of Definition 6.8 is $(\varepsilon_{sr}^n, \varepsilon_{sr}^h)$-controlled. Define the coarse-stage consensus readout*

$$\mathcal{C}_r(y) := (n_r(y), h_r(y))$$

and the fine-then-coarse readout

$$\mathcal{C}_{s\to r}^{\mathrm{fine}}(x) := (\rho_{sr}(n_s(x)), \chi_{sr}(h_s(x))).$$

Then, for every fine state $x \in Q_s$, the product readout distance between $\mathcal{C}_{s\to r}^{\mathrm{fine}}(x)$ and $\mathcal{C}_r(\rho_{sr}(x))$ is bounded by

$$\max\{\varepsilon_{sr}^n, \varepsilon_{sr}^h\}.$$

Thus reconciling at the fine stage and then coarse-graining gives the same macroscopic law data as coarse-graining first and reconciling at the coarse stage, up to the declared control errors. If the exact naturality clauses of Definition 6.1 hold, then $\varepsilon_{sr}^n = \varepsilon_{sr}^h = 0$. If the errors are cofinally vanishing, then at every fixed macroscopic stage the supremum product-pseudometric distance between the two readouts tends to zero along the refinement tail. Equality of inverse-limit cylinder values additionally requires a compatible limiting family and a separated quotient or other completeness/separation hypotheses; it does not follow from vanishing pseudometric defects alone.

Proof. The normal-form component of the product readout is bounded by the first inequality in Definition 6.8, and the obstruction component is bounded by the second. Taking the maximum gives the stated product bound. Exact naturality is precisely the special case

$$\rho_{sr} \circ n_s = n_r \circ \rho_{sr}, \qquad \chi_{sr} \circ h_s = h_r \circ \rho_{sr},$$

so both errors vanish there. Cofinal vanishing means that, for any fixed coarse stage and any desired readout tolerance, all sufficiently fine presentations put the two readouts within that tolerance uniformly over the fine state. This proves asymptotic zero distance and nothing stronger without the additional limit compatibility and separation clauses stated in the theorem. □

Remark 6.10 (Scope of the coarse-graining theorem). Theorem 6.9 is not a claim that arbitrary renormalization maps commute with arbitrary repair laws. Its conclusion requires control of the two square defects ε_{sr}^{n} and ε_{sr}^{h}, derived or estimated from the chosen coarse-graining channel, recovery map, decoder, and obstruction readout. The exact refinement theorem is the zero-defect case; approximate RG matching is theorem-grade only when these defects are explicitly controlled.

7 Record Algebras and the Operator Observation Layer

Inputs used here. From the fixed-cutoff collar package we use only the observer-accessible finite-dimensional algebra on one completed compare/write/verify slice, the declared pointer and overlap-sector projectors read on that same slice, and the trace-distance control on restored accessible states when restoration is invoked. No continuum lift and no broader observer-metaphysical assumption is used here.

Definition 7.1 (Exact record algebras and approximate record presentations)**.** *Fix one completed observer-accessible slice at cycle t, and let $\mathcal{A}_t^{\mathrm{acc}}$ be the corresponding finite-dimensional accessible algebra. An* exact record presentation *is a family of orthogonal projectors $\{\widehat{P}_a(t)\}_a \subset Z(\mathcal{A}_t^{\mathrm{acc}})$ whose generated algebra*

$$\mathcal{Z}_{\mathrm{rec}}(t) := \mathrm{Alg}(\{\widehat{P}_a(t)\}_a)$$

is finite and commutative.

An approximate record presentation *on the same declared readout slots is a family of projectors $\{P_a(t)\}_a \subset \mathcal{A}_t^{\mathrm{acc}}$ together with an exact record presentation $\{\widehat{P}_a(t)\}_a$ such that*

$$\delta_{\mathrm{rec}}(t) := \max_a \|P_a(t) - \widehat{P}_a(t)\|$$

is finite.

Theorem 7.2 (Record algebra, declared Born–Lüders instrument, and quantitative stability)**.** *Fix one completed observer-accessible slice at cycle t, supply a density state ρ_t, use its trace valuation for event probabilities, and declare the selective outcome instrument to be the Lüders map for the displayed record projectors.*

1. *The exact record projectors generate a finite commutative central algebra $\mathcal{Z}_{\mathrm{rec}}(t) \subset Z(\mathcal{A}_t^{\mathrm{acc}})$.*

2. *For every event E in the finite event algebra generated by $\mathcal{Z}_{\mathrm{rec}}(t)$,*

$$\mathbb{P}_t(E) = \mathrm{Tr}(\rho_t \widehat{P}_E(t)), \qquad \rho_t|_E = \frac{\widehat{P}_E(t)\rho_t\widehat{P}_E(t)}{\mathrm{Tr}(\rho_t\widehat{P}_E(t))}.$$

   The conditional state is defined only when $\mathbb{P}_t(E) > 0$; the probability formula also applies to zero-weight events.

3. *Conditional on observing E with nonzero weight and applying the declared Lüders instrument, an immediate reread with the same projector has probability 1. The same conclusion after an intervening accepted operation requires the explicit persistence hypothesis that the resulting state remains supported in $\widehat{P}_E(t)$.*

4. *If $\{P_a(t)\}_a$ is an approximate record presentation with modulus $\delta_{\mathrm{rec}}(t)$, then*

$$\|[P_a(t), P_b(t)]\| \le 4\,\delta_{\mathrm{rec}}(t)$$

for all declared record projectors $P_a(t), P_b(t)$. For every declared elementary record event a and every restored accessible state $\widetilde{\rho}_t$ satisfying

$$\|\widetilde{\rho}_t - \rho_t\|_1 \le \varepsilon,$$

one has

$$\left|\mathrm{Tr}\big(\widetilde{\rho}_t P_a(t)\big) - \mathrm{Tr}\big(\rho_t \widehat{P}_a(t)\big)\right| \le \varepsilon + \delta_{\mathrm{rec}}(t).$$

Proof. Because the exact record projectors lie in the center of $\mathcal{A}_t^{\mathrm{acc}}$, they generate a finite commutative central algebra. The supplied trace valuation and declared Lüders instrument give the two formulas in (2); the projector algebra does not by itself select that instrument. This proves (1) and (2) under the displayed premises. The finite-matrix steps behind (1) and (2) are machine-checked in the companion Lean 4 event-algebra development [11]: the commutative span of a projective partition and its containment in the center of the partition commutant, the trace-preserving expectation onto that span, preservation of the partition Born statistics, the Lüders fixed-point law, and compatibility of partition averaging with conditioning are theorem-level exports with declaration-level axiom audits. It is the *partition-averaged* conditional state that equals the normalized block projector. Conditioning alone retains the state inside a higher-rank block; for example, a pure state supported on a rank-two block remains pure, whereas the normalized block projector has rank two.

After the declared conditioning on a nonzero-weight event E, the state lies in the range of $\widehat{P}_E(t)$, so an immediate reread of the same projector has probability 1. More generally, the same calculation applies after an intervening operation when its output state is explicitly assumed to remain in that range. This gives (3); absence of a touched-support marker by itself is not used as a mathematical substitute for support preservation.

For (4), write

$$[P_a(t), P_b(t)] = [P_a(t) - \widehat{P}_a(t), P_b(t)] + [\widehat{P}_a(t), P_b(t) - \widehat{P}_b(t)],$$

because $[\widehat{P}_a(t), \widehat{P}_b(t)] = 0$. Since every projector has operator norm at most 1,

$$\|[P_a(t), P_b(t)]\| \le 2\|P_a(t) - \widehat{P}_a(t)\| + 2\|P_b(t) - \widehat{P}_b(t)\| \le 4\,\delta_{\mathrm{rec}}(t).$$

For the probability bound,

$$\left|\mathrm{Tr}\big(\widetilde{\rho}_t P_a(t)\big) - \mathrm{Tr}\big(\rho_t \widehat{P}_a(t)\big)\right| \le \left|\mathrm{Tr}\big((\widetilde{\rho}_t - \rho_t) P_a(t)\big)\right| + \left|\mathrm{Tr}\big(\rho_t (P_a(t) - \widehat{P}_a(t))\big)\right|.$$

The first term is bounded by $\|\widetilde{\rho}_t - \rho_t\|_1 \|P_a(t)\| \le \varepsilon$, and the second by $\|\rho_t\|_1 \|P_a(t) - \widehat{P}_a(t)\| \le \delta_{\mathrm{rec}}(t)$. Hence the total error is at most $\varepsilon + \delta_{\mathrm{rec}}(t)$. □

The theorem gives a fixed-cutoff observation surface. It does not show that richer branches keep physically relevant pointer surfaces close to one central record algebra, or that the same control survives refinement and continuation.

8 Finite Observable Nets on the Consensus Tower

The record algebra extends, in the accompanying Lean development [12], to a machine-checked interface family in which the observer is a typed object. The base object is a consensus tower: a regulator-indexed family of finite matrix models carrying observer and record fibres, observer-indexed record orders, commutative public record subalgebras of the private matrix algebra, selected states and generators, and functorial refinement maps with compatibility laws. Every statement in this section is a finite structure over declared data; physical regions, instruments, clocks, spacetime and continuum objects require the higher layers.

The operational observerhood receipt. An operational observer over a consensus tower carries six complete clause groups and the own-observer half of a seventh: (i) a bounded accessible star subalgebra containing the observer's public record algebra; (ii) self-readback, a map into the public record algebra fixing it; (iii) durable records annihilated by the observer's generator; (iv) record-conditioned interventions preserving the accessible interface; (v) no-regress prediction in the observer's record order; (vi) continuation, commuting with the refinement maps; and (vii) readable evidence, a readout equal to the trace pairing of each record element with the observer's state and fixed by every intervention. The cross-observer half of (vii) is a separate fixed-regulator receipt over two observers and one access cut: distinct labels, accessible algebras agreeing with the cut, every committed record restricting to the same element at the owner-region meet, agreeing own readouts and one nonzero common restriction. A proper-meet witness on diagonal algebras with corner-evaluation restrictions has record representatives that differ before restriction and agree nontrivially after it, and mutating one shared corner breaks the receipt. Refinement-natural cross-observer evidence and higher-overlap coherence are not supplied. The base witness is a one-regulator tower with private algebra $M_2(\mathbb{C})$, diagonal public algebra, three records in strict chain order, Hamiltonian generator $\mathrm{diag}(1,2)$ and intervention by conjugation with $\mathrm{diag}(1,-1)$; the generator is nonzero and moves the raising matrix unit, so durability is a computed fixed-point fact. Four negative controls each break one field: an erasing readback, the identity generator, a resetting prediction and a constant readout. The receipt selects no unique observer, attaches no instrument and supplies no source realization of the tower data.

Access cuts and the common-origin theorem. A finite regional observer net enriches one consensus tower with a finite region order, per-region local star subalgebras, declared contravariant restrictions, elementwise commutation on declared-disjoint regions, regional expectations, and idempotent local repair. An access cut over such a net assigns each observer one declared support region and one accessible star subalgebra squeezed between the observer's public record algebra and the regional algebra of the declared region, with the declared restrictions below that region preserving accessibility; stability under the local repair idempotents is a separate strengthening receipt. The observer-local algebra at a region is the meet of the regional local algebra with the accessible algebra, and the regional receipts (isotony, elementwise commutation on declared-disjoint regions, restriction stability, selected-state expectation compatibility, regional-repair fixed points, and remote nondisturbance) transport to this meet. Unique descent into the observer-local algebra is conditional on a separate accessible-glue receipt; ambient unique descent alone does not place a glued section in the accessible algebra, and the universal singleton-family packet keeps the receipt type inhabited without a nontrivial gluing claim.

A finite event world binds one packet to each event, with fixed finite event, chart, and packet carriers, declared active masks, chart ownership tying chart supports to observer regions, packet-derived support, record, and coordinate readouts, and the visibility law that an active visible event has support inside the seeing chart's region. The common-origin theorem returns, for every active visible chart/event pair (i,e), three conjuncts from one packet p_e:

$$\mathrm{supp}(\mathsf{p}_e) \preceq R_{O(i)}, \qquad \mathrm{rec}(\mathsf{p}_e) \in \mathcal{L}_{O(i)}\big(\mathrm{supp}(\mathsf{p}_e)\big) \cap \mathcal{P}_{O(i)}, \qquad x_i(e) = \mathrm{geo}_i(\mathsf{p}_e),$$

with $O(i)$ the declared chart owner, $R_{O(i)}$ its observer region, $\mathcal{L}_{O(i)}$ its observer-local algebra, and $\mathcal{P}_{O(i)}$ its public record algebra: event support below the owner's region, record observable in the meet of the observer-local algebra at that support with the owner's public algebra, and the soldering chart coordinate as a readout of the same packet. An event precedence adapter carries strict order data on event slots, tied to the owner's record order on labeled events and

identified with no repair schedule, modular parameter, clock reading, proper time, or global time function; where the events carry certified commit provenance, the source-derived construction below generates that order, and an exact adapter is its verifier rather than a choice of it. A selected coarse-to-fine refinement (slot maps for events, charts, and packets, with naturality for masks, ownership, regions, supports, packets, records, and visibility, plus a declared recharting map) transports the full common-origin conclusion, and raw event candidates with a declared semantic-coincidence setoid descend to the event quotient; a reflexive symmetric relation without transitivity admits no such setoid and is closed to the quotient constructor. The witness world reuses the dimension-two cut, whose interposition chain is strict (scalar public algebra inside the diagonal accessible algebra inside the private matrix algebra), and its record readout separates the two packets.

Source-derived event precedence from mismatch provenance. Event precedence need not be declared where the source supplies provenance. In the companion machine-checked package, every quotient-visible register version carries the identifier of its last semantic writer, and each commit certifies the register versions on which its acceptance genuinely depends, with frame and stamp laws fixing unwritten registers and recording written ones. The `causalSupp` field is supplied semantic data: Lean checks that it lies inside the read set, while producer and independent-verifier receipts must establish that it is sound and complete for genuine dependence. It is not inferred by a kernel-checked dependence analysis. One commit is a direct semantic parent of another exactly when the child certifies a register that the parent wrote, the child's pre-commit snapshot names the parent as that register's writer, and the child's pre-commit value equals the parent's post-commit value. A writer label without the write and value witnesses does not create an edge. The generated informational order is the transitive closure of these edges. Under one abstract ancestry-rank witness that increases across every edge, the generated order is irreflexive, transitive, and asymmetric, and it inhabits the same strict-order interface the event precedence adapter consumes. It is the least strict transitive relation containing the direct parent edges, so an adapter that contains every edge and asserts no comparability beyond the generated order carries exactly the generated order. Geometry does not use the numerical values of this acyclicity witness; it recomputes canonical source height from authenticated parenthood. On the committed chain witness world, a two-commit log generates the declared adapter order in full, so the declaration there is a verifier rather than a choice.

An overlap's mismatch score reads only the committed values on its support, so the score is writer-blind: two snapshots with identical values carry identical scores on every overlap, and a two-register countermodel realizes both parent attributions on value-identical snapshots. A static residual therefore determines no causal arrow; provenance is load-bearing for direction. A commit whose write set misses the support leaves the score unchanged, and a repair whose certified support does not cite the injected version acquires no injection-to-response edge. Under dependency-complete read sets, any two proposals able to move one seam score overlap read-against-write in both directions and land in one aggregate conflict component. Any aggregation map that is constant on conflict components assigns them one aggregate label; an empty-read countermodel shows the completeness clause is load-bearing. A four-commit log realizes the minimal branching interval: one injection, two responses each certifying exactly the injected register, and one answer certifying both outputs generate the four-element Boolean diamond with incomparable responses, while a parent-child pair alone spans the two-element interval. The serialization of the two responses is invisible to this order, because the second response's snapshot contains the first response's output without certifying it. Executed histories carry three further machine-checked controls. One adjacent independent swap preserves the authenticated direct-parent relation and its transitive closure when

the two executions both satisfy fresh-ID and duplicate-free hypotheses and the commits' write sets avoid each other and each other's certified supports. The general equivalence-generated swap chain preserves only raw writer citation, not arbitrary labeled payloads. Executor stutters are removed before commits are formed. A dependent pair marks the boundary: swapping a writer past a reader that certifies the written register changes the edge set. On such histories the append-only acyclicity rank is derived from the execution position rather than declared. Independently of that witness, the geometry layer computes canonical source height from authenticated parenthood alone, zero at roots and one plus the maximum direct-parent height otherwise, proves that it is the attained longest authenticated-parent-chain length, and proves strict increase on every parent edge and generated ancestry. Under fresh identifiers, duplicate-free execution, and visible-support hypotheses, a commit writing only outside the visible register carrier changes no visible mismatch and supplies no outgoing direct raw or authenticated parent edge to a visibly supported child; incoming edges into the hidden commit are not excluded. A freshly injected mismatch persists at full strength through a continuation whose commits never write its overlap support. A refinement receipt assumes that its map carries every direct parent edge into the coarse reflexive precedence; under that assumption it transports the generated order and its causal intervals through coincidence merges. The reversed assignment on the committed diamond-to-chain witness inhabits no such receipt. The generated order is informational. No theorem here supplies invariance over the full admissible implementation class, event separation, physical causal faithfulness, or a Lorentzian continuum.

The generated structure meets two established programs. In causal set theory, a locally finite partial order represents proto-causality [1, 2]. The OPH order is finite and therefore locally finite, but its proved interpretation is semantic read-from dependence. It is thus causal-set-like rather than a physically attached causal set. The exact-embedding continuum route requires an order embedding whose image has the approximately Poisson count density needed for a causal-set faithful embedding, together with manifoldlikeness and dimension [3, 4, 5]. In a past- and future-distinguishing continuum, causal order fixes conformal geometry [7, 8]; the volume datum fixes the conformal factor. The standard Lorentz-invariant continuum approximation uses Poisson sprinkling [6]; the OPH commit log is not a Poisson sprinkling, and its cardinality is not a spacetime volume. Neither continuum existence nor the causal-set Hauptvermutung follows from a finite order. In distributed computing the generated relation is an authenticated read-from dependence suborder and analogue of happened-before [9]; record or program order contributes only when a later commit genuinely consumes the earlier record. Its OPH-specific content is the binding to quotient-visible mismatch, certified read supports, writer-blind static scores, and the strict-order interface of the finite event world.

Every finite strict partial order compiles exactly into an abstract authenticated log, with rank derived from predecessor cardinality. In this generic compiler the authenticated-parent relation is the whole supplied order, including pairs outside its Hasse reduction. This is an expressivity theorem for the grammar, not a dynamics or execution-history selection theorem. The exact Boolean-diamond inhabitant shows that the interface is not restricted to chains. Placement of such logs on the rank-three source carrier with a real time axis, one-way and two-way cone compatibility, the source-direction Einstein implication and a conservative source-record family with a flat causal-order limit are developed in the spacetime layer under its own placement, speed, count–volume and calibration premises [32]. None of them follows from the generated order alone.

Character-block coexistence of regional algebras. For every $n \geq 2$ the full matrix algebra $M_n(\mathbb{C})$ admits no unital complex-algebra homomorphism into the scalars: the matrix ring is simple, so such a homomorphism is injective, while the off-diagonal matrix unit is a nonzero element

whose square is zero. No unital complex-algebra homomorphism into the scalars accepts an anticommuting pair of units. On this interface, at any regulator of nonzero dimension, a region above a commutative-algebra region carries no anticommuting pair, and every full matrix factor of dimension at least two above a scalar overlap region is excluded: the tensor-factor local-algebra picture with trivial overlap is impossible. The inhabitant locating the boundary is a dimension-four net with three regions: a bottom overlap region carrying the scalar algebra and two declared-disjoint maximal regions carrying the block algebras $\mathrm{diag}(a, a, A)$ and $\mathrm{diag}(B, b, b)$ with arbitrary two-by-two blocks. Both regional algebras are noncommutative, neither contains the other, their elements commute pairwise, and each admits a character onto the scalars through its one-dimensional corner block, which supplies the star-homomorphic restriction retractions the interface demands. The block shape is forced: a noncommutative regional algebra compatible with a scalar drop must carry a one-dimensional block, and the witness realizes that minimal shape. The declared overlap algebra is a proper subalgebra of the set intersection of the two block algebras; the regional repair maps are identities. The construction provides no genuine coverage semantics for a source-produced net.

Hamiltonian rigidity of the private block. The reversible private dynamics behind these interfaces is forced by continuity [12]. On one full finite matrix block, every complex-linear Leibniz endomorphism is an inner derivation, and every star-compatible one is the commutator action $x \mapsto (-i)(Hx - xH)$ of a self-adjoint generator, unique up to an additive real multiple of the identity. A real-parameter group of star automorphisms with pointwise continuous orbit maps is conjugation by $\exp(t(-i)H)$ for one time-independent self-adjoint H: continuity upgrades to differentiability at zero because the short-interval average of the norm-continuous matrix family is invertible. On a general finite private algebra, the center of any unital star subalgebra is spanned by a minimal orthogonal central resolution of the identity, unique up to permutation; the subalgebra is a finite direct sum of full matrix blocks; and a pointwise-continuous automorphism group fixes every block and acts by one Hamiltonian per block, with block fixing derived and the block swap realized on the square two-block algebra. The parameter is a real number, and no physical clock or source-selected Hamiltonian is claimed. The native seam repair lies outside the reversibility hypothesis: its executed channel strictly attenuates a coherent mode, and its local generator is a positive Liouville operator on operator space [13].

Finite covariant-net consequences and limitations. The finite observer net determines a category of event regions and causal embeddings together with a covariant observable functor [12]. Its causal embeddings are proof-only order data, so the category is thin and carries no Lorentzian geometry. In particular, these carriers provide no region reflection, antiunitary conjugation, distinguished vacuum, spin structure, symmetry-group action, Lorentzian Cauchy-embedding class, translation action, or spectrum condition. Consequently they lack the data needed to formulate CPT, spin–statistics, relative-Cauchy stress response, or Haag–Ruelle scattering. These are limitations of the mathematical carrier, not negative theorems about QFTs built from richer data.

The same module encodes seven structural questions and kernel-checks their partition by mathematical conclusion. Four of them (CPT, spin–statistics, relative-Cauchy stress response, and particle scattering) lack defining geometric or spectral data on this carrier. Full field/action reconstruction likewise cannot be inferred from the finite static identities alone. Of the two remaining questions, the restricted endomorphism surrogate collapses as described next, while the finite KMS identity coexists with the net- and public-algebra counterexamples below. Thus the seven-way classification is a statement about what follows from the specified carrier.

The finite sector theorem has a narrower exact content. Every bijective localized endomorphism

of the witness net is inner and hence equivalent to the identity under the declared finite equivalence relation. This does not constitute DHR reconstruction: the carrier supplies no source-selected vacuum representation, quasi-local net, transportability structure, or reconstructed gauge group. The selected tower states nevertheless induce mathematical GNS representations of the completed colimit, as described below. Innerness of this restricted class supplies no charge or flavor quantum number.

Two further finite results delimit thermality and field reconstruction. The oriented-face curvature has kernel exactly the port gradients, and its five-neighbor seam action is gauge invariant exactly for conserved seam currents. This is a static finite action identity; it supplies no temporal dynamics, charge–current continuity map, source-selected physical current, physical matter or Spin action, spacetime carrier, or continuum limit. On a private finite block, the normalized Gibbs state satisfies the algebraic KMS identity. On the constructed example net, however, every functional is KMS for every grading-preserving flow at every complex time, while every pointwise-continuous group of star automorphisms of the public record algebra is the identity, a statement that leaves continuous row-stochastic public semigroups untouched. The finite Gibbs identity therefore does not establish thermality of the observer net.

Natural powers give one exact constraint on a fixed-cutoff scattering construction: in every Hausdorff topological group, $g^n \to a$ forces $g = 1$. Hence a nontrivial exact unitary update at fixed finite cutoff cannot itself settle to an ordinary matrix-topology large-time limit. The same module proves $(g^n)^{-1}g^n = 1$, so relative evolution can converge even when the individual powers do not. Any fixed-cutoff direct-power construction that uses U^n itself as its asymptotic carrier cannot use full convergence of that sequence; it must change the asymptotic carrier or dynamics, or use a projected or weaker convergence notion. At fixed finite dimension, full weak-operator convergence does not evade the obstruction because the standard finite-dimensional operator topologies coincide. Comparison dynamics, selected projected scalar or observable readouts, an infinite-dimensional weak limit after a continuum or infinite-volume passage, open-system evolution, and finite-time or recurrence-aware readouts lie outside this no-go. The theorem constructs no S-matrix, asymptotic state, optical theorem, cross section, or interacting continuum limit.

Restriction naturality on authenticated event regions. Let a finite regional net have local algebras $\mathcal{A}_r(U)$, contravariant restrictions $R^r_{V,U}$ for $V \subseteq U$, algebra refinements j_{rs}, and regional repair maps P^r_V. Restriction is additional algebraic data; isotony alone does not supply it. Suppose repair is natural,

$$j_{rs}P^r_V = P^s_{V_s}j_{rs},$$

where V_s is the refined region. The following compatibility is a separate hypothesis on the restricted domain:

$$R^r_{V,U}X = P^r_V X \qquad (X \in \mathcal{A}_r(U)). \tag{1}$$

Proposition 8.1 (Finite restriction and refinement square)**.** *Under these hypotheses,*

$$j_{rs}R^r_{V,U}X = R^s_{V_s,U_s}j_{rs}X. \tag{2}$$

Let each event e of an authenticated semantic log carry a declared region U_e, with $U_e \subseteq U_f$ on every authenticated direct-parent edge. Generated reflexive precedence then defines restrictions from the algebra at f to that at e, with identity and contravariant composition. If a certified event map preserves those direct edges in target generated precedence and satisfies $U'_{F(e)} = (U_e)_s$, the same square holds for these event restrictions.

Proof. For the algebra square, substitute (1), apply repair naturality, and substitute the target compatibility. Inclusion extends from direct edges to generated precedence by transitivity; the restriction identity and tower law give identity and composition. The event map preserves generated precedence by induction on an authenticated path, so the target restriction is typed. Its region compatibility identifies its algebra square with (2). □

An inhabited example decorates the committed four-event Boolean diamond and three-event chain by a two-dimensional diagonal algebra. The injection event has the scalar algebra, and every response and answer has the full diagonal algebra. Restriction to the root sends $\mathrm{diag}(x_0, x_1)$ to $x_0 I$; in particular, $\mathrm{diag}(1, 0)$ is changed. The authenticated coarsening maps the diamond labels $(0, 1, 2, 3)$ to $(0, 1, 1, 2)$ and merges two distinct responses. Its region decorations agree, and the restriction square is inhabited. The reversed assignment $(2, 1, 1, 0)$ violates authenticated precedence. The finite identity, composition, compatibility, and coarsening statements are machine checked [12].

The algebra regulator in this example is constant. Region decoration and edge preservation are explicit data; neither physical regions nor a source-selected refinement family are inferred. These are restrictions of commutative finite algebras, without an assertion of homomorphic scalar restriction for matrix factors. No normed limit restriction, physical locality, spacetime identification, or physical time-slice property follows.

Joint carriers of retained observer pairs. Two regional diamonds carry source-counted correlation data of retained observer pairs from one bounded companion run: a two-slot conditional-expectation net for the pair 86/88 on a joint carrier of dimension 182, and one for the support-disjoint pair 86/247 on a joint carrier of dimension 169. Both transport onto one constant consensus-tower stage by the same specified pattern, with observer 86's checkpoint family as commutative public layer, the uniform selected state and the template nontriviality receipts. An exact shared-observer theorem relates them: the two source-derived joint paths have equal left label components at every step, equal left occupation counts and equal induced diagonal hinge states, each equal to the source marginal state of observer 86. At the anchored-net level the carriers are not identified, and no map between the two anchored nets is constructed [12].

That carrier-level obstruction is removed below the net level by the common finite carrier $(\mathrm{obs86}, \mathrm{obs88}, \mathrm{obs247}) \in \mathrm{Fin}(13) \times \mathrm{Fin}(14) \times \mathrm{Fin}(13)$. Its coordinate projections recover both complete 32-row source paths and their occupation tables. The diagonal triple counted state is positive with trace one, and explicit linear positive trace-preserving marginals recover both counted pair states. Four triple checkpoint projectors have normalized partial traces recovering both anchored checkpoint partition members, and on each of the 31 actual adjacent source transitions a triple reindexing transports the source projector and intertwines with both pair marginals. All three final-to-initial source counts are exactly zero, so the endpoint is not wrapped to the first row. This is a common state, path and checkpoint coupling, not a canonical gluing theorem: the pair of marginals is noninjective on triple matrices, so the row alignment selects the coupling.

The one-step time-slice interface. `TimeIndexedNetEvolution` is a finite one-step interface: a finite index with successor, a region family, per-step ambient star automorphisms mapping each regional algebra to its successor, and admitted data in the current regional algebra whose evolved images generate the successor algebra. A full inhabitant exists over the 86/247 net on cyclic Fin(32): it follows the 31 actual adjacent source-row pairs and adds one terminal-to-initial bookkeeping automorphism that is not a source transition, with observer 86's lifted source generators as data and the left region at every step. The natural single-projector walk candidate is a negative control:

its evolved singleton cannot generate the top regional algebra. On this net, an evolution family that maps each regional algebra onto itself forces a step-invariant region family, every net-compatible slice family over the walk-step evolution is constant, and a one-step region change of any inhabitant exchanges the two slot regions. A slot-alternating inhabitant composes the walk step with the internal slot exchange and alternates observer 86 and observer 247 generators; its region family is not step invariant, and the exchange factor is a constructed symmetry, not a source transition. The interface has no multi-step cocycle, Cauchy geometry, clock or source-selected dynamics, and implies no time-slice theorem.

Joins of the two carriers. At the diagonal layer, which carries every source-counted state, the two carriers glue along the 13-dimensional observer-86 hinge. Fiber-uniform sections of slot marginalization embed the hinge into the 182- and 169-dimensional layers, and the join is the pushout over the hinge. Both embeddings are injective and agree on the hinge, the join has dimension $338 = 182 + 169 - 13$, strictly below the direct-sum dimension 351, and the intersection of the two images is exactly the embedded hinge. A hinge restriction retracts the join, under which the two transported counted states restrict to the same observer-86 occupation law; the ambient anchorings and the 31 walk-step transports descend to one join evolution intertwined by both embeddings. Region by region, the record layer of each diamond is exactly the set of diagonal restrictions of the specified regional algebra, with dimensions $1, 13, 14, 182$ and $1, 13, 13, 169$; the induced net on the join is isotone, both embeddings are net morphisms, the conditional expectations restrict to the record layer and the join evolution preserves every induced region. Two negatives delimit this layer: the gluing submodule is not multiplicatively closed, so the pushout carries no algebra structure, and the record layer is commutative while the left regional algebra is not.

At the operator layer, adjoining the unused observer's identity gives injective unital star homomorphisms of the full M_{182} and M_{169} pair algebras into M_{2366}, whose images intersect in exactly $M_{13} \otimes I_{14} \otimes I_{13}$ and generate the triple algebra; the shared hinge is noncommutative [16]. Both embeddings intertwine all 31 adjacent path-read permutations, and ordered products give finite interval composition with no invented terminal-to-initial edge. Partial traces recover the counted pair states and are not multiplicative. For the 86/247 carrier the boundary between the layers is an exact asymmetry: the record carrier embeds into the joint matrix algebra as an injective unital star homomorphism compatible with the regional algebras and expectations, while the left-slot conditional expectation is positive, unital and trace preserving but not multiplicative. No map from the joint algebra to the record carrier is multiplicative on the left regional algebra and restricts to the record-layer morphism, and no unital algebra homomorphism carries that regional algebra into the record carrier at all. The tensor split and path permutations are supplied adapters; physical regions, preparations, a clock, Cauchy geometry and a state-preserving tower morphism are not derived, and observer 86/88 source supports overlap, so tensor slot locality does not establish physical support disjointness.

The normed completion of the colimit. The consensus tower supplies a directed family of finite matrix algebras with star homomorphisms as connecting maps. Its filtered colimit carries an inductive-limit seminorm, the infimum over declared refinements of one representative of the refined stage norm, independent of the representative by contractivity. The stage norms of one representative lie in a finite set fixed by the spectrum of a^*a, so the infimum is attained, and a class has seminorm zero exactly when some declared refinement annihilates its representative. The seminorm is therefore a norm satisfying the C*-identity, and the uniform completion is a C*-algebra. The closures of the limit local algebras are isotone, independent of the presenting regulator,

elementwise commuting on declared-disjoint regions, and equal along the order-theoretic Cauchy embeddings, each instantiated on a finite witness net with a norm-one element. Two controls fix the boundary: a connecting map annihilating the unit gives a nonzero stage element of limit seminorm zero, and a diagonal scaling that is not a star map increases a matrix-unit norm. The construction is algebraic and order-theoretic and supplies no time-indexed physical evolution or physical attachment.

A selected-state Hilbert-space representation of the completed net. Equip the completed colimit with the canonical C*-spectral order. For every positive complex-linear functional ω normalized by $\omega(1) = 1$, the GNS construction gives a complex Hilbert space $\mathcal{H}_\omega$ and a unital star representation π_ω of the completed colimit by bounded operators. The completion image Ω_ω of the algebra unit is explicit and obeys

$$\|\Omega_\omega\| = 1, \qquad \langle \Omega_\omega, \pi_\omega(a)\Omega_\omega\rangle = \omega(a), \qquad \overline{\{\pi_\omega(a)\Omega_\omega : a\}} = \mathcal{H}_\omega.$$

Sending every completed regional algebra to its image under the same representation preserves isotony, refinement independence, elementwise locality for the declared-disjoint regions, and the order-theoretic Cauchy-embedding equality. Images are used rather than identified with their sources because normalization does not imply that the representation is faithful.

For the specified tower interface, the functional need not be added as an unrelated input. Let ξ_r be an explicitly supplied observer family compatible with every regulator refinement, and let ρ_{r,ξ_r} be the selected density state carried by the tower. State naturality makes

$$\omega_\xi(\iota_r(X)) := \mathrm{Tr}(\rho_{r,\xi_r} X)$$

well defined on the algebraic colimit. The finite state-expectation bound extends it continuously to the completed colimit, where it is positive for the spectral order and normalized. Hence ω_ξ supplies the GNS Hilbert space and representation above. The one-regulator witness instantiates this construction, and its GNS unit class is nonzero with norm one. This yields a mathematical Hilbert-space representation of the selected-state completed net.

Taken alone, this construction's coherent observer family and finite density states are declared structural data, not a state selected by an experimental source. Neither they nor Ω_{ω_ξ} are identified as a physical vacuum. No Hamiltonian or physical dynamics, energy–momentum, field content, Lorentzian attachment, continuum or infinite-volume limit, interacting theory, or physical time-slice theorem follows [12].

An exact finite two-site Hamiltonian benchmark. As a separate finite-dimensional test of Hamiltonian structures outside the completed-net attachment, take

$$\mathcal{H}_\mathrm{I} = \mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4, \qquad H_\mathrm{I} = \frac{1}{2}(1 - Z \otimes Z).$$

The full matrix algebra acts faithfully by bounded operators on $\mathcal{H}_\mathrm{I}$. Exact matrix proofs give $H_\mathrm{I}^* = H_\mathrm{I}$, $H_\mathrm{I} \geq 0$, and $H_\mathrm{I}^2 = H_\mathrm{I}$. The explicit unit vector $|00\rangle$ and density $\rho_{00} = |00\rangle\langle 00|$ obey

$$\||00\rangle\| = 1, \qquad H_\mathrm{I}|00\rangle = 0, \qquad H_\mathrm{I}\rho_{00} = \rho_{00}H_\mathrm{I} = 0, \qquad \mathrm{Tr}(\rho_{00}) = 1.$$

Thus ρ_{00} is a normalized stationary ground-state density; the orthogonal unit state $|11\rangle$ also has zero energy, so the ground space is degenerate and no unique vacuum is selected. The propagator $U(t) = \exp(-itH_\mathrm{I})$ is unitary and obeys $U(s+t) = U(s)U(t)$; its conjugation flow $\alpha_t(X) = U(t)XU(-t)$

is multiplicative, fixes ρ_{00}, and preserves commutation between jointly evolved left- and right-slot observables. The bond is genuinely interacting in the exact finite sense that H_{I} is not any sum $H_L \otimes 1 + 1 \otimes H_R$ of one-site Hamiltonians; in particular it belongs to neither one-site matrix-algebra image. The evolution is nontrivial: for $X_L = X \otimes 1$,

$$\left.\frac{d}{dt}\right|_{t=0} \alpha_t(X_L) = -i[H_{\mathrm{I}}, X_L] \neq 0,$$

with the displayed matrix entry equal to i. Deleting only the $Z \otimes Z$ interaction leaves the scalar Hamiltonian $H_{\mathrm{del}} = \frac{1}{2}1$, makes the full conjugation flow static, and makes the same infinitesimal generator zero. These are zero-residual identities on the declared four-dimensional carrier.

Taken alone, this is an exact coupled two-spin lattice benchmark, not an interacting quantum field theory and not a Hamiltonian attachment to $(\mathcal{H}_{\omega_\xi}, \pi_{\omega_\xi})$ above. Its tensor factors are declared sites rather than Lorentzian regions; H_{I}, its normalization, and the state are supplied rather than selected by an OPH source; and t has no physical clock or energy calibration. Equal-time commutation here preserves one finite tensor-product commutator and is not a microcausality theorem. No continuum or infinite-volume limit, field algebra, Lorentz covariance, spectrum condition, renormalization flow, particle interpretation, scattering construction, detector readout, or physical time-slice theorem follows.

The finite Hamiltonian, regional diamond, and selected GNS on one carrier. The two preceding constructions compose in one typed finite model. Use the canonical basis equivalence $\mathrm{Fin}(2) \times \mathrm{Fin}(2) \simeq \mathrm{Fin}(4)$ to place the two-spin algebra on a one-regulator consensus tower T_{I}. Its selected density is exactly ρ_{00}, and its stored generator is not the constant tower adaptor's zero map but

$$\delta_{\mathrm{I}}(X) = -i[H_{\mathrm{I}}, X].$$

On the same carrier, the scalar/left/right/top conditional-expectation diamond has commuting declared-disjoint slot algebras, and its left and right algebras jointly generate $M_4(\mathbb{C})$. Let ω_{I} be the tower-selected functional, let $(\mathcal{H}_{\mathrm{I}}^{\mathrm{GNS}}, \pi_{\mathrm{I}}, \Omega_{\mathrm{I}})$ be its GNS triple, and define

$$h_{\mathrm{I}} = \iota(H_{\mathrm{I}}), \qquad \widehat{H}_{\mathrm{I}} = \pi_{\mathrm{I}}(h_{\mathrm{I}}), \qquad \widehat{U}_{\mathrm{I}}(t) = \pi_{\mathrm{I}}(\iota(U(t)))\,.$$

The exact composite representation from $M_4(\mathbb{C})$ through the colimit completion to bounded GNS operators is injective. Consequently the ground-vector equations do not arise from a collapsed quotient:

$$\|\Omega_{\mathrm{I}}\| = 1, \qquad \widehat{H}_{\mathrm{I}} \neq 0, \qquad \widehat{H}_{\mathrm{I}}^* = \widehat{H}_{\mathrm{I}}, \qquad \widehat{H}_{\mathrm{I}}\Omega_{\mathrm{I}} = 0.$$

The Hamiltonian belongs to the represented top region, while the represented left and right Pauli observables retain their regional memberships and commute. The mapped propagators are unitary, form an additive one-parameter group, and obey the exact intertwining identity

$$\pi_{\mathrm{I}}(\iota(\alpha_t(X))) = \widehat{U}_{\mathrm{I}}(t)\pi_{\mathrm{I}}(\iota(X))\,\widehat{U}_{\mathrm{I}}(-t).$$

Interaction remains load bearing after representation: $\pi_{\mathrm{I}}(\iota(-i[H_{\mathrm{I}}, X_L])) \neq 0$, and this first-order response is outside the represented left algebra, whereas the same represented response is zero after deleting the coupling.

The left conditional expectation E_L is trace preserving but does not preserve the selected pure state:

$$E_L(1 \otimes Z) = 0, \qquad \mathrm{Tr}(\rho_{00}(1 \otimes Z)) = 1.$$

Thus the construction is a finite regional diamond together with co-located interacting C*-dynamics on its ambient algebra, represented on the tower-selected GNS Hilbert space. The dynamics need not preserve each regional algebra: indeed, the nonzero first-order response above leaves the represented left algebra. The failure of state preservation is a property of that particular trace-preserving expectation, not a failure of a global state to restrict to the regional algebras. It therefore supplies a useful incompatibility diagnostic but is not itself the criterion separating a finite net from a physical QFT. The regulator, factorization, density, Hamiltonian, and flow parameter are declared rather than source produced or calibrated; the ground sector is degenerate; and Lorentzian localization, a spectrum condition, continuum or infinite-volume control, renormalization flow, field and particle content, scattering, detector readback, and a physical time-slice theorem are absent.

A source-counted finite history Hamiltonian on the selected GNS space. The retained eight-history packet supplies a separate finite attachment in which the state and diagonal action entries are source counted rather than freely chosen. For each binary path $g = (s_0, s_1, s_2)$, let $n_g > 0$ be its exact window count among the 1754 retained windows and let

$$S_g = \mathbf{1}_{s_0 \neq s_1} + \mathbf{1}_{s_1 \neq s_2}.$$

On $M_8(\mathbb{C})$, define

$$\rho_{\text{hist}} = \operatorname{diag}\left(\frac{n_g}{1754}\right), \qquad H_{\text{hist}} = \operatorname{diag}(S_g).$$

The source receipts prove every n_g positive and prove the displayed state-change formula for S_g. Exact matrix proofs then give

$$\rho_{\text{hist}} \geq 0, \quad \operatorname{Tr}\rho_{\text{hist}} = 1, \quad H_{\text{hist}} \geq 0, \quad H_{\text{hist}}^* = H_{\text{hist}}, \quad [\rho_{\text{hist}}, H_{\text{hist}}] = 0,$$

with H_{hist} nonzero and non-scalar. The selected energy retains the counted source value

$$\operatorname{Tr}(\rho_{\text{hist}} H_{\text{hist}}) = \frac{197}{1754}.$$

Place this pair in one consensus-tower stage with stored generator $\delta_{\text{hist}}(X) = -i[H_{\text{hist}}, X]$. The selected density constructs a normalized GNS triple $(\mathcal{H}_{\text{hist}}, \pi_{\text{hist}}, \Omega_{\text{hist}})$, and the composite representation of $M_8(\mathbb{C})$ is injective. Thus the represented Hamiltonian is nonzero and self-adjoint. Its cyclic matrix coefficient is exactly 197/1754, which also proves

$$\pi_{\text{hist}}(H_{\text{hist}})\Omega_{\text{hist}} \neq 0.$$

The matrix unit E_{01} supplies a nonclassical control:

$$\left(-i[H_{\text{hist}}, E_{01}]\right)_{01} = i,$$

so the represented commutator response is nonzero. The exponential propagators are unitary, obey the additive group law, intertwine exactly with the represented Heisenberg flow, and leave ρ_{hist} stationary.

The same carrier has an exact tensor-local history decomposition. The source encoding identifies its basis with $((s_0, s_1), s_2) \in (\mathbb{Z}/2)^3$. Tensor-factor embeddings define the one-slot algebras $\mathcal{A}_0, \mathcal{A}_1, \mathcal{A}_2$, the generated adjacent interval algebras $\mathcal{A}_{01} = \mathcal{A}_0 \vee \mathcal{A}_1$ and $\mathcal{A}_{12} = \mathcal{A}_1 \vee \mathcal{A}_2$, and $\mathcal{A}_{012} = M_8(\mathbb{C})$. The inclusions are isotone and the separated endpoint algebras obey exact locality,

$$[\mathcal{A}_0, \mathcal{A}_2] = 0.$$

Writing the two adjacent domain-wall projectors as

$$H_{01} = \mathrm{diag}(\mathbf{1}_{s_0 \neq s_1}), \qquad H_{12} = \mathrm{diag}(\mathbf{1}_{s_1 \neq s_2}),$$

gives the residual-free local decomposition

$$H_{\mathrm{hist}} = H_{01} + H_{12}, \qquad H_{01} \in \mathcal{A}_{01}, \qquad H_{12} \in \mathcal{A}_{12}.$$

Both bonds are positive, and their separately counted source energies are

$$\mathrm{Tr}(\rho_{\mathrm{hist}} H_{01}) = \frac{94}{1754}, \qquad \mathrm{Tr}(\rho_{\mathrm{hist}} H_{12}) = \frac{103}{1754}.$$

For $\delta_{\mathrm{hist}}(X) = -i[H_{\mathrm{hist}}, X]$, locality removes the remote bond exactly, so

$$\delta_{\mathrm{hist}}(\mathcal{A}_0) \subseteq \mathcal{A}_{01}, \qquad \delta_{\mathrm{hist}}(\mathcal{A}_2) \subseteq \mathcal{A}_{12}.$$

These are first-commutator finite-support statements, not a continuum finite-speed theorem. The single source functional $\omega(X) = \mathrm{Tr}(\rho_{\mathrm{hist}} X)$ restricts to compatible states on every displayed algebra: for each inclusion $\mathcal{A}_U \subseteq \mathcal{A}_V$, $\omega_V|_{\mathcal{A}_U} = \omega_U$. The injective selected-GNS representation transports the endpoint locality, the two bond memberships, the Hamiltonian decomposition, and the generator-support statements without collapse. This compatible restriction is the state compatibility required of the finite net; it does not assert a source-state-preserving conditional expectation from the full algebra onto every region.

This is a source-counted finite three-history-slot local dynamical system on its selected GNS Hilbert space, not a physical QFT. The retained run was hash-pinned, but the empirical law and repair-count postprocessing were not jointly preregistered. The packet fixes the diagonal density and the two repair-change contributions; the full $M_8(\mathbb{C})$ quantum extension, tensor-factor locality reading, off-diagonal coherence probe, one-regulator tower, and real flow parameter are mathematical adapters. The nonzero energy of Ω_{hist} prevents its promotion to a vacuum or ground vector. The three successive history coordinates are not calibrated physical times or Lorentzian regions, and there is no physical clock or energy calibration, spectrum condition, continuum or infinite-volume control, renormalization flow, field/particle interpretation, scattering, detector readback, or physical time-slice theorem.

The rational record layer carries a finite slot-indexed conditioning family. For every slot region $S \subseteq \{0, 1, 2\}$, conditioning the committed window law on the slots in S defines a map E_S on $\mathbb{Q}$-valued record observables that is rational-linear, unital, pointwise positive, localized, idempotent, and tower-compatible, $E_S \circ E_T = E_S$ for $S \subseteq T$. The full-region map is the identity because the three slots separate the eight histories, the empty-region map is the constant empirical mean, and every E_S preserves that mean. The domain walls are fixed points of their own interval maps, every regional map preserves the bond means 94/1754 and 103/1754 and the action mean 197/1754, and the late-slot conditional weights are 383/415 in the $(s_0, s_1) = (0, 0)$ cell and 2/89 in the $(0, 1)$ cell.

The matrix obstruction has a fixed target and hypotheses. No complex-linear map from $M_8(\mathbb{C})$ into this particular $\mathcal{A}_{01}$ can both satisfy the $\mathcal{A}_{01}$-bimodule law and preserve the committed state ω: such a map would assign the slot-two projector a single conditional weight over $\mathcal{A}_{01}$, whereas the two cells demand 383/415 and 2/89. This does not rule out ordinary AQFT nets of inclusions with state restriction, other or enlarged target algebras, ancillary boundary memory, generalized expectations, or non-bimodule positive or completely positive channels. The finite-probability conditioning identities and wall coefficients transport to represented diagonal elements on the selected GNS carrier (Lean module `SourceHistoryExpectationNet` [12]).

Boundary-fibre circularity control. Over the bare consensus tower (finite presentation and quotient states, a mismatch functional, a strictly descending step relation, an idempotent normal form, a protected boundary map and coarse maps), a signature family that factors through the protected boundary on consistent states and is complete on consistent states forces the boundary fibre: equal boundary values imply equal consistent states. Instantiating the signature as the protected boundary itself makes the factoring hypothesis trivial and makes the completeness hypothesis definitionally identical to the conclusion; the degeneration is recorded as a definitional equivalence, so the circularity is itself a theorem, and the result is substantive only for a signature constructed independently of the boundary and proved complete on its own grounds. A selected settled branch (one normal-form-fixed, coherent quotient point per regulator) is an explicit premise with a concrete inhabitant and an explicit failing candidate, and an event and geometry readout fragment carries normal-form-invariant, coarsely natural packet readouts that separate two consistent states on the same concrete tower [12].

9 Distributed Presentations of One Finite Universe

The implementation closure clause in Definition 5.21 has a distributed form. A worker partition is allowed to accelerate or package the computation, but it is not a new physical ingredient. The physical question is whether the worker run presents the same finite quotient repair system as the monolithic carrier.

Definition 9.1 (Finite global OPH carrier)**.** *At refinement stage r, a finite global OPH carrier consists of a finite patch graph*

$$G_r = (V_r, E_r),$$

finite patch state sets S_i, interface alphabets I_e, endpoint maps $\pi_{i,e} : S_i \to I_e$, an implementation-hiding group Γ_r, a boundary/sector map B_r, a mismatch potential Φ_r, accepted quotient repairs $\to_r$, and an observer-readable map

$$\mathrm{Read}_r : Q_r \to \mathcal{Y}_r, \qquad Q_r := \left(\prod_{i \in V_r} S_i\right) / \Gamma_r.$$

Let $C_r \subseteq Q_r$ be the quotient consistency set. On the finite branch covered by Theorems 3.25 and 5.2, the accepted quotient repair relation has a unique normal-form map

$$n_r : Q_r \to C_r.$$

A one-universe input is the pair $\mathbf{U}_r = (\mathfrak{U}_r, q_0)$, where $\mathfrak{U}_r$ is the carrier data above and $q_0 \in Q_r$. A worker partition is not part of $\mathfrak{U}_r$.

Definition 9.2 (Worker presentation and physical projection)**.** *Let $\kappa : V_r \to W$ assign every patch to one authoritative worker. The cut set is*

$$E_\kappa^{\mathrm{cut}} := \{\{i, j\} \in E_r : \kappa(i) \neq \kappa(j)\}.$$

A distributed presentation for κ has worker-owned states, ghost or halo copies, message queues, transaction records, retries, worker identifiers, checkpoints, event logs, and scheduler state. Let $\widetilde{Q}_{r,\kappa}$ be its quotient by purely local worker bookkeeping that leaves authoritative physical states unchanged. The physical projection

$$p_{r,\kappa} : \widetilde{Q}_{r,\kappa} \to Q_r$$

keeps only the authoritative patch states, quotients by Γ_r, and discards queues, retry metadata, worker labels, and external clocks. Let

$$D_r := \bigsqcup_{\kappa} \{\kappa\} \times \widetilde{Q}_{r,\kappa}, \qquad P_r(\kappa, \tilde{q}) := p_{r,\kappa}(\tilde{q}).$$

A distributed readout is physical when it factors as

$$\widetilde{\mathrm{Read}}_r = \mathrm{Read}_r \circ P_r.$$

Definition 9.3 (Admissible distributed event)**.** *For $d, d' \in D_r$, a distributed event $d \Rightarrow d'$ is **admissible** when one of the following holds.*

(a) ***Linearizable physical commit.*** *There is a nonempty monolithic accepted repair path*

$$P_r(d) \to_r^* P_r(d')$$

whose repair word is recorded as the event's linearization witness.

(b) ***Physical stutter.*** *$P_r(d') = P_r(d)$. Message delivery, halo refresh, prepare, abort, checkpoint, worker start or stop, idempotent replay, and repartition metadata are allowed only in this class unless they also carry a physical commit witness.*

(c) ***Certified rollback.*** *There is an earlier committed state d_j in the same run history such that $P_r(d') = P_r(d_j)$, and the rollback certificate names that committed projection root. Transparent restart from a committed frontier is a stutter.*

Proposition 9.4 (Normal form is constant on an accepted reachability cone)**.** *If $q \to_r^* q'$, then*

$$n_r(q') = n_r(q).$$

Proof. The state q' is reachable from q. The state $n_r(q')$ is a normal form reachable from q', hence also reachable from q. Theorem 3.25 and Theorem 5.2 give the unique quotient normal form reachable from q, so $n_r(q') = n_r(q)$. □

Theorem 9.5 (Distributed realization of one finite OPH universe)**.** *Fix a finite global carrier $\mathbf{U}_r = (\mathfrak{U}_r, q_0)$ satisfying Theorems 3.25 and 5.2. Let*

$$d_0 \Rightarrow d_1 \Rightarrow \cdots \Rightarrow d_m$$

be a finite distributed execution in D_r with $P_r(d_0) = q_0$, and assume every event is admissible in the sense of Definition 9.3. Then, for every t,

$$q_0 \to_r^* P_r(d_t), \qquad n_r(P_r(d_t)) = n_r(q_0).$$

If the final projection is a monolithic normal form, then

$$P_r(d_m) = n_r(q_0).$$

For every physical observable or observer readout $M : Q_r \to Y$,

$$M(P_r(d_m)) = M(n_r(q_0))$$

whenever $P_r(d_m)$ is normal. In particular, $\widetilde{\mathrm{Read}}_r(d_m) = \mathrm{Read}_r(n_r(q_0))$ for every distributed readout that factors through P_r.

Proof. Induct on t. The claim is true at $t = 0$. Suppose it holds at t. For a linearizable physical commit, admissibility gives

$$P_r(d_t) \to_r^* P_r(d_{t+1}),$$

so $q_0 \to_r^* P_r(d_{t+1})$, and Lemma 9.4 gives

$$n_r(P_r(d_{t+1})) = n_r(P_r(d_t)) = n_r(q_0).$$

For a physical stutter, $P_r(d_{t+1}) = P_r(d_t)$, so both statements are unchanged. For a certified rollback, $P_r(d_{t+1})$ is the projection of an earlier committed state; the induction hypothesis gives reachability from q_0 and equality of normal forms for that projection. This proves the two displayed identities for every t.

If $P_r(d_m)$ is a monolithic normal form, then it is the unique quotient normal form reachable from q_0, hence $P_r(d_m) = n_r(q_0)$. The observable and readout statements follow by applying M or Read_r to this equality and using $\widetilde{\mathrm{Read}}_r = \mathrm{Read}_r \circ P_r$. □

Corollary 9.6 (Partition, schedule, and restart invariance)**.** *Any two admissible distributed executions of the same finite carrier $\mathbf{U}_r$, with the same initial quotient state q_0, have the same terminal quotient observables and observer-readable outputs once their final projections are monolithic normal forms. The statement is independent of partition κ, worker count, scheduler choices, restart history, and repartition metadata.*

Proof. Apply Theorem 9.5 to each execution. Both final projections equal $n_r(q_0)$, so every quotient observable and every readout factoring through the projection has the same value. □

Proposition 9.7 (Restart stabilization separates safety from liveness)**.** *Assume Q_r is finite, every physical commit strictly descends the accepted potential until normality, every rollback returns to an earlier committed projection, only finitely many progress-erasing rollbacks occur, and after the last such rollback the scheduler is fair enough to commit an enabled repair whenever the projected state is not normal. Then the distributed run reaches a monolithic normal form after finitely many physical commits.*

Proof. Safety is Theorem 9.5. For liveness, ignore stutters and the finitely many progress-erasing rollbacks before the last one. After that time, each nonnormal projected state takes a strict descending accepted repair after finitely many scheduler steps. The set of possible potential values below the post-rollback value is finite, so strict descent can occur only finitely many times before a normal projected state is reached. □

Theorem 9.8 (Distributed refinement cube)**.** *Let $r \preceq s$. Suppose $\rho_{sr} : Q_s \to Q_r$ is a repair morphism, the corresponding holonomy maps form a cochain morphism, and distributed presentations at both stages are exact in the sense of Theorem 9.5. If the lifted restriction $\widetilde{\rho}_{sr} : D_s \to D_r$ commutes with physical projection,*

$$P_r \circ \widetilde{\rho}_{sr} = \rho_{sr} \circ P_s,$$

then distributed execution and coarse restriction commute on normal forms and readouts:

$$P_r(\widetilde{\rho}_{sr}(d_m^s)) = n_r(\rho_{sr}(q_0^s)) = \rho_{sr}(n_s(q_0^s))$$

whenever the fine final projection is normal. The same statement holds for holonomy readouts via χ_{sr}.

Proof. The fine distributed theorem gives $P_s(d_m^s) = n_s(q_0^s)$. Projection compatibility gives

$$P_r(\widetilde{\rho}_{sr}(d_m^s)) = \rho_{sr}(n_s(q_0^s)).$$

Theorem 6.5 identifies this value with $n_r(\rho_{sr}(q_0^s))$. The holonomy statement is exactly the cochain-morphism identity of Definition 6.6. □

Certificate contract. A run that claims Theorem 9.5 must let a verifier reconstruct its premises without trusting worker narration: the global carrier and its hash, the monolithic graph, the initial quotient state, the partition map, cut-interface records with restriction maps, a global observer registry, a committed event log with a linearization witness for each physical commit, stutter and rollback records naming committed projection roots, and a final monolithic normal form with its recomputed readout. Missing, stale, shard-local or synthetic replacements for these fields fail closed.

9.1 Quotient chart transport and a neutral readout

The distributed certificate above proves that a worker presentation is one finite quotient system. A separate step is needed before observer rows from different shards may be read as a common neutral geometry.

Definition 9.9 (Common quotient chart atlas)**.** *For a shard s, let Σ_s be raw records, let Γ_s be the groupoid of declared presentation-only moves such as gauge representative changes and local port relabelings, and set*

$$\overline{Q}_s := \Sigma_s/\Gamma_s, \qquad X_s := n_s(\overline{Q}_s),$$

where n_s is the schedule-independent normal-form map. An interface atlas is a family of domains $U_{st} \subseteq X_s$, $U_{ts} \subseteq X_t$, and bijections

$$\tau_{ts} : U_{st} \to U_{ts}$$

satisfying identity, inverse, and cocycle laws, with zero closed-path holonomy on graph-shaped systems. A channel registry assigns each channel c a metric space (Z_c, d_c), weight $w_c > 0$, and local features $F_{s,c} : D_{s,c} \subseteq X_s \to Z_c$. The channel descends through the atlas when both domain membership and values are transported:

$$x \in D_{s,c} \iff \tau_{ts}x \in D_{t,c}, \qquad F_{t,c}(\tau_{ts}x) = F_{s,c}(x).$$

Theorem 9.10 (Quotient-visible neutral readout)**.** *Under Definition 9.9, the relation generated by interface transport on $\bigsqcup_s X_s$ is an equivalence relation and defines*

$$Q_{\mathrm{vis}} := \left(\bigsqcup_s X_s\right) / \sim_\tau .$$

The canonical maps $X_s \to Q_{\mathrm{vis}}$ agree with transport, and zero closed-path holonomy makes them injective. Every compatible family of local channel maps descends uniquely to partial maps $F_c : Q_{\mathrm{vis}} \dashrightarrow Z_c$. Therefore, on the complete channel domain

$$Q_\star = \{x : F_c(x) \text{ exists for every } c \in \mathcal{C}_\star\},$$

fix $p \geq 1$ and assume that $\mathcal{C}_\star$ is finite. More generally, an infinite declared channel set is admitted only if

$$\sum_{c \in \mathcal{C}_\star} w_c\, d_c(F_c(x), F_c(y))^p < \infty \qquad \text{for every } x, y \in Q_\star.$$

The product formula

$$d_{\mathrm{neu}}(x,y) = \left(\sum_{c \in \mathcal{C}_\star} w_c \, d_c(F_c(x), F_c(y))^p \right)^{1/p}$$

is then a finite-valued pseudometric. It is a metric only after quotienting by zero-distance feature collisions, or after proving that the declared channel family separates points.

Proof. Identity, inverse, and cocycle laws give reflexivity, symmetry, and transitivity. The transport compatibility of local features is exactly the universal-property condition for descent through the quotient. The weighted product distance is independent of the representative and finite by the finite-channel or pairwise-summability premise. Its triangle inequality is Minkowski's inequality applied to the channel metrics. The only possible failure of identity of indiscernibles is equality of all declared feature values, which is removed by quotienting feature-collision classes or by a joint-separation proof. □

Pairwise available-channel comparison does not give a metric: different pairs can share different channels and violate the triangle inequality. Presentation invariance requires a bijection of Q_{vis} together with channel isometries, and Euclidean claims require the double-centered Gram test $B = -\frac{1}{2} H(D^{\circ 2}) H \succeq 0$. The section certifies a common quotient metric or pseudometric only; a Riemannian or Lorentzian identification belongs to the spacetime layer.

10 Noisy Fair-Block Approximate Consensus

The exact consensus theorem supplies the quotient normal-form target. A noisy implementation needs one more quantitative certificate: complete fair blocks must contract expected distance to that target. This section records the conditional bridge from local noisy repair to long-run observer-facing approximate consensus.

Theorem 10.1 (Global noisy approximate consensus under fair-block contraction)**.** *Let (Q, d_Q) be the observer-facing quotient state space of a fixed exported OPH patch federation, and let $\mathcal{N} \subset Q$ be the exact quotient normal-form set supplied by the finite repair theorem on that quotient. Define*

$$D(q) := d_Q(q, \mathcal{N}) = \inf_{n \in \mathcal{N}} d_Q(q, n).$$

Let $q_{t+1} = \widetilde{T}_{i_t,t}(q_t)$ be a noisy asynchronous repair process adapted to a filtration $(\mathcal{F}_t)_t$. Assume:

(G1) ***Exact quotient target.*** *The ideal repair relation on Q has the exact OPH normal-form package: finite exact descent, the semantic-complete transactional local-diamond theorem with its concrete premise receipt, and repair completeness, so every fixed initial quotient state has a schedule-independent exact normal form in $\mathcal{N}$.*

(G2) ***Fair asynchronous blocks.*** *There are stopping times*

$$0 = \tau_0 < \tau_1 < \tau_2 < \cdots$$

such that each interval $[\tau_m, \tau_{m+1})$ contains enough local repairs to expose and act on every active mismatch class required by the exact transaction/confluence and repair-completeness certificate, with uniformly bounded length $\tau_{m+1} - \tau_m \le L$. Write the noisy block map as

$$\widetilde{B}_m := \widetilde{T}_{i_{\tau_{m+1}-1}, \tau_{m+1}-1} \circ \cdots \circ \widetilde{T}_{i_{\tau_m}, \tau_m}.$$

(G3) ***Uniform block contraction toward normal form.*** *There are constants $0 < \lambda < 1$ and $\varepsilon \geq 0$ such that, for every block m and every reachable quotient state q at time τ_m,*

$$\mathbb{E}\Big[D\Big(\widetilde{B}_m(q)\Big) \mid \mathcal{F}_{\tau_m}\Big] \leq \lambda D(q) + \varepsilon.$$

(G4) ***Controlled within-block excursions.*** *There are constants $A \geq 1$ and $\beta \geq 0$ such that, for every $t \in [\tau_m, \tau_{m+1})$,*

$$\mathbb{E}[D(q_t) \mid \mathcal{F}_{\tau_m}] \leq A\, D(q_{\tau_m}) + \beta.$$

Then, at fair-block times,

$$\mathbb{E}[D(q_{\tau_m})] \leq \lambda^m D(q_0) + \frac{1-\lambda^m}{1-\lambda}\,\varepsilon,$$

and hence

$$\limsup_{m\to\infty} \mathbb{E}[D(q_{\tau_m})] \leq \frac{\varepsilon}{1-\lambda}.$$

If the block endpoints τ_m are deterministic, then at all intermediate asynchronous times,

$$\limsup_{t\to\infty} \mathbb{E}[D(q_t)] \leq A\,\frac{\varepsilon}{1-\lambda} + \beta.$$

Thus the process has a controlled block-indexed expected tube, and the same certificate gives the displayed calendar-time tube on the deterministic-block branch. On that branch, if $\varepsilon = \beta = 0$, then $D(q_t) \to 0$ in L^1, hence also in probability, along the full asynchronous run.

Proof. Let $D_m := D(q_{\tau_m})$. Assumption (G3), applied to the realized state q_{τ_m}, gives

$$\mathbb{E}[D_{m+1} \mid \mathcal{F}_{\tau_m}] \leq \lambda D_m + \varepsilon.$$

Taking expectations,

$$\mathbb{E}[D_{m+1}] \leq \lambda\, \mathbb{E}[D_m] + \varepsilon.$$

Iterating the scalar recursion gives

$$\mathbb{E}[D_m] \leq \lambda^m D(q_0) + \varepsilon \sum_{r=0}^{m-1} \lambda^r = \lambda^m D(q_0) + \frac{1-\lambda^m}{1-\lambda}\varepsilon.$$

Taking $m \to \infty$ yields the fair-block limsup bound, including when the endpoints are stopping times. For the remaining statements assume that the endpoints are deterministic. Then the index $m = m(t)$ containing a fixed calendar time t is deterministic, and Assumption (G4) gives

$$\mathbb{E}[D(q_t)] \leq A\, \mathbb{E}[D(q_{\tau_m})] + \beta.$$

Substitution of the fair-block estimate and then taking the limsup over all intermediate times gives

$$\limsup_{t\to\infty} \mathbb{E}[D(q_t)] \leq A\,\frac{\varepsilon}{1-\lambda} + \beta.$$

If $\varepsilon = \beta = 0$, then $\mathbb{E}[D(q_{\tau_m})] \leq \lambda^m D(q_0) \to 0$. Since $D \geq 0$, convergence in expectation to zero implies convergence in probability at block times. Assumption (G4) then gives $\mathbb{E}[D(q_t)] \leq A\, \mathbb{E}[D(q_{\tau_m})]$ inside each block, so the same L^1 and probability convergence holds along the full asynchronous run. □

Remark 10.2 (Stopping-time blocks can bias a calendar-time readout). The deterministic-endpoint condition cannot be omitted from the displayed calendar-time constant. For an exact finite counterexample, let the block-endpoint distance be a two-state chain $D_m \in \{0,1\}$, with

$$\Pr(D_{m+1}=1 \mid D_m=0)=\tfrac{1}{4}, \qquad \Pr(D_{m+1}=1 \mid D_m=1)=\tfrac{3}{4}.$$

It obeys (G3) with $\lambda=1/2$, $\varepsilon=1/4$, and its stationary mean is $1/2$. Keep the distance constant within each block, assigning length one to a zero block and length two to a unit block. Thus $L=2$, $A=1$, and $\beta=0$, even pathwise inside each block. The calendar-time chain has three states $0, 1_{\text{first}}, 1_{\text{second}}$, with transition matrix

$$\begin{pmatrix} 3/4 & 1/4 & 0 \\ 0 & 0 & 1 \\ 1/4 & 3/4 & 0 \end{pmatrix}.$$

Its unique stationary law is uniform, so its limiting expected distance is $2/3 > \varepsilon/(1-\lambda) = 1/2$. Sampling the block containing a fixed time favors longer blocks. A random containing-block index therefore cannot be substituted into the estimate for a fixed block index. Random block endpoints need their own occupation-time or conditional sampling bound to obtain a calendar-time constant.

Corollary 10.3 (Approximate observer-facing schedule independence)**.** *Let $M : Q \to Y$ be an observer-facing readout into a metric space (Y, d_Y), and suppose M is L_M-Lipschitz. Fix an exact sector ζ whose exact normal-form set is the singleton $\mathcal{N}_\zeta = \{n_\zeta\}$, and apply Theorem 10.1 on that sector, so that $D(q) = d_Q(q, n_\zeta)$, with deterministic block endpoints. Then*

$$\limsup_{t\to\infty} \mathbb{E}[d_Y(M(q_t), M(n_\zeta))] \le L_M \left(A\,\frac{\varepsilon}{1-\lambda} + \beta \right).$$

For two noisy asynchronous schedules q_t and q'_t started in the same exact singleton sector and satisfying the same certificate,

$$\limsup_{t\to\infty} \mathbb{E}[d_Y(M(q_t), M(q'_t))] \le 2L_M \left(A\,\frac{\varepsilon}{1-\lambda} + \beta \right).$$

Proof. The Lipschitz condition gives

$$d_Y(M(q_t), M(n_\zeta)) \le L_M d_Q(q_t, n_\zeta) = L_M D(q_t).$$

The first bound follows from Theorem 10.1. The two-schedule bound follows from the triangle inequality through $M(n_\zeta)$ and applying the first estimate to both schedules. □

Corollary 10.4 (High-probability noisy consensus tube)**.** *Assume the block-distance process also admits the pathwise decomposition*

$$D_{m+1} \le \lambda D_m + \varepsilon + \xi_{m+1}, \qquad \mathbb{E}[\xi_{m+1} \mid \mathcal{F}_{\tau_m}] = 0, \qquad |\xi_{m+1}| \le b$$

almost surely. Then, for every $a > 0$,

$$\Pr\left[D_m > \lambda^m D_0 + \frac{1-\lambda^m}{1-\lambda}\varepsilon + a \right] \le \exp\left(-\frac{a^2(1-\lambda^2)}{2b^2} \right).$$

Proof. Unrolling the recursion gives

$$D_m \le \lambda^m D_0 + \frac{1-\lambda^m}{1-\lambda}\varepsilon + \sum_{r=1}^{m} \lambda^{m-r}\xi_r.$$

If $b = 0$, the noise vanishes almost surely and the displayed exceedance probability is zero; the exponential bound is understood in that limiting sense. Suppose henceforth that $b > 0$. The final term is a weighted martingale sum with increments bounded by $|\lambda^{m-r}\xi_r| \le \lambda^{m-r} b$. Azuma–Hoeffding gives

$$\Pr\left[\sum_{r=1}^{m} \lambda^{m-r}\xi_r > a\right] \le \exp\left(-\frac{a^2}{2\sum_{r=1}^{m} \lambda^{2(m-r)} b^2}\right).$$

Since $\sum_{r=1}^{m} \lambda^{2(m-r)} \le (1-\lambda^2)^{-1}$, substitution yields the claimed bound. □

Proposition 10.5 (Finite fair-block contraction certificate)**.** *Let Q be finite, let $\mathfrak{B}_{\mathrm{fair}}$ be a finite list of noisy fair-block types, and let $K_B(q, q')$ be the Markov kernel induced by block type B. Require that the type B_m is chosen at block start and is $\mathcal{F}_{\tau_m}$-measurable, and that the implemented block has the certified conditional law*

$$\Pr(q_{\tau_{m+1}} = q' \mid \mathcal{F}_{\tau_m}) = K_{B_m}(q_{\tau_m}, q') \quad \textit{almost surely.}$$

Randomized choices may be included in the block-start filtration, provided this conditional-law identity remains valid. If there are constants $0 < \lambda < 1$ and $\varepsilon \ge 0$ such that

$$\sum_{q' \in Q} K_B(q, q') D(q') \le \lambda D(q) + \varepsilon \qquad \forall q \in Q, \quad \forall B \in \mathfrak{B}_{\mathrm{fair}},$$

then Assumption (G3) of Theorem 10.1 holds for any run whose fair blocks satisfy this implementation contract.

Proof. By the certified conditional-law identity, conditioning on the entire block-start filtration (including the chosen type B), the conditional expectation of D after the block is exactly $\sum_{q'} K_B(q, q') D(q')$. The displayed inequality is therefore Assumption (G3), uniformly over all reachable states and fair-block types. □

Selecting a block type using its as-yet-unobserved output does not meet this contract. For example, for one fair bit U, the two maps with outputs U and $1 - U$ each have expected distance $1/2$. Choosing the map whose output is one gives expected distance one. Their marginal kernels therefore do not certify that selected implementation. Adaptation inside a block is permitted only when its complete adapted law is the certified kernel.

Finite certificate route and constants. For a finite exported packet net, Proposition 10.5 gives the route from the finite quotient Q and its exact normal forms $\mathcal{N}$ to a distance table $D(q)$, noisy fair-block kernels K_B and a (λ, ε) certificate. Here λ is the net contraction of one completed fair block, ε the per-block irreducible recovery, record, readout, calibration and environmental noise, A and β bound within-block expansion and noise, and L is the fairness horizon. In collar implementations ε may absorb Petz-domain truncation, Fawzi–Renner recovery error, approximate central-record error, detector noise and coarse-graining defect. The theorem does not follow from the exact finite repair theorem alone, which supplies only the target $\mathcal{N}$.

11 Interfaces to the higher layers

The layer supplies five typed outputs, each consumed under the receiving layer's own premises.

1. **Public normal form and records.** The quotient normal form, its boundary data and the central record algebra of Theorem 7.2 are the public world that the observer layer reads, retains and acts on.

2. **Holonomy residue.** Cycle and crossed-module holonomy classes, with their refinement-limit classes, are the invariants on which the matter hypothesis builds defects (Section 4.1).

3. **Implementation hiding.** The quotient by changes that preserve overlap data, with its fixed-cutoff edge sectors, is the input of the gauge layer (Section 5.2).

4. **Informational order.** Authenticated provenance generates a finite order and its source height. The spacetime layer attaches causal, metric and clock meaning to it only through placement, count–volume, dimension and calibration premises [32].

5. **Refinement and implementation control.** Natural restriction systems, controlled coarse-graining squares, distributed presentations and fair-block certificates relate finite stages and implementations to one quotient object.

Approximate control is collar-local: Theorem A.1 gives the exact-Markov modulus and the one-shot recovery bound, and Theorem 7.2 the record-stability bound. Long-run noisy control requires a fair-block certificate; Ref. [10] adds a receipt-only endpoint estimate from residual and inverse-observation moduli, which supplies no fair-block contraction. The protocol and code extensions of Appendix C are separate from the core theorems: a Byzantine-agreement reading needs its certificate, quorum and timing contract, and a quantum-code reading needs a code subspace, error family, recovery map and, for distance claims, a topological-code certificate.

A Quantum/Algebraic Lift: Markov-Collar Splice Theorem

This appendix records the algebraic splice statement relating the finite patch-net model to the OPH collar formalism.

For this collar lemma, write the support-local algebra-state-record reduct of an observer patch as

$$O_{\mathrm{red}} = (P, \mathcal{A}(P), \rho, R),$$

where P is the support-screen patch, $\mathcal{A}(P)$ the local von Neumann algebra, ρ the local state, and R the record algebra. The full operational observer also carries overlap interface algebras and restriction maps, allowed update and repair instruments, and checkpoint data used for continuation.

Theorem A.1 (Markov-collar splice theorem, exact and controlled)**.** *Suppose a collar tripartition A-B-D has the EC-aligned exact Markov decomposition*

$$\rho_{ABD} = \bigoplus_{\alpha} p_\alpha \, \rho^{(\alpha)}_{Ab^\alpha_L} \otimes \rho^{(\alpha)}_{b^\alpha_R D}$$

over the preselected edge factors. This displayed form is the hypothesis: by the spacetime and Einstein paper's Markov-split alignment analysis it is strictly stronger than $I(A : D \mid B)_\rho = 0$,

which by HJPW yields such a factorization only over a state-dependent split of B. Let $\sigma^{(\alpha)}_{b^\alpha_R D'}$ be any family of normalized environment states compatible with the same right-boundary sectors. Define

$$\rho'_{ABD'} = \bigoplus_\alpha p_\alpha \, \rho^{(\alpha)}_{Ab^\alpha_L} \otimes \sigma^{(\alpha)}_{b^\alpha_R D'}.$$

Then for every observable X supported on $A \cup b_L$,

$$\mathrm{Tr}(X\rho'_{ABD'}) = \mathrm{Tr}(X\rho_{ABD}).$$

Fix one finite-dimensional collar model and let

$$\mathfrak{M}_{A:B:D} := \{\tau_{ABD} : I(A:D \mid B)_\tau = 0\},$$

with exact-Markov distance modulus

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) := \sup\left\{ \inf_{\tau \in \mathfrak{M}_{A:B:D}} \|\omega - \tau\|_1 : I(A:D \mid B)_\omega \le \varepsilon \right\}.$$

Then

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0 \qquad (\varepsilon \downarrow 0).$$

On a fixed faithful collar class with lower floor $\lambda_* > 0$, this qualitative modulus can be sharpened to a collar-local rate

$$\delta^{\mathrm{M},\lambda_*}_{A:B:D}(\varepsilon) \le C_{A:B:D,\lambda_*} \, \varepsilon^{\theta_{A:B:D,\lambda_*}},$$

by the compact real-analytic Lojasiewicz inequality applied to $I(A:D \mid B)$. The constants depend on the fixed collar model and floor; they are not a dimension-free stability theorem for arbitrary tripartite systems. Let $\mathfrak{M}^{\mathrm{EC}}_{A:B:D} \subset \mathfrak{M}_{A:B:D}$ be the states that also satisfy the Markov-split alignment hypothesis on the preselected edge factors. For a declared controlled family $\mathcal{F}$, define

$$\delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon) := \sup\left\{ \inf_{\tau \in \mathfrak{M}^{\mathrm{EC}}_{A:B:D}} \|\omega - \tau\|_1 : \omega \in \mathcal{F},\ I(A:D \mid B)_\omega \le \varepsilon \right\}.$$

Convergence of the generic exact-Markov modulus does not imply convergence of this aligned modulus. Hence, if $\omega \in \mathcal{F}$, $I(A:D \mid B)_\omega \le \varepsilon$, and the controlled family separately satisfies $\delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon) \to 0$, choose $\widetilde{\omega}_\varepsilon \in \mathfrak{M}^{\mathrm{EC}}_{A:B:D}$ so that

$$\|\omega - \widetilde{\omega}_\varepsilon\|_1 \le \delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon),$$

the corresponding exact splice $\widetilde{\omega}'_\varepsilon$ satisfies

$$\left|\mathrm{Tr}(X\omega) - \mathrm{Tr}(X\widetilde{\omega}'_\varepsilon)\right| \le \|X\|_\infty \, \delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon)$$

for every observable X supported on $A \cup b_L$.

Independently, if $I(A:D \mid B)_\omega \le \varepsilon$, then there exists a recovery map $\mathcal{R}_{B\to BD}$ such that

$$\|\omega_{ABD} - (\mathrm{id}_A \otimes \mathcal{R}_{B\to BD})(\omega_{AB})\|_1 \le 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

Proof. The exact splice statement is the usual blockwise factorization argument:

$$\mathrm{Tr}(X\rho'_{ABD'}) = \sum_\alpha p_\alpha \, \mathrm{Tr}\Big(X \, \rho^{(\alpha)}_{Ab^\alpha_L}\Big) \, \mathrm{Tr}\Big(\sigma^{(\alpha)}_{b^\alpha_R D'}\Big).$$

Each right factor is normalized, so the value agrees with the same computation for ρ_{ABD}.

For the controlled statement, compactness of the fixed finite-dimensional state space and continuity of conditional mutual information imply $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0$: otherwise one could find a sequence with $I(A:D \mid B) \to 0$ staying a fixed trace distance away from every exact Markov state, contradicting convergence of a subsequence to an exact Markov limit point. This generic fact does not prove the separately assumed aligned-modulus convergence: the splice identity requires $\widetilde{\omega}_\varepsilon$ to be EC-aligned over the preselected edge factors, while small conditional mutual information supplies closeness only to the full exact Markov set. Once such an EC-aligned $\widetilde{\omega}_\varepsilon$ is chosen with the declared family-specific bound, the exact splice identity gives

$$\left|\mathrm{Tr}(X\omega) - \mathrm{Tr}(X\widetilde{\omega}'_\varepsilon)\right| = \left|\mathrm{Tr}[X(\omega - \widetilde{\omega}_\varepsilon)]\right| \le \|X\|_\infty \, \delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon).$$

The final inequality is trace/operator-norm duality applied to the separately supplied EC-aligned modulus. It is not the Fawzi–Renner recovery bound [31], which guarantees a recovered comparison state rather than this EC-aligned exact Markov state. □

Exact splice identities are used in two regimes only: literal EC-aligned exact Markovity, or a controlled collar family on one fixed finite-dimensional model with $\delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon) \to 0$. Fawzi–Renner recovery supplies a recovered comparison state, and the generic fixed-collar modulus supplies an exact-Markov comparison on a possibly different split.

For a faithful finite state ρ_{ABD} with $K_X = -\log \rho_X$, the exact finite identity

$$\mathrm{Tr}\, \rho_{ABD}(K_{AB} + K_{BD} - K_B - K_{ABD}) = I_\rho(A:D \mid B)$$

is an expectation identity only; it does not make conditional mutual information a state-independent local stress source. Near a full-rank exact Markov state σ, a perturbation $\rho(t) = \sigma + tX + O(t^2)$ has $\frac{\mathrm{d}}{\mathrm{d}t} I_{\rho(t)}(A:D \mid B)|_{t=0} = 0$, while a fixed-reference modular-energy variation is generically linear in t. Recovery bounds therefore prove no local stress-source theorem.

B Fixed-Cutoff Realization, Quotient Repair, and Edge Centers

This appendix makes the fixed-cutoff realization and the collar boundary data explicit at the finite patch-net level. On the collar branch the local repair step is read from exact Markov splice or a declared Petz/Fawzi–Renner recovery move; the recovery move gives a recovered comparison state, while exact splice requires exact Markovity or a controlled fixed-collar replacement modulus. Representative repair maps are lifts of the resulting quotient-local update.

B.1 Quotient Repair and UV Underdetermination

At fixed cutoff, each regulator cell x carries a finite-dimensional factor $\mathfrak{h}_x$, patch algebras are finite type-I algebras, and gauge-as-gluing is realized as a compact boundary redundancy action on cut data. The physical repair law therefore belongs on the overlap-invariant quotient rather than on hidden representatives. If $q : \Sigma \to \Sigma/\Gamma$ is the quotient by boundary redundancy and $\overline{T}_i$ is the physical quotient update, a representative-level map T_i is only required to be a lift satisfying

$$q \circ T_i = \overline{T}_i \circ q.$$

Hence $q(T_i(\gamma \cdot s)) = q(T_i(s))$ for gauge-equivalent inputs: quotient descent is structural, and representative-level covariance is bookkeeping. The accepted recovery-derived moves must satisfy the repair-completeness, support-local disjoint-commutation, nested-collar restriction-compatibility and Petz-domain clauses on the declared branch.

Proposition B.1 (Ancilla-stable UV underdetermination)**.** *Let a fixed-cutoff OPH realization be stabilized by finite ancillary factors K_P in a fixed product state, with observable patch algebras embedded as $\mathcal{A}(P) \otimes \mathbf{1}_{K_P}$ and repair dynamics acting trivially on the ancillas. Then observable expectations on the physical subalgebras, overlap data, the local-Gibbs branch, the collar conditional mutual information $I(A : D \mid B)$, the Fawzi–Renner remainder, the collar Markov modulus, and the quotient normal form are unchanged. Thus the fixed-cutoff theorem package determines the UV branch only modulo such ancillary stabilization together with gauge or implementation hiding, not a unique microscopic presentation.*

Proof. Product ancillas leave physical observables unchanged, cancel additively inside conditional mutual information, and are inert under the repair maps. Hence every invariant listed above is unchanged. □

B.2 Derived Boundary Data and Ordinary EC

Proposition B.2 (Derived boundary gluing datum)**.** *Choose a finite regulator chart for the patches meeting along a connected cut Σ. Because the local overlap algebras are finite-dimensional matrix algebras, any overlap-consistent recharting is an inner automorphism and is implemented by a unitary on the cut Hilbert space. The compact closure of the subgroup generated by these recharting unitaries is a compact boundary redundancy group K_Σ. If triple-overlap defects are central, the projective composition law lifts to a compact central extension $\widehat{K}_\Sigma$; on the ordinary branch one simply sets $\widehat{K}_\Sigma = K_\Sigma$. A genuinely noncentral 2-group defect is the only obstruction to reducing the overlap transition system to an ordinary compact group action.*

Theorem B.3 (Derived EC decomposition)**.** *Under the fixed-cutoff regulator realization above, and on the ordinary or central-defect branch, the collar Hilbert space is*

$$\mathcal{H}_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\widehat{K}_\Sigma} \cong \bigoplus_\alpha \left(\mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\alpha} \right),$$

and the center of the collar algebra is generated by the block projectors:

$$Z(\mathcal{A}(B_\delta)) = \bigoplus_\alpha \mathbb{C} \cdot \mathbf{1}_\alpha.$$

The right half-collar carries the contragredient representation because it sees inverse transport across the same cut.

This is the finite patch-net origin of the collar center used by the Markov, record and observer constructions. EC provides the kinematic block structure; exact Markovity is an additional state hypothesis.

B.3 Higher-Gauge Replacement on the Genuinely Noncentral Branch

Proposition B.4 (Derived higher-gauge cut datum)**.** *On the genuinely noncentral branch, weak overlap gluing on a connected cut Σ is encoded by a compact crossed module*

$$\mathbb{K}_\Sigma = (H_\Sigma \xrightarrow{\partial_\Sigma} G_\Sigma, \triangleright)$$

with defect class

$$q_\Sigma \in \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma),$$

and compact higher-gauge change system

$$\mathcal{T}_\Sigma = C^1(N_\Sigma, H_\Sigma) \rtimes C^0(N_\Sigma, G_\Sigma).$$

Theorem B.5 (Higher-gauge EC decomposition and defect transport)**.** *On the genuinely noncentral branch,*

$$\mathcal{H}_{B_\delta}^{2g} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\mathcal{T}_\Sigma} \cong \bigoplus_\lambda (\mathcal{H}_{b_L^\lambda} \otimes \mathcal{H}_{b_R^\lambda}),$$

and

$$Z(\mathcal{A}_{2g}(B_\delta)) = \bigoplus_\lambda \mathbb{C} \cdot \mathbf{1}_\lambda.$$

The full crossed-module orbit q_Σ is also invariant under local rechartings and classifies fixed-cutoff genuinely noncentral gluing data. The higher associator is removable iff q_Σ lies in the image of

$$\check{H}^1(N_\Sigma, G_\Sigma) \longrightarrow \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma), \qquad [g] \longmapsto [(g, 1)].$$

That map need not be injective. Strict endpoint-only ordinary transport additionally requires at least one allowed strict representative with trivial represented loop holonomy.

Corollary B.6 (Exact Markov plus split alignment adds the state factorization)**.** *On either the ordinary/central branch described in the main consensus theorem or the genuinely noncentral higher-gauge branch, if in addition*

$$I_\omega(A_\delta : D_\delta \mid B_\delta) = 0$$

and *the state satisfies the Markov-split alignment hypothesis (its HJPW decomposition of $\mathcal{H}_{B_\delta}$ can be chosen to be the EC decomposition itself; see the spacetime and Einstein paper's Markov-split alignment definition), or one passes to the explicitly stated idealized recoverability limit that supplies both conditions, then*

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_\alpha p_\alpha \left(\rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta} \right).$$

EC therefore gives the kinematic block decomposition, while exact Markovity plus split alignment is the extra state input that gives the EC-aligned HJPW normal form. Exact Markovity alone yields the normal form only over a state-dependent HJPW split, which the spacetime and Einstein paper's Bell-pair example shows can be transposed relative to the EC factors.

Proposition B.7 (Certified parenthesization-invariant union-collar gluing)**.** *Fix a finite union collar U built from two overlapping local repair collars on the declared fixed-cutoff branch. Assume one of the following payload receipts:*

(i) *an exact aligned Markov construction, including the HJPW split-alignment condition, whose nested collar recovery maps satisfy the commuting-square and pentagon identities;*

(ii) *a specified canonical recovery construction, such as a fixed Petz/Markov splice, together with those commuting-square and pentagon identities; or*

(iii) *a primitive aggregate state ω_U whose restrictions to every constituent and nested collar equal the declared local payloads.*

On the ordinary or central-defect branch, assume in addition that changes of representative act through the declared boundary-redundancy action. On the genuinely noncentral branch, assume the corresponding coherence identities hold in the crossed-module change system $\mathcal{T}_\Sigma$. Then the physical quotient-local glued state on U is independent of parenthesization.

Proof. In cases (i) and (ii), the commuting-square identities identify every two-step restriction and the pentagon identity identifies all iterated parenthesizations. In case (iii), every parenthesization is a restriction of the same primitive state ω_U. The additional quotient clause removes only the declared boundary representative action or its crossed-module analogue, so all parenthesizations define one quotient-local state. □

Remark B.8 (Marginals and sectors do not determine the aggregate state). Central sector labels and compatible proper marginals do not imply any of the three payload receipts. The uniform even-parity and odd-parity distributions on three bits have identical one- and two-bit marginals and different tripartite states. The states

$$|\mathrm{GHZ}_{\pm}\rangle = \frac{|000\rangle \pm |111\rangle}{\sqrt{2}}$$

give the quantum counterpart: all two-party reductions agree while the global states differ. Approximate recoverability therefore carries a coherence defect in addition to its local recovery error.

Corollary B.9 (Physical observables are invariant on one quotient-local glued state)**.** *Let U be a finite union collar on the declared fixed-cutoff branch, and let ω_U, ω'_U be two microscopic representatives of the same quotient-local glued state from Proposition B.7. Then every physical observable X on the collar fixed-point / quotient-local algebra has the same expectation in both representatives:*

$$\mathrm{Tr}(X\omega_U) = \mathrm{Tr}(X\omega'_U).$$

In particular the same holds for the central sector projectors and for any observer-accessible record observable generated from them on that same declared surface.

Proof. On the ordinary or central-defect branch, Proposition B.7 says the two representatives differ only by the boundary-redundancy action inside one fixed sector block. The fixed-point collar algebra and its central block projectors are invariant under that action, so their expectation values agree. On the genuinely noncentral branch, the same proposition says the two representatives differ only inside one $\mathcal{T}_\Sigma$-orbit, and the quotient-local physical algebra $\mathcal{A}_{\mathrm{phys}}(U)$ of Definition 5.15 is defined precisely on that orbit space. Therefore the induced physical state and all expectations of physical observables agree there as well. □

C Protocol and Code Extensions

The core theorems are finite rewriting statements. Stronger readings of the same formalism need their own certificates; the table lists them.

Desired statement	Required certificate
Overlap network is a code	finite constraint-code data of Proposition 2.3
Global repair is a physical normal-form map	$\mathrm{Rep}_\lambda = \overline{\mathrm{nf}}_\lambda : Q \to Q$, total, idempotent, schedule-independent, landing in C_Q
Repair converges	finite exact descent; confluence additionally needs semantic-complete transactions, coherent canonical aggregate gluing and repair completeness
Distance equals min-cut	topological-code certificate with homological logicals and matching systole/min-cut geometry
Corrects t corrupted carriers	code projector, error family, Knill–Laflamme condition and certified $t < d/2$ distance bound
Exponential convergence	declared transfer or channel operator with stationary projection and spectral gap
Long-run noisy approximate consensus	fair-block contraction certificate $(\lambda, \varepsilon, A, \beta, L)$; deterministic block endpoints for the calendar-time constant
Cross-view BFT safety and liveness	the certificate, lock, quorum, authentication and timing contract of Definition C.1 and (A1)–(A6)

Byzantine agreement under a prepared-lock contract

The Byzantine fault model is that of Lamport, Shostak and Pease [21].

Definition C.1 (QBFT-style protocol)**.** *A consensus protocol is* QBFT-style *in this analysis if it satisfies the following five structural properties. Same-view safety uses (P1)–(P2), cross-view safety additionally uses the lock and monotone-view parts of (P1) and (P4), and the liveness clause uses (P3)–(P5) together with the timing and connectivity assumptions below.*

(P1) ***One-vote-per-view.*** *Each nonfaulty node supports at most one value per view number. If the protocol has prepare, prepared-certificate acceptance, and commit phases, its messages in all three phases refer to that same value. A node that has supported a value in view v ignores any later conflicting request in view v. View participation is monotone: after entering view w, a nonfaulty node never signs a prepare, prepared-certificate acceptance, or commit message for $v < w$, and never adopts a durable lower-view lock.*

(P2) ***Certificate semantics.*** *A raw prepared certificate* $\mathrm{PC}(v, x)$ *contains q distinct, unforgeable, authenticated prepare votes for value x in view v. A lock certificate* $\mathrm{LC}(v, x)$ *contains that* $\mathrm{PC}(v, x)$ *together with q distinct authenticated acceptance acknowledgements. These acknowledgements are provisional: they do not change the acceptor's voting lock. A raw prepared certificate without q acceptance acknowledgements cannot justify a new view or a commit vote. A nonfaulty validator that receives the assembled* $\mathrm{LC}(v, x)$ *verifies it and records the durable lock (v, x) before signing a commit vote. A decision certificate contains q distinct authenticated commit votes, each referring to the same valid* $\mathrm{LC}(v, x)$*. In the classical exact-size case $q = 2f + 1$. For larger validator sets at fixed fault budget, the threshold must scale so that (A6) holds. A nonfaulty observer finalises a value only after accepting such a valid decision certificate.*

(P3) ***DLS-style view-change.*** *If no certificate is produced within a timeout, every nonfaulty node increments the view number by one and a new leader is selected by a fixed fair deterministic rule. Once the post-GST timeout bound of (A1) is active, the pacemaker brings all nonfaulty nodes through successive views and reaches a nonfaulty leader within at most $f + 1$ consecutive views.*

(P4) ***Prepared-certificate dissemination, commit lock, and justified new view.*** *The assembler disseminates a valid* $\mathrm{PC}(v, x)$ *and obtains the* q *authenticated acceptance acknowledgements of* $\mathrm{LC}(v, x)$*, then disseminates the assembled lock certificate before collecting commits. Each validator's authenticated view-change message carries its highest valid lock certificate, if any. A new leader collects* q *such messages. If it issues a proposal, that proposal carries the value of their highest-view lock certificate; it may carry a fresh value only if none is present. The proposal carries the complete new-view justification. Every nonfaulty validator verifies the signatures, quorum size, and highest-certificate selection. An unlocked validator may support a fresh proposal only when the verified justification contains no lock certificate. Otherwise it votes only for its current lock value, except that the value selected by a valid* q*-message new-view justification supersedes its local assembled-lock-certificate lock. This authorizes the prepare vote without adopting a lower-view durable lock; receipt of the assembled current-view lock certificate records the replacement before commit. This rule applies without a local test for whether an earlier commit set reached* q*: it reconciles every predecision lock, while the proof below shows that quorum intersection makes the selected value compatible whenever a decision certificate exists.*

(P5) ***Post-GST nonfaulty progress.*** *In a sufficiently long post-GST view led by a nonfaulty node, that leader collects the required new-view messages and emits the proposal selected by (P4). Every nonfaulty validator that receives a valid proposal promptly sends its prepare vote. The nonfaulty leader disseminates the resulting valid prepared certificate; every nonfaulty validator verifies it and returns a provisional acceptance acknowledgement. After assembling the lock certificate, the leader disseminates it; every nonfaulty recipient verifies it, records or advances its durable lock, and sends its commit vote. After accepting a valid decision certificate it finalises and relays that certificate. Each of these fixed protocol phases completes within* $O(\Theta)$*, with* Θ *defined in (A1).*

The Istanbul BFT / QBFT protocol family [23, 24] motivates this pattern; (P1)–(P5), including their exact certificate, pacemaker, lock, and progress semantics, must be checked against an implementation rather than inferred from the family name.

Assumptions A1–A6.

(A1) **Partial synchrony [22].** Fixed but initially unknown bounds Δ (per-edge message delay) and Φ (local processing time) hold after GST. Together with the certified routes of (A4), they supply an effective end-to-end protocol-phase bound Θ that includes route length, processing, and the fixed number of message steps in one phase. Pacemaker timeouts eventually exceed the corresponding per-view bound. *Safety* holds without extra timing assumptions; *liveness* holds after the Global Stabilisation Time (GST).

(B2) **Byzantine fault model.** At most f observers behave arbitrarily; the remaining $n - f$ are nonfaulty.

(C3) **Classical exact sizing for this fixed-quorum theorem.** $n = 3f + 1$, so that $q = 2f + 1$ certificates have a nonfaulty overlap witness. The usual resilience condition $n \geq 3f + 1$ is necessary for the fault model, but when $n > 3f + 1$ a fixed $q = 2f + 1$ quorum is insufficient for the overlap used in the safety proof; one must either impose (A6) directly or choose a threshold q with $2q \geq n + f + 1$.

(D4) **Strong quorum connectivity.** Every quorum Q with $|Q| = q$ is strongly connected within G: for any $u, v \in Q$ there is a directed path in G contained entirely in Q. This is strictly stronger than requiring the overlap graph of quorums to be connected, and is needed to propagate signed votes within a quorum.

(E5) **Message authentication.** All messages carry unforgeable digital signatures.

(F6) **OPH quorum overlap.** Any two quorums Q_a, Q_b of size q satisfy $|Q_a \cap Q_b| \geq f + 1$. For $q = 2f + 1$ this is guaranteed by the exact sizing $n = 3f + 1$ in (A3); for general n it is the separate threshold condition $2q \geq n + f + 1$, not a consequence of $n \geq 3f + 1$ alone. The liveness clause additionally requires $q \leq n - f$, so a certificate can be formed without Byzantine participation.

Theorem C.2 (QBFT safety and liveness)**.** *Under (A1), (A2), and (A4)–(A6), any consensus protocol satisfying (P1)–(P5) of Definition C.1 and run over the OPH observer graph satisfies:*

(i) ***Cross-view safety.*** *No two nonfaulty observers finalise conflicting patch states, whether in the same view or in different views.*

(ii) ***Liveness.*** *From the first post-GST view in which the pacemaker timeout bound is active, every nonfaulty observer finalises within $O((f+1)\Theta)$ wall-clock time, provided $q \leq n - f$. No bound from GST to timeout activation is claimed.*

(iii) ***Certificate-contract threshold.*** *Joint feasibility of the safety threshold $2q \geq n + f + 1$ and Byzantine-independent liveness threshold $q \leq n - f$ forces $n \geq 3f + 1$; the exact-size choice $n = 3f + 1, q = 2f + 1$ attains both.*

Sizing. Assumption (A3) is the classical exact-size way to discharge (A6): when $n = 3f + 1$ and $q = 2f + 1$ the required overlap is automatic, and for general n (A6) must be checked directly. The prepared-lock rule proves safety only for the protocol just defined, never from the name "QBFT." Under these assumptions no two nonfaulty observers finalise conflicting patch states.

Proof. Same-view safety. Suppose O_a and O_b finalise $s_a \neq s_b$ in one view. By (P2), each required a certificate of q votes: sets Q_a, Q_b. By (A6), $|Q_a \cap Q_b| \geq f + 1$. In the classical exact-size case this is the inclusion-exclusion calculation $|Q_a \cap Q_b| \geq (2f + 1) + (2f + 1) - (3f + 1) = f + 1$. By (A2), at most f are Byzantine, so $Q_a \cap Q_b$ contains a nonfaulty O^*. By (P1), O^* voted for at most one value. Contradiction. The identical overlap argument applies to two prepared certificates in the same view, using their prepare-vote quorums.

Cross-view safety. Suppose $\mathrm{DC}(v, x)$ and $\mathrm{DC}(w, y)$ are conflicting decision certificates with $v < w$. The first certificate refers to a valid $\mathrm{LC}(v, x)$. Let C_x be its q committers. The nonfaulty subset $L_x \subseteq C_x$ has size at least $q - f$, and every member of L_x received the assembled lock certificate and recorded (v, x) before committing. Every size-q prepare or view-change quorum P intersects that actual lock set because

$$|L_x \cap P| \geq (q - f) + q - n = 2q - n - f \geq 1$$

by (A6).

Assume for contradiction that a raw prepared certificate for a value different from x exists above view v, and choose one with least view $r > v$. Its proposal carries a valid q-message new-view justification J_r. Since $|L_x| \geq q - f$ and $|J_r| = q$, choose a nonfaulty $O^* \in L_x \cap J_r$. Validator O^* signed a commit for $\mathrm{LC}(v, x)$, and later sent its authenticated view-r change. Monotone view

participation in (P1) orders its view-v lock and commit before that view-r message: it cannot send a view-r change and subsequently return to commit in view v. Its view-change message therefore reports $\mathrm{LC}(v, x)$, or a later valid lock certificate. By minimality of r, every raw prepared certificate underlying such a later reported lock certificate below view r carries x. Hence O^*'s report carries x.

The justification J_r therefore cannot be certificate-free. It also cannot contain a conflicting highest lock certificate, because that certificate contains a conflicting raw prepared certificate in a view below r, contradicting the minimality of r. Rule (P4) consequently selects x, so nonfaulty validators do not prepare the alleged conflicting value in view r. Thus no higher conflicting prepared certificate exists, and (P2) excludes the alleged $\mathrm{DC}(w, y)$.

This induction is also the lock-transfer invariant for view change. Every valid q-message new-view justification after a decision contains a nonfaulty report from the decision's $q - f$ commit-lock set, and its highest valid report carries the decided value.

Liveness and certificate threshold. From the first post-GST view whose pacemaker timeout is active, (A1) makes every failed-view timeout and every fixed protocol phase $O(\Theta)$. Rule (P3) reaches a nonfaulty leader within at most $f + 1$ consecutive views. Since $q \leq n - f$, nonfaulty validators can supply all q new-view, prepare, prepared-certificate acceptance, and commit messages without Byzantine participation.

Consider the valid q-message justification used by the successful new view. If it contains no lock certificate, (P4) selects a fresh value. Otherwise it selects its unique highest-view lock certificate; same-view uniqueness follows from the prepare-quorum overlap. Every nonfaulty validator may replace a local assembled-lock-certificate predecision lock with that valid selection, including a lock whose local commit set never reached a decision certificate. If a decision certificate exists, the $q - f$ nonfaulty commit-lock set proved above intersects the q-message justification. Its report forces the selected lock certificate to carry the decided value, since the safety induction excludes a conflicting certificate. Thus at least q nonfaulty validators can support the selected proposal in either case, even when some of their orphan locks were omitted from the new-view messages.

Rule (P5) executes the proposal, prepare, provisional acceptance, lock-certificate dissemination, commit, and decision-relay phases, so every nonfaulty observer finalises within $O((f + 1)\Theta)$ from that activation point. Eventual timeout activation yields eventual liveness, but the theorem does not bound its delay after GST.

Finally, $q \leq n - f$ and $2q \geq n + f + 1$ imply

$$2(n - f) \geq n + f + 1,$$

hence $n \geq 3f + 1$ (with the integer convention specified in (A6)). At $n = 3f + 1$, $q = 2f + 1$ attains both inequalities. This is a threshold statement for the declared certificate contract, not a universal resilience claim for every authenticated protocol model.

Note on FLP. Fischer, Lynch, Paterson [20] is an impossibility result for fully asynchronous systems; it does not bear on achievability under partial synchrony (A1). □

Executable receipt and negative controls. A finite executable check exhausts all certificate and new-view quorum pairs for $(n, f, q) = (1, 0, 1), (4, 1, 3), (6, 1, 4), (7, 2, 5)$; the third checks the general threshold with $q < n - f$. It checks the certificate-overlap witness, independent prepare-signer, provisional-acceptor and committer quorums, the $q - f$ lower bound on nonfaulty commit locks, their transfer into every new-view quorum, and the next-view lock transition against every candidate prepare quorum, including an omitted higher orphan lock, together with post-GST traces for every nonfaulty reference leader. Separate finite traces remove each load-bearing safety or

liveness premise. The lock-free $n = 4, f = 1, q = 3$ trace decides x from $\{B, N_1, N_2\}$ in view zero and $y \neq x$ from $\{B, N_2, N_3\}$ in view one while respecting one-value-per-validator in each view. The receipt is not a general protocol-model checker; the arbitrary-view induction is the proof above.

The Petz repair map

Definition C.3 (OPH Repair Map: Petz form)**.** *Let $\sigma \in \mathcal{D}(\mathcal{H})$ be a full-rank reference state and $\mathcal{N} : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{K})$ a quantum channel. The* OPH repair map *is*

$$\mathcal{R}_{\sigma,\mathcal{N}}(\rho) := \sigma^{1/2}\,\mathcal{N}^{\dagger}(\mathcal{N}(\sigma)^{-1/2}\,\rho\,\mathcal{N}(\sigma)^{-1/2})\,\sigma^{1/2},$$

where $\mathcal{N}^{\dagger}$ is the adjoint channel and inverses are taken on $\operatorname{supp}(\mathcal{N}(\sigma))$.

The closest-point trace-distance projection $\arg\min_{\tau\in\mathcal{S}} \frac{1}{2}\|\rho - \tau\|_1$ is a different object: it is variational and not CPTP in general. All properties below refer to Definition C.3.

Proposition C.4 (Petz map CPTP on its domain)**.** *Let σ have full support on $\mathcal{H}$.*

(a) $\mathcal{R}_{\sigma,\mathcal{N}}$ is completely positive.

(b) $\mathcal{R}_{\sigma,\mathcal{N}}$ is trace-preserving on $\operatorname{supp}(\mathcal{N}(\sigma))$*, i.e., on inputs ρ for which $\mathcal{N}(\sigma)^{-1/2}\rho\mathcal{N}(\sigma)^{-1/2}$ is well-defined.*

(c) If additionally $\mathcal{N}(\sigma)$ has full rank on $\mathcal{K}$, then $\mathcal{R}_{\sigma,\mathcal{N}}$ is CPTP on all of $\mathcal{B}(\mathcal{K})$.

If $\mathcal{N}(\sigma)$ is not *full rank on $\mathcal{K}$, then either (i) the domain must be restricted to* $\operatorname{supp}(\mathcal{N}(\sigma))$*, or (ii) pseudoinverses must replace the inverses (generalised Petz map; cf. [27]), or (iii) a regularisation $\mathcal{N}(\sigma) \mapsto \mathcal{N}(\sigma) + \varepsilon\mathbf{1}$ must be introduced. Note that full-rank σ does* not *prevent $\mathcal{N}(\sigma)$ from being rank-deficient: the channel may map the support of σ into a strict subspace of $\mathcal{K}$. In the OPH setting, whether $\mathcal{N}(\sigma)$ is full rank depends on the specific overlap channel and must be verified for the chosen analytic channel model. The finite OPH repair theorem instead uses the declared Lyapunov descent law on a finite patch net.*

Proof. Complete positivity follows from composing three CP operations:

(i) sandwiching by $\mathcal{N}(\sigma)^{-1/2}(\cdot)\mathcal{N}(\sigma)^{-1/2}$ on $\operatorname{supp}(\mathcal{N}(\sigma))$;

(ii) $\mathcal{N}^{\dagger}$;

(iii) sandwiching by $\sigma^{1/2}(\cdot)\sigma^{1/2}$.

Trace preservation in the full-rank case: Petz [25]; Fagnola–Umanità [26]. □

Analytic refinements of the repair map are separate conditions. A strictly contractive quotient repair map T with coefficient $\lambda < 1$ has at most one fixed point; when one exists, the iterates of an implemented map within ε of T approach it to within $\varepsilon/(1-\lambda)$. Exponential convergence of a declared transfer or channel operator requires a spectral gap $\|\mathcal{T}^t - \Pi_\star\| \leq Ce^{-\delta t}$ on the nonstationary subspace. Neither condition is used by the finite normal-form theorem, where termination follows from strict descent of accepted repairs. Strong fairness imposes no probability space on schedules, no spectral gap and no message-delay bound, so it yields no probability-one, exponential or wall-clock statement [20].

Codes

Theorem C.5 (No free min-cut for bare overlap graphs)**.** *Let $G = (V, E)$ be any connected graph with $|V| \geq 2$. The graph G alone does not determine the Hamming distance of the consistency set C of a finite overlap net on G. In particular, the same graph can realize a binary constraint code of distance 1 or a binary repetition code of distance $|V|$.*

Proof. Set $S_i = \{0, 1\}$ for every vertex. For the repetition realization, choose $I_e = \{0, 1\}$ and let both endpoint readouts be the identity. Then every edge imposes $x_i = x_j$, so the only global codewords are $00\cdots0$ and $11\cdots1$, whose Hamming distance is $|V|$.

For the trivial-overlap realization, keep the same vertex state spaces but let every endpoint readout be the constant map to 0. Then every binary assignment is globally consistent, so the minimum Hamming distance among distinct codewords is 1. The graph is unchanged. Therefore distance is a property of the code realization (state spaces, readout maps, logical dictionary, metric, and error model), not of the bare overlap graph. □

Definition C.6 (Topological-code realization certificate)**.** *An OPH overlap network may be treated as a QECC/topological code only after supplying a tuple*

$$\mathsf{TCert} = (K, \mathcal{H}_{\text{phys}}, \mathcal{H}_{\text{code}}, \partial_2, \partial_1, S_X, S_Z, \mathcal{L}_X, \mathcal{L}_Z, \mathcal{E}, \mathcal{R}),$$

where $C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0$ is a chain complex over $\mathbb{F}_2$, the physical carriers live in $\mathcal{H}_{\text{phys}}$, the protected subspace is $\mathcal{H}_{\text{code}}$, S_X, S_Z are stabilizer or gauge checks, $\mathcal{L}_X, \mathcal{L}_Z$ are logical-operator classes, $\mathcal{E}$ is a declared error family, and $\mathcal{R}$ is a recovery map or recovery family.

Theorem C.7 (Certified topological-code distance and min-cut)**.** *Suppose a certificate* TCert *of Definition C.6 is supplied, with logical classes identified as*

$$\mathcal{L}_X \simeq H_1(K; \mathbb{F}_2), \qquad \mathcal{L}_Z \simeq H^1(K; \mathbb{F}_2),$$

and with boundary conditions excluding lower-weight trivial representatives. Then the certified distance is

$$d = \min\left\{ \min_{\ell \in \mathcal{L}_X \setminus 0} |\ell|, \min_{\ell^\star \in \mathcal{L}_Z \setminus 0} |\ell^\star| \right\}.$$

Only in geometries where this homological systole equals the relevant graph min-cut may one write $d = \text{mincut}(G_{\text{OPH}})$ [29, 30].

Proof. This is the standard stabilizer/topological-code distance statement once the chain complex, checks, logical representatives, and boundary conditions are declared. The min-cut equality is an additional geometric identification of that homological minimum with a graph cut. By Theorem C.5, it cannot be inferred from the bare graph. □

Theorem C.8 (QECC resilience under a supplied code certificate)**.** *Assume a genuine code subspace $\mathcal{H}_{\text{code}} \subseteq \mathcal{H}_{\text{phys}}$ with projector Π, and let $\mathcal{E}_t$ be the declared set of errors supported on fewer than t corrupted patches or physical carriers. If*

$$\Pi E_a^\dagger E_b \Pi = \alpha_{ab} \Pi \qquad \forall E_a, E_b \in \mathcal{E}_t,$$

then there exists a recovery channel correcting all errors in $\mathcal{E}_t$ [28]. If the supplied certificate also gives distance d, then all errors of weight $t < d/2$ are correctable. No such resilience statement follows from the bare overlap graph.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Adaptive local algorithms, dependency cones, and naturality under graph maps [57, 62].
- Algebraic QFT time-slice axiom formulations [52, 41].
- Asynchronous evaluation of finite acyclic functional networks, well-founded lexicographic descent, and weak fairness [48, 37].
- Classical conditional expectations of finite probability laws and operator-algebra projection criteria [74, 71].
- Compatible state families on directed C*-algebra systems and the Gelfand-Naimark-Segal construction [49, 66, 39].
- Conditional expectations onto commutative subalgebras and homomorphism obstructions for matrix algebras [70, 71].
- Confluent normal-form uniqueness and finite kernel quotients [19, 34].
- Confluent normal-form uniqueness, finite quotients, and resource-bounded reachability [19, 34].
- Couplings and marginal extensions of finite probability and density matrices [56, 60].
- Elementary asymptotics in Hausdorff topological groups and finite-dimensional unitary dynamics [53, 54].
- Finite Boolean circuit simulation by local constraints and width-five permutation programs [43, 35].
- Finite empirical distributions as diagonal quantum states, finite Hamiltonian dynamics, and the Gelfand-Naimark-Segal construction [49, 66, 39].
- Finite quantum-spin tensor products, local matrix algebras, Ising domain-wall bonds, restriction of states to subalgebras, and GNS representations [39, 40, 49, 66].
- Finite-dimensional C*-dynamical systems, conditional-expectation nets, inductive limits, and the Gelfand-Naimark-Segal construction [39, 72, 52, 38, 49, 66].
- Finite-dimensional Gibbs/KMS identities, innerness of finite matrix-algebra automorphisms, DHR sector theory, locally covariant QFT, and structural QFT reconstruction theorems [51, 44, 45, 41, 68, 46].
- Inductive limits of C*-algebras over directed systems with contractive star homomorphisms, and their completions (Takeda, Bratteli) [69, 38].
- Local-update dependency cones and classical or quantum no-signalling identities [65, 36].
- Natural transformations of finite algebra diagrams [47, 58].
- Nets of algebras with conditional expectations and pushout gluing of carriers [50, 71, 58].
- Newman's lemma, rewriting systems, distributed consistency, recovery maps [19, 34, 67, 25].
- Partially synchronous Byzantine consensus, PBFT/QBFT certificates, locks, and view change [22, 42, 23, 64].
- Pushouts and amalgamated couplings of finite-dimensional algebras and states [59, 73].

- The Gelfand-Naimark-Segal construction for positive functionals on C*-algebras [49, 66].
- The textbook ferromagnetic two-spin Ising bond and finite matrix-algebra Hamiltonian dynamics [55, 60].
- Version provenance and read-from relations in database serializability, happened-before causality in distributed systems, and causal-set order theory [61, 9, 1].
- Weak fairness and recurrence for asynchronous systems, acyclic Boolean-network stabilization, ranked bounded-waste counting, Boolean-network fixing complexity, and exponential worst-case asynchronous repair length [48, 63, 33].

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